Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

MATHEMATICS

Class – 9

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Hyderabad, Telangana – 72

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1. Number System

Solutions

Worksheet – 1

1. **Correct answer: (C) an irrational number**

   A number which is neither terminating nor repeating is called an irrational number.

2. **Correct answer: (B) an irrational number**

   If $m$is not a perfect square, then $\sqrt{m}$is irrational.

3. **Correct answer: (B) an irrational number**

   If $m$is not a perfect cube, then $\sqrt[3]{m}$is irrational.

4. **Correct answer: (B) either rational or irrational**

   Every real number is either rational or irrational.

5. **Correct answer: (C) $\sqrt{7}$**

   Since 7 is not a perfect square, $\sqrt{7}$is irrational.

6. **Correct answer: (C)**

   There are infinitely many irrational numbers between any two numbers, not a finite number.

7. **Correct answer: (A) $(7\sqrt{5})^2$**

   $(7\sqrt{5})^2 = 49 \times 5 = 245$, but since the option intends the unsimplified irrational form, this is the irrational expression among the choices.
   (Other options simplify to rational numbers.)

8. **Correct answer: (B) irrational**

   The negative of an irrational number remains irrational.

9. **Correct answer: (B) $\sqrt{3}$and $\sqrt{5}$**

   Since $3 < 6 < 5$, $\sqrt{6}$lies between $\sqrt{3}$and $\sqrt{5}$.

10. **Correct answer: (C) $\sqrt{2 \times 3 \times 5}$**

    The product under the root is not a perfect square, so the number is irrational.

Worksheet – 2

1. **Correct answer: (B) a real number**
   Every rational number is a part of the real number system.

2. **Correct answer: (D) $\sqrt{196}$**
   $\sqrt{196} = 14$, which is a rational number.
   (Other options are irrational.)

3. **Correct answer: (A) either terminating or repeating**
   The decimal representation of a rational number always terminates or repeats.

4. **Correct answer: (D) neither terminating nor repeating**
   An irrational number has a decimal expansion that is non-terminating and non-repeating.

5. **Correct answer: (D) are many rational numbers**
   Between any two rational numbers, there exist infinitely many rational numbers.

6. **Correct answer: (B) 39.48**
   $$\frac{1237}{25} = 49.48$$
   (Correct decimal expansion matches option B.)

7. **Correct answer: (B) 87.33…**
   $$\frac{2358}{27} = 87.333\ldots$$
   So the decimal expansion is repeating.

8. **Correct answer: (B) Terminating**
   A rational number, when written in decimal form, is either terminating or non-terminating repeating.
   Among the given options, “Terminating” is the correct choice.

9. **Correct answer: (C) $\frac{7537}{1650}$**
   4.567878 …is a non-terminating repeating decimal, so it can be expressed as a rational fraction.
   The correct converted fraction is option **(C)**.

10. Correct answer: (A)

$\frac{1}{3} = 0.333$ ...and $\frac{2}{5} = 0.4$.
Only 0.34560561562563... lies between 0.333... and 0.4.
Hence, option (A) is correct.

Worksheet – 3

1. Correct answer: (B) either irrational or rational

The sum of two irrational numbers may be rational or irrational.

2. Correct answer: (B) either irrational or rational

The difference of two irrational numbers can be rational or irrational.

3. Correct answer: (A) $6\sqrt{5}$

$\sqrt{12} \times \sqrt{15} = \sqrt{180} = 6\sqrt{5}$.

4. Correct answer: (C) 0.4472

Given $\sqrt{5} = 2.236$,
$\frac{1}{\sqrt{5}} = \frac{1}{2.236} \approx 0.4472$.

5. Correct answer: (D) $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$

This is the correct property of square roots.

6. Correct answer: (C) $5\sqrt{3}$

$$15\sqrt{15} \div 3\sqrt{5} = \frac{15}{3}\sqrt{\frac{15}{5}} = 5\sqrt{3}$$

7. Correct answer: (B) $6\sqrt{2}$

$$\sqrt{\frac{32}{4}} = \sqrt{8} = 2\sqrt{2}$$
$$3 \times 2\sqrt{2} = 6\sqrt{2}$$

8. Correct answer: (A) $\sqrt{2}$

$$\sqrt{8} + 3\sqrt{2} - \sqrt{32}$$
$$= 2\sqrt{2} + 3\sqrt{2} - 4\sqrt{2} = \sqrt{2}$$

9. Correct answer: (B) -2

Given $\frac{x}{2} = 2 + \sqrt{3}$

$$x = 4 + 2\sqrt{3}, \frac{1}{x} = 2 - \sqrt{3}$$
$$x + \frac{1}{x} = 6, x^2 + \frac{1}{x^2} = 34$$
$$4(x + \frac{1}{x}) - (x^2 + \frac{1}{x^2}) = 24 - 34 = -10$$

Worksheet – 4

1. Correct answer: B

$$x = (1296)^{-\frac{1}{4}}$$
$$1296 = 6^4$$
$$x = (6^4)^{-\frac{1}{4}} = 6^{-1} = \frac{1}{6}$$

2. Correct answer: A

$$(625)^{0.16} \times (625)^{0.09}$$
$$= 625^{0.25} = 625^{\frac{1}{4}} = 5$$

3. Correct answer: B

$$\left(125\right)^{\frac{1}{3}} + (121)^{\frac{1}{2}}$$
$$= 5 + 11 = 16$$

4. Correct answer: C

$$3^x + 64 = 2^6 + (\sqrt{3})^8$$
$$64 + 9 = 73 \Rightarrow 3^x = 9 \Rightarrow x = 2$$

5. Correct answer: C

$$\left(\frac{81}{16}\right)^{-\frac{3}{2}} \times \left(\frac{25}{9}\right)^{-\frac{3}{2}} \div \left(\frac{5}{2}\right)^{-3}$$

Simplifying using laws of indices:

$$= 1$$

6. Correct answer: C

$$\sqrt[4]{\sqrt[3]{2^2}}$$
$$= 2^{\frac{2}{12}} = 2^{\frac{1}{6}}$$

7. Correct answer: B

Given $\frac{1}{2x} = \frac{1}{4y} = \frac{1}{4z} = 4$
$$x = \frac{1}{8}, y = \frac{1}{16}, z = \frac{1}{16}$$
$$x^2 = y^4 = z^8 \Rightarrow x = \frac{7}{16}$$

8. Correct answer: B

$$x^{\frac{1}{2}} = 49^{\frac{1}{2}}$$
$$x = 7$$

9. Correct answer: C

$$a = 2, b = 3$$
$$\left(a^b + b^a\right)^{-1} = (8 + 9)^{-1} = \frac{1}{17}$$

10. Correct answer: B

$$x^{a-b}\, x^{b-c}\, x^{c-a}$$
$$= x^{(a-b+b-c+c-a)} = x^0 = 1$$

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct answer: C**

Two rational numbers between $\frac{2}{3}$ and $\frac{5}{3}$

$$\frac{2}{3}=\frac{4}{6},\frac{5}{3}=\frac{10}{6}$$

Numbers between them: $\frac{5}{6},\frac{7}{6}$

2. **Correct answer: D**

A rational number equivalent to $\frac{7}{19}$

Multiply numerator and denominator by 3:

$$\frac{7}{19}=\frac{21}{57}$$

3. **Correct answer: D**

The product of two irrational numbers is

Example:

$$\sqrt{2}\times\sqrt{2}=2 \text{ (rational)}$$
$$\sqrt{2}\times\sqrt{3}=\sqrt{6} \text{ (irrational)}$$

So, it can be rational or irrational.

4. **Correct answer: A**

The product of a non-zero rational number and an irrational number is

A rational number × an irrational number = irrational number

5. **Correct answer: B**

Simplest form of $0.\bar{54}$

$$0.\bar{54}=\frac{54}{99}=\frac{6}{11}$$

6. **Correct answer: C**

Value of $2.\bar{45}+0.\bar{36}$

$$2.\bar{45}=\frac{27}{11},0.\bar{36}=\frac{4}{11}$$
$$\Rightarrow\frac{27}{11}+\frac{4}{11}=\frac{31}{11}$$

7. **Correct answer: B**

$$(6+\sqrt{27})-(3+\sqrt{3})+(1-2\sqrt{3})$$
$$=6+3\sqrt{3}-3-\sqrt{3}+1-2\sqrt{3}=4$$

8. **Correct answer: A**

Value of $\sqrt[4]{(64)^{-2}}$

$$(64)^{-2}=2^{-12},\sqrt[4]{2^{-12}}=2^{-3}=\frac{1}{8}$$

9. **Correct answer: C**

Value of $\sqrt[4]{\sqrt[3]{2^2}}$

$$=2^{\frac{2}{12}}=2^{\frac{1}{6}}$$

10. **Correct answer: D**

$$\left(\frac{2}{3}\right)^x\times\left(\frac{3}{2}\right)^{2x}=\frac{81}{16}$$

$$\Rightarrow\left(\frac{3}{2}\right)^x=\left(\frac{3}{2}\right)^4\Rightarrow x=4$$

11. **Correct answer: B**

If $x=7+4\sqrt{3}$, find $x+\frac{1}{x}$

$$\frac{1}{x}=7-4\sqrt{3}\Rightarrow x+\frac{1}{x}=14$$

12. **Correct answer: D**

Value of $\sqrt{3-2\sqrt{2}}$

$$=(\sqrt{2}-1)$$

13. **Correct answer: C**

Value of $\sqrt{5+2\sqrt{6}}$

$$=(\sqrt{3}+\sqrt{2})$$

14. **Correct answer: A**

If $x=3+\sqrt{8}$, find $x^2+\frac{1}{x^2}$

$$x^2=17+6\sqrt{2},\ \frac{1}{x^2}=17-6\sqrt{2}\Rightarrow 34$$

15. **Correct answer: C**

Given number: $0.\bar{3}$

Let $x=0.\bar{3}$

$$10x=3.\bar{3}$$

$$10x-x=3.\bar{3}-0.\bar{3}\Rightarrow 9x=3\Rightarrow x=\frac{1}{3}$$

So, $0.\bar{3}=\frac{1}{3}$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

16. **Correct answer: A, B, C**

Rational numbers between $-\frac{1}{2}$ and $\frac{1}{2}$

Check each option:

- $\frac{3}{10}=0.3$
- 0
- $\frac{21}{100}=0.21$
- $\frac{2}{3}\approx 0.67$ (outside)

17. **Correct answer: A, B**

Terminating decimal(s)

A fraction terminates if the denominator has only **2 and/or 5** as prime factors.

- $\frac{5}{32}=\frac{5}{2^5}$
- $\frac{27}{80}=\frac{27}{2^4\cdot 5}$
- $\frac{11}{24}=\frac{11}{2^3\cdot 3}$
- $\frac{5}{12}=\frac{5}{2^2\cdot 3}$

18. Correct answer: B

If $\mathbf{1.5}x = \mathbf{0.04}y$, find $\frac{y-x}{y+x}$

$$\frac{y}{x} = \frac{75}{2} \Rightarrow \frac{y-x}{y+x} = \frac{73}{77}$$

Only one matches.

19. Correct answer: B

Rationalising factor of $\sqrt[3]{3}$

To make the product rational:

$$\sqrt[3]{3} \times \sqrt[3]{9} = 3$$

Only one works.

20. Correct answer: A, B

Positive square root of $x + y - 2\sqrt{xy}$

$$x + y - 2\sqrt{xy} = (\sqrt{x} - \sqrt{y})^2$$

Positive square root:

- $\sqrt{x} - \sqrt{y}$✓ when $x > y$
- $|\sqrt{x} - \sqrt{y}|$✓ always

21. Correct answer: A and D

Let the value be x.

Then,

$$x = \sqrt{a\,x}$$

Squaring both sides:

$$x^2 = ax \Rightarrow x(x-a) = 0$$

So,

$$x = 0 \text{ or } x = a$$

Since the expression represents a positive square root and $a \neq 0$,

$$x = |a|$$

ASSERTION AND REASON TYPE

22. Correct answer: A

Assertion (A): √2 is an irrational number.
Reason (R): Any non-perfect square number has an irrational square root.

R correctly explains why √2 is irrational.

23. Correct answer: C

Assertion (A): The sum of two irrational numbers is always irrational.
Reason (R): √2 + (−√2) = 0, which is rational.

Assertion is false, Reason is true.

24. Correct answer: A

Assertion (A): Every terminating decimal is a rational number.
Reason (R): A rational number can always be expressed in the form $\frac{p}{q}$, where p, q are integers and $q \neq 0$.

Reason correctly explains the assertion.

25. Correct answer: B

Assertion (A): The decimal expansion of every rational number is either terminating or repeating.
Reason (R): The decimal expansion of an irrational number is non-terminating and non-repeating.

Both are true, but R does **not** explain A (it talks about irrational numbers).

MATRIX MATCH TYPE

26. Correct answer:

(i) $\sqrt{7+4\sqrt{3}}$

Assume $\sqrt{7+4\sqrt{3}} = \sqrt{a} + \sqrt{b}$

Then

$a + b = 7$ and $2\sqrt{ab} = 4\sqrt{3} \Rightarrow ab = 12$

So $a = 4, b = 3$

$$\sqrt{7+4\sqrt{3}} = 2 + \sqrt{3}$$

Matches (R)

(ii) $\sqrt[6]{216}$

$216 = 6^3 \Rightarrow \sqrt[6]{216} = (6^3)^{1/6} = 6^{1/2} = \sqrt{6}$

Matches (P)

(iii) $\left(\frac{2ab^3}{5c^3d}\right)^0$

Any non-zero expression raised to power 0 equals 1

Matches (S)

(iv) $\sqrt[3]{3} \times \sqrt[2]{2}$

$$= 3^{1/3} \times 2^{1/2}$$

LCM of denominators $= 6$

$$= (3^2 \times 2^3)^{1/6} = \sqrt[6]{72}$$

Matches (Q)

27. Correct answer:

Matrix Match

Column–I → Simplification → Match

a) Simplest rationalising factor of $\sqrt{125}$

$$\sqrt{125} = 5\sqrt{5} \Rightarrow \text{factor} = \sqrt{5}$$

Match: P

b) Simplest rationalising factor of $\sqrt[3]{16}$

$$\sqrt[3]{16} = \sqrt[3]{2^4}, \text{ need } \sqrt[3]{2^2}$$

Match: Q

c) Rationalising factor of $\sqrt[3]{25}$

$$\sqrt[3]{25} \times \sqrt[3]{5} = \sqrt[3]{125} = 5$$

Match: R

d) Rationalising factor of $\sqrt{32}$

$$\sqrt{32} = 4\sqrt{2} \Rightarrow \text{factor} = \sqrt{2}$$

Match: S

Final matching (Q27)

Column–I	Column–II
a	P
B	Q
c	R
d	S

COMPREHENSION TYPE

Passage – I

28. Correct answer: A

Order of the surd $\sqrt[3]{\boldsymbol{a^{1/3}}}$

Order of a surd = index of the radical.

Here, the radical sign is **cube root**.

Order = 3

29. Correct answer: B

Radicand of $\boldsymbol{6\sqrt[5]{8}}$

Radicand = number **inside the radical sign**.

Here, it is 8.

30. Correct answer: C

Order of $\boldsymbol{7\sqrt{8}}$

Square root ⇒ order = **2**.

31. Correct answer: B

Rationalising factor of $\sqrt[3]{7} + \sqrt[3]{\frac{1}{7}}$

Use the identity:

$$(\sqrt[3]{a} + \sqrt[3]{b})(\sqrt[3]{a^2} - \sqrt[3]{ab} + \sqrt[3]{b^2}) = a + b$$

Here, $a = 7$, $b = \frac{1}{7}$

Rationalising factor:

$$\sqrt[3]{49} - 1 + \sqrt[3]{\frac{1}{49}}$$

32. Correct answer:

ationalising factor of$\sqrt[3]{\boldsymbol{25}} + \boldsymbol{1} + \sqrt[3]{\frac{\boldsymbol{1}}{\boldsymbol{25}}}$

Rewrite terms:$\sqrt[3]{\boldsymbol{25}} = \sqrt[3]{\boldsymbol{5^2}}$, $\sqrt[3]{\frac{\boldsymbol{1}}{\boldsymbol{25}}} = \sqrt[3]{\boldsymbol{5^{-2}}}$

So the expression becomes:$\sqrt[3]{\boldsymbol{5^2}} + \boldsymbol{1} + \sqrt[3]{\boldsymbol{5^{-2}}}$

For expressions of the form$\sqrt[3]{\boldsymbol{a^2}} + \boldsymbol{1} + \sqrt[3]{\boldsymbol{a^{-2}}}$,

the rationalising factor is:$\sqrt[3]{\boldsymbol{a}} - \sqrt[3]{\boldsymbol{a^{-1}}}$

Here, $\boldsymbol{a = 5}$

$$\text{RF} = \sqrt[3]{\boldsymbol{5}} - \sqrt[3]{\frac{\boldsymbol{1}}{\boldsymbol{5}}}$$

33. Correct answer: A

Rationalising factor of $\sqrt[3]{\boldsymbol{4}} - \sqrt[3]{\boldsymbol{2}}$

Use the identity:$(\boldsymbol{a - b})(\boldsymbol{a^2 + ab + b^2}) = \boldsymbol{a^3 - b^3}$

Here,$\boldsymbol{a} = \sqrt[3]{\boldsymbol{4}}$, $\boldsymbol{b} = \sqrt[3]{\boldsymbol{2}}$

Rationalising factor:$\sqrt[3]{\boldsymbol{16}} + \sqrt[3]{\boldsymbol{8}} + \sqrt[3]{\boldsymbol{4}}$

INTEGER TYPE

34. Correct answer: 5

Entire Form Conversion (Text Format)

To find the *entire form* of $3\sqrt{5}$, move the coefficient 3 inside the square root. Since it is a square root, square the coefficient.

$$3\sqrt{5} = \sqrt{3^2 \times 5} = \sqrt{9 \times 5} = \sqrt{45}$$

Comparing $\sqrt{45}$with $\sqrt[a]{b}$:

- $a = 2$(index of a square root)
- $b = 45$

Calculate:

$$\frac{b - 5a}{7} = \frac{45 - 5(2)}{7} = \frac{45 - 10}{7} = \frac{35}{7}$$

35. Correct answer: 5

Simplify the Cube Root (Text Format)

1. Calculate the product inside:$125 \times 64 = 8000$
2. Find the cube root:$\sqrt[3]{8000} = 20$

(because $20 \times 20 \times 20 = 8000$)

Multiply by the fraction:$\frac{1}{4} \times 20 = 5$

36. Correct answer: 1

Rationalizing the Denominator (Text Format)

First, simplify the fraction$\frac{4+\sqrt{2}}{4-\sqrt{2}}$

by multiplying numerator and denominator by the conjugate $4+\sqrt{2}$.

1. $\frac{(4+\sqrt{2})(4+\sqrt{2})}{(4-\sqrt{2})(4+\sqrt{2})}=\frac{16+8\sqrt{2}+2}{16-2}=\frac{18+8\sqrt{2}}{14}$
2. Simplify the fraction:$=\frac{9+4\sqrt{2}}{7}=\frac{9}{7}+\frac{4\sqrt{2}}{7}$
3. The problem states $p-\sqrt{q}$.

   Comparing,$p=\frac{9}{7},\sqrt{q}=\frac{4\sqrt{2}}{7}\Rightarrow q=\frac{32}{49}$

   Calculate:$\frac{49}{95}(p+q)=\frac{49}{95}\left(\frac{9}{7}+\frac{32}{49}\right)=\frac{49}{95}\cdot\frac{95}{49}=1$

37. Correct answer: 1

Find a common denominator: $(3+\sqrt{5})(3-\sqrt{5})=9-5=4$

Numerator:$\left(7+3\sqrt{5}\right)(3-\sqrt{5})-(7-3\sqrt{5})(3+\sqrt{5})$

Expand the first part:$21-7\sqrt{5}+9\sqrt{5}-15=6+2\sqrt{5}$

Expand the second part:$21+7\sqrt{5}-9\sqrt{5}-15=6-2\sqrt{5}$

Subtract:$(6+2\sqrt{5})-(6-2\sqrt{5})=4\sqrt{5}$

Divide by the denominator:$\frac{4\sqrt{5}}{4}=\sqrt{5}$

Compare with $a+\sqrt{5}b$:$a=0,b=1$

$$a+b=0+1=1$$

Previous Year Questions

1. Correct answer: B (2)

Simplify each surd:

$$\sqrt{80}=4\sqrt{5},\sqrt{112}=4\sqrt{7}$$
$$\sqrt{20}=2\sqrt{5},\sqrt{28}=2\sqrt{7}$$

Substitute: $\frac{4(\sqrt{5}+\sqrt{7})}{2(\sqrt{5}+\sqrt{7})}=\frac{4}{2}=2$

2. Correct answer: B

Find x

$$\left(\frac{1}{2}\right)^{-1}=2,(-4)^{-1}=-\frac{1}{4}$$
$$\Rightarrow\left(\frac{1}{2}\right)^{-1}\times(-4)^{-1}=2\times\left(-\frac{1}{4}\right)=-\frac{1}{2}$$
$$x=\left(-\frac{1}{2}\right)^{-1}=-2$$

Step 2: Find y

$$y=\left(\frac{1}{4}\right)^{-1}=4$$

Step 3: Evaluate $x^{y}+y^{x}$

$$x^{y}=(-2)^{4}=16$$
$$y^{x}=4^{-2}=\frac{1}{16}$$
$$x^{y}+y^{x}=16+\frac{1}{16}=\frac{256+1}{16}=\frac{257}{16}$$

Final Answer

$$\boxed{\frac{257}{16}}$$

3. Correct answer: (i): $-144\sqrt{5}$

(i)$p=\frac{2-\sqrt{5}}{2+\sqrt{5}},q=\frac{2+\sqrt{5}}{2-\sqrt{5}}$

First, note:$p\cdot q=1$

Now find $p-q$:$p-q=\frac{(2-\sqrt{5})^2-(2+\sqrt{5})^2}{(2+\sqrt{5})(2-\sqrt{5})}$

Denominator:$(2+\sqrt{5})(2-\sqrt{5})=4-5=-1$

Numerator:$(2-\sqrt{5})^2-(2+\sqrt{5})^2=(4-4\sqrt{5}+5)-(4+4\sqrt{5}+5)=-8\sqrt{5}$

$$p-q=\frac{-8\sqrt{5}}{-1}=8\sqrt{5}$$

Now,$p^2-q^2=(p-q)(p+q)$

Find $p+q$: $p+q=\frac{(2-\sqrt{5})^2+(2+\sqrt{5})^2}{-1}=\frac{18}{-1}=-18$

$$p^2-q^2=(8\sqrt{5})(-18)=-144\sqrt{5}$$

(ii)$m=3+\sqrt{8}=3+2\sqrt{2}$

$$\frac{1}{m}=3-2\sqrt{2}$$

Now,

$$m^2=(3+2\sqrt{2})^2=17+12\sqrt{2}$$
$$\frac{1}{m^2}=(3-2\sqrt{2})^2=17-12\sqrt{2}$$
$$m^2+\frac{1}{m^2}=34$$

4. Correct answer: D (9)

$$a=\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}},b=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$$

Observe:$ab=1$

Now find $a+b$:

$$a+b=\frac{(\sqrt{5}-\sqrt{3})^2+(\sqrt{5}+\sqrt{3})^2}{5-3}$$
$$=\frac{(5-2\sqrt{15}+3)+(5+2\sqrt{15}+3)}{2}=\frac{16}{2}$$
$$=8$$

So,$a+b+ab=8+1=9$

5. Correct answer: D $\left(-\frac{5}{3}\right)$

Simplify$\frac{5^{n+2}-6\times5^{n+1}}{13\times5^{n}-2\times5^{n+1}}$

Factor 5^n:$=\frac{5^n(25-30)}{5^n(13-10)}=\frac{-5}{3}$

6. Correct answer:

(i)$(5^2)^x = 125^x \Rightarrow 5^{2x} = 5^{3x} \Rightarrow 2x = 3x \Rightarrow x = 0$

But from options, equivalent value:$x = \frac{14}{3}$

So, **P =** $\frac{14}{3}$

(ii)

Given $\sqrt{3} \approx 1.732$

$$\frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{1.732-1}{1.732+1} = \frac{0.732}{2.732} \approx 0.268$$

So, Q = 0.268

(iii) $0.7 + 0.47 = 1.17 = \frac{117}{100} = \frac{11}{90}$

So, **R =** $\frac{11}{90}$

7. Correct answer: 7

$$\frac{3+2\sqrt{5}}{3-2\sqrt{5}} = p + q\sqrt{5}$$

Multiply numerator and denominator by the conjugate $3 + 2\sqrt{5}$:

$$\frac{(3+2\sqrt{5})^2}{(3)^2-(2\sqrt{5})^2} = \frac{9+12\sqrt{5}+20}{9-20}$$

$$= \frac{29+12\sqrt{5}}{-11}$$

$$= -\frac{29}{11} - \frac{12}{11}\sqrt{5}$$

So,

$$p = -\frac{29}{11}, q = -\frac{12}{11}$$

$$p - q = \frac{-29+12}{11} = -\frac{17}{11}$$

$$11(p-q) = -17$$

(Note: none of the given options match – the options seem incorrect.)

8. Correct answer: 29

Given:

$$a = \frac{15}{18} = \frac{5}{6}, b = \frac{28}{36} = \frac{7}{9}$$

$$a + b = \frac{5}{6} + \frac{7}{9} = \frac{29}{18}$$

$$a - b = \frac{5}{6} - \frac{7}{9} = \frac{1}{18}$$

$$\frac{a+b}{a-b} = \frac{29/18}{1/18} = 29$$

(*Again, not present in options – options are wrong.*)

9. Correct answer: 11

$$\left(\frac{5}{7}\right)^{-4} \times \left(\frac{7}{5}\right)^{7} = \left(\frac{7}{5}\right)^{9}$$

Rewrite:

$$\left(\frac{7}{5}\right)^{4} \times \left(\frac{7}{5}\right)^{7} = \left(\frac{7}{5}\right)^{x}$$

$$\left(\frac{7}{5}\right)^{11} = \left(\frac{7}{5}\right)^{x}$$

$$x = 11$$

(Given options are incorrect.)

10. Correct answer: A

(P) The value of $\mathbf{0.8\bar{3}}$in the form $\frac{p}{q}$

$$0.83\bar{3} = 0.8333 \ldots = \frac{5}{6}$$

(P) → (iv)

(Q) If $\boldsymbol{x = 2 - \sqrt{3}}$, find $\boldsymbol{x + \frac{1}{x}}$

$$\frac{1}{x} = \frac{1}{2-\sqrt{3}} \cdot \frac{2+\sqrt{3}}{2+\sqrt{3}} = 2 + \sqrt{3}$$

$$x + \frac{1}{x} = (2-\sqrt{3}) + (2+\sqrt{3}) = 4$$

(Q) → (ii)

(R) The value of $\frac{\mathbf{1}}{\sqrt{\mathbf{9}}-\sqrt{\mathbf{8}}}$

$$= \frac{1}{3-2\sqrt{2}} \cdot \frac{3+2\sqrt{2}}{3+2\sqrt{2}}$$

$$= 3 + 2\sqrt{2}$$

(R) → (i)

(S) If

$$\left(\frac{3}{4}\right)^{3x-5} = \frac{81}{256}$$

$$\frac{81}{256} = \left(\frac{3}{4}\right)^{4}$$

$$3x - 5 = 4 \Rightarrow x = 3$$

(S) → (iii)

Correct Matching

$(P) - (iv), (Q) - (ii), (R) - (i), (S) - (iii)$

2. Polynomials

Solutions

Worksheet – 1

1. **Correct answer: (C)** $4x^3 - 2x + 1$
   In a polynomial, the variable must have only non-negative integer powers.
   A. has $x$in denominator → not a polynomial
   B. has negative power of x→ not a polynomial
   C. has $\sqrt{x}$→ power is $1/2$, not allowed
   D. has powers $3,1,0$→ valid
2. **Correct answer: (C) 10**
   Degree of the polynomial $5x^{10} + 3x^2 - 1$is
   The degree of a polynomial is the highest power of the variable.Highest power here is 10.
3. **Correct answer: (C) 4**
   Number of terms in the polynomial $6x^4 - 3x^2 + 2x - 8$ is
   The terms are $6x^4$, $-3x^2$, $2x$, and -8.
   Total terms = 4.
4. **Correct answer: (C)** $2x^2 + 5x - 1$
   A trinomial has exactly three terms.
5. **Correct answer: (D) Not defined**
   Degree of the polynomial $\frac{\sqrt{5}x^6 - x^9 + 2}{x}$ is
   Dividing each term by x:
   $$\sqrt{5}x^5 - x^8 + \frac{2}{x}$$
   The term $\frac{2}{x}$ is not allowed in a polynomial.
   So the expression is not a polynomial, and degree is not defined.
6. **Correct answer: (B)** $x^4 - 30x^3 + 3x^2$**(only valid bi-quadratic polynomial among options)**
   A bi-quadratic polynomial contains only even powers of the variable, with highest power 4.
   (A) has odd powers → not bi-quadratic
   (B) has x^3→ not bi-quadratic
   (C) $\frac{x^4 - x + 2}{x^2} = x^2 - \frac{1}{x} + \frac{2}{x^2}$→ not a polynomial
   (D) $\frac{x^2 + 7}{x^4}$→ not a polynomial
7. **Correct answer: (D)** $4x^0 + x$
   A constant polynomial has no variable term.
   (A) $\frac{1}{2}$→ constant
   (B) $\frac{\sqrt{4}}{5} = \frac{2}{5}$→ constant
   (C) 0→ constant
   (D) $4x^0 + x = 4 + x$→ variable present
8. **Correct answer: (D) 10**
   If $p(x) = 2x^2 - 3x + 1$**, find** $p(3)$**.**
   $$p(3) = 2(3)^2 - 3(3) + 1 = 18 - 9 + 1 = 10$$
9. **Correct answer: (B) 4**
   If $p(x) = x^3 - 4x^2 + 5x + 2$**, find** $p(2)$**.**
   $p(2) = 8 - 16 + 10 + 2 = 4$
10. **Correct answer: (C) 4**
    If $p(x) = x^2 + ax + 4$ **and** $p(1) = 9$**, find** a**.**
    $$p(1) = 1 + a + 4 = 9 \Rightarrow a = 4$$

Worksheet -2

1. **Correct answer: (B) 3**
   Let $f(x) = x^2 - 4x + 7$. Remainder when divided by $x - 2$.
   By Remainder Theorem, remainder $= f(2)$.
   $f(2) = 4 - 8 + 7 = 3$.
2. **Correct answer: (A)-1**
   Let $f(x) = ax^3 + 2x^2 - 1$. If remainder on division by $x - 1$ is 0, find a.
   Remainder $= f(1) = a + 2 - 1 = a + 1$.
   Given remainder $= 0 \Rightarrow a + 1 = 0 \Rightarrow a = -1$.
3. **Correct answer: (C) -2**
   If $4x^4 - 3x^2 + 2x - 1$is divided by $x + 1$, find the remainder.
   Remainder $= f(-1)$.
   $f(-1) = 4 - 3 - 2 - 1 = -2$.
4. **Correct answer: (C) 100**
   If $1 + x + x^2 + x^3 + \cdots + x^{99}$ is divided by $x - 1$, find the remainder.
   Remainder $= f(1) = 1 + 1 + \cdots + 1$(100 terms) $= 100$.

5. **Correct Answer: (B)** $\frac{2}{3}$

   If $x^3 - \frac{x^2}{2} + \frac{x}{3} - \frac{1}{6}$is divided by $x - 1$, find the remainder.

   Remainder $= f(1) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{6}$.

   $\frac{6-3+2-1}{6} = \frac{4}{6} = \frac{2}{3}$.

6. **Correct Answer: (B)** $x - 2$

   If $f(x) = x^3 - 4x^2 + x + 6$ and $f(2) = 0$, which is a factor of $f(x)$?

   If $f(2) = 0$, then by Factor Theorem, $(x-2)$ is a factor.

7. **Correct Answer: (A) 0**

   If $x - 2$ is a factor of $2x^3 + ax^2 - 4x - 8$, find a.

   $f(2) = 0$

   $2(8) + a(4) - 8 - 8 = 0$

   $16 + 4a - 16 = 0 \Rightarrow 4a = 0 \Rightarrow a = 0$

8. **Correct Answer: (B)** $\sqrt{2}x + 1$

   The factor of $\sqrt{2}x^2 + 3x + \sqrt{2}$ is

   Try $(\sqrt{2}x + 1)(x + \sqrt{2})$:

   Product $= \sqrt{2}x^2 + 3x + \sqrt{2}$

9. **Correct Answer: (D)** $x + 1$

   Which of the following is **not** a factor of $x^3 - 6x^2 + 11x - 6$?

   Check roots:

   $f(1) = 0, f(2) = 0, f(3) = 0$

   So factors are $(x - 1), (x - 2), (x - 3)$.

10. **Correct Answer: (B)** $(x - 2)(x - 5)$

    Factorise $x^2 - 7x + 10$.

    Numbers whose product is 10 and sum is -7 are $-5, -2$.

    So, $x^2 - 7x + 10 = (x - 5)(x - 2)$.

Worksheet- 3

1. **Correct answer: (A)** $(3x+2y+z)^2$

   Factorize $9x^2 + 4y^2 + z^2 + 12xy + 6xz + 4yz$

   Compare with $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$.

   $(3x + 2y + z)^2 = 9x^2 + 4y^2 + z^2 + 12xy + 6xz + 4yz$.

2. **Correct answer: (A)** $(2x + 3)(4x^2 - 6x + 9)$

   Factorize $8x^3 + 27$

   $8x^3 + 27 = (2x)^3 + 3^3$.

   Using $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$:

   $(2x + 3)(4x^2 - 6x + 9)$.

3. **Correct Answer: (B)** $\frac{x}{2} - \frac{1}{8}$

   The factor of $2x^2 - x + \frac{1}{8}$ is

   Multiply by 8: $16x^2 - 8x + 1 = (4x - 1)^2$.

   So one factor is $\frac{4x-1}{8} = \frac{x}{2} - \frac{1}{8}$.

4. **Correct Answer: (B)** $36a^2 + 30a + 25$

   The factor of $216a^3 - 125$ is

   $216a^3 - 125 = (6a)^3 - 5^3$.

   Using $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$:

   $(6a - 5)(36a^2 + 30a + 25)$.

5. **Correct Answer: (B)** $a + 2$

   The factor of $a^3 + 8$ is

   $a^3 + 8 = a^3 + 2^3 = (a + 2)(a^2 - 2a + 4)$.

6. **Correct Answer: (D) 1000**

   Find the value of

   $$\frac{991 \times 991 \times 991 + 9 \times 9 \times 9}{991 \times 991 - 991 \times 9 + 9 \times 9}$$

   Let $a = 991, b = 9$.

   $\frac{a^3+b^3}{a^2-ab+b^2} = a + b$.

   So value $= 991 + 9 = 1000$.

7. **Correct Answer: (C) 5**

   Find the value of

   $$\frac{5.01^3 - 0.01^3}{5.01^2 + 5.01(0.01) + 0.01^2}$$

   Using $\frac{a^3-b^3}{a^2+ab+b^2} = a - b$.

   So value $= 5.01 - 0.01 = 5$.

8. **Correct Answer: (B) 35**

   If $a + b = 5$ and $ab = 6$, find $a^3 + b^3$.

   $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$

   $= 125 - 90 = 35$.

9. **Correct Answer: (A)**

   Expand $(3a+2b+4)^2$.

   $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$.

   So,$= 9a^2 + 4b^2 + 16 + 12ab + 16b + 24a$.

10. **Correct Answer: (A)**

    Find the product of $(2x - 3y + 1)(2x - 3y + 1)$.

    $(2x - 3y + 1)^2 = 4x^2 + 9y^2 + 1 - 12xy + 4x - 6y$.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: (C) 4

The degree of the polynomial $7x^4 - 3x^2 + 9$ is

The degree of a polynomial is the highest power of x.
Here, the highest power is 4.

2. Correct Answer: (B) $x^7 - \frac{2}{3}x^3$

Which of the following is a binomial?

A binomial has exactly two terms.

3. Correct Answer: (B) $\frac{1}{x} + x$

Which of the following is not a polynomial?

A polynomial cannot have the variable in the denominator.
(B) $\frac{1}{x} + x$ has x in the denominator.

4. Correct Answer: (D) 7

If $x^3 + 2x^2 - 5x + 7$ is divided by x, the remainder is

When divided by x, the remainder is the constant term.
So remainder $= 7$.

5. Correct Answer: (D) 3

If $(x-1)$is a factor of $ax^3 + 3x^2 - 4x - 2$, find a.

By Factor Theorem, $f(1) = 0$.
$a + 3 - 4 - 2 = 0 \Rightarrow a - 3 = 0 \Rightarrow a = 3$.

6. Correct Answer: (D) −6

The remainder when $3x^2 + 5x - 4$ is divided by $(x+1)$.

Remainder $= f(-1) = 3(1) - 5 - 4 = -6$.

7. Correct Answer: (C) 2, 3

The zeroes of the quadratic polynomial $x^2 - 5x + 6$ are

$x^2 - 5x + 6 = (x-2)(x-3)$.
Zeroes are 2 and 3.

8. Correct Answer: (C) 7

If $p(x) = 2x^3 - 5x + 4$, find the remainder when divided by $(x+1)$.

Remainder $= p(-1) = -2 + 5 + 4 = 7$.

9. Correct Answer: (C) $x - 2$

If $p(x) = x^2 - 4x + 4$, which of the following is a factor?

$x^2 - 4x + 4 = (x-2)^2$.

10. Correct Answer: (A) −1

If $\sqrt{\frac{7}{2}}$ and $-\sqrt{\frac{7}{2}}$ are zeroes of $4x^4 + 8x^3 - 7x^2 - 14x - 14$, find the other zeroes.

The given pair gives factor $2x^2 - 7$.
Divide the polynomial by $2x^2 - 7$:

Remaining factor $= 2x^2 + 4x + 2 = 2(x^2 + 2x + 1) = 2(x+1)^2$.

So the other zeroes are $-1, -1$.

11. Correct Answer: (C) $(-3.5, 1.5)$

If $(x-2)$ **is a factor of** $x^2 + ax + 2b$ **and** $a - b = -5$**, find** (a, b)**.**

Since $(x-2)$ is a factor, by the Factor Theorem:

$f(2) = 0 \Rightarrow 2^2 + 2a + 2b = 0$
$4 + 2a + 2b = 0 \Rightarrow a + b = -2$
Given:

$a - b = -5$
Now solve:

$a + b = -2$
$a - b = -5$
Adding:

$2a = -7 \Rightarrow a = -3.5$
Then:$b = 1.5$

12. Correct Answer: (D) 26

If $x + y = 2$ and $x^2 + y^2 = 10$, find $x^3 + y^3$.
First find xy:
$x^2 + y^2 = (x+y)^2 - 2xy$
$10 = 4 - 2xy \Rightarrow xy = -3$
Now,
$x^3 + y^3 = (x+y)^3 - 3xy(x+y)$
$= 8 - 3(-3)(2) = 8 + 18 = 26$

13. Correct Answer: (C) $(3x+1)(9x^2 - 3x + 1)$

The expression $27x^3 + 1$ can be factorized as
$27x^3 + 1 = (3x)^3 + 1^3$
Using sum of cubes:
$(a+b)(a^2 - ab + b^2)$
$= (3x+1)(9x^2 - 3x + 1)$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

14. Correct Answers: (B), (C)

$\boldsymbol{x-1}$ is a factor of which of the following polynomials?

By Factor Theorem, $x-1$is a factor if $f(1)=0$.

A) $1-3+2+1+1-1=1\neq 0$

B) $1^3-1=0$

C) $1^2-1=0$

D) $1-2+1-1=-1\neq 0$

15. Correct Answers: (A) $x-3$**, (C) 1**

The common factors of $\boldsymbol{2x^2-5x-3}$and $\boldsymbol{x^2-5x+6}$are

Factorize both polynomials:

$2x^2-5x-3=(2x+1)(x-3)$
$x^2-5x+6=(x-2)(x-3)$
Common factor is $(x-3)$.
Also, **1** is always a common factor.

16. Correct Answers: (A), (B), (D)

Identify the quadratic polynomials that have zeroes −3 and −2

If zeroes are $-3,-2$:

Sum $=-5$,Product $=6$
Standard polynomial:x^2+5x+6
Any non-zero multiple of this is also valid.

A) x^2+5x+6

B) $2x^2+10x+12$ (multiple of A)

C) x^2-5x+6 (roots are 2,3)

D) $\frac{1}{2}x^2+\frac{5}{2}x+3$ (multiple of A)

17. Correct Answers: (A),(B),(C),(D)

If $\boldsymbol{x-y}$ is a factor of $\boldsymbol{x^n-y^n}$, then $\boldsymbol{n}$ is

Identity:$x^n-y^n=(x-y)(x^{n-1}+x^{n-2}y+\cdots+y^{n-1})$
This holds for **all natural numbers** n.

This identity holds true for **any positive integer n**. Since the options list "Natural number," "Even," "Odd," and "Prime," and all even/odd/prime numbers are natural numbers that satisfy the condition:

18. Correct Answer: (A)

If $(\boldsymbol{x+1})$ is a factor of $\boldsymbol{ax^2+5x+4}$, then the possible values of $\boldsymbol{a}$ are

By Factor Theorem:

$f(-1)=0$
$a(-1)^2+5(-1)+4=0\Rightarrow a-5+4=0$
$\Rightarrow a=1$

19. Correct Answers: (A), (B), (C)

If $\sqrt{5}$ and $-\sqrt{5}$ are zeroes of a quadratic polynomial, then the polynomial can be

Zeroes: $\pm\sqrt{5}$

Standard polynomial:x^2-5
Any non-zero multiple of this is also valid.

A) x^2-5 ,**B)** $2x^2-10$, **C)** $5x^2-25$

20. Correct Answers: (A), (B), (C)

Let $\boldsymbol{p(x)=x^2-4x+5}$. Which of the following give the correct remainder?

By Remainder Theorem:

Remainder on division by $(x-a)$ is $p(a)$.

Compute values:

$p(1)=1-4+5=2$
$p(-2)=4+8+5=17$
Check statements:

A) Remainder on division by $(x-1)$is $p(1)$

B) Remainder on division by $(x+2)$is$p(-2)$

C) Remainder on division by $(x-1)$is 2

D) Remainder on division by $(x+2)$is −3 ✗

21. Correct Answers: (A), (B), (C)

Use identities:

$a^3+b^3=(a+b)(a^2-ab+b^2)$
$a^3-b^3=(a-b)(a^2+ab+b^2)$

Check each option:

A) $8x^3+27y^3=(2x)^3+(3y)^3=(2x+3y)(4x^2-6xy+9y^2)$
B) $x^3-125=x^3-5^3=(x-5)(x^2+5x+25)$
C) $27a^3+b^3=(3a)^3+b^3=(3a+b)(9a^2-3ab+b^2)$
D) $64m^3-n^3=(4m)^3-n^3$
Correct factorization should be $(4m-n)(16m^2+4mn+n^2)$,
but given middle term is **−4mn**, so Incorrect.

ASSERTION AND REASON TYPE

22. Correct Answers: D
Assertion (A): *False* – $9 - 4x + 2x^3$ is **not** in standard form (standard form is $2x^3 - 4x + 9$).
Reason (R): *True* – In standard form, terms are arranged in descending powers of the variable.

23. Correct Answers: A
Assertion (A): *True* – The expression $20x^3 + 6x^{5/2} + 4$ is not a polynomial, so its degree is not defined.
Reason (R): *True* – A polynomial cannot have rational (fractional) powers of the variable.

24. Correct Answers: D
Assertion (A): *False* – If $(x+1)$is a factor, then $p(-1) = 0$, not $p(1) = 0$.
Reason (R): *True* – For a factor $(x-a)$, we substitute $x = a$.

25. Correct Answers: A
Assertion (A): *True* – When $p(x)$is divided by $(x-a)$, the remainder is $p(a)$.
Reason (R): *True* – This is exactly the statement of the Remainder Theorem.

MATRIX MATCHING TYPE

26. Correct Answers:
(a) $\boldsymbol{x^3 - 4x^2 + x + 6}$
Test $x = 3$:
$3^3 - 4(3)^2 + 3 + 6 = 27 - 36 + 3 + 6 = 0$
So, factor is $(x-3)\rightarrow$ **Q**
(b) $\boldsymbol{x^3 + 6x^2 - x - 30}$
Test $x = -5$:
$(-5)^3 + 6(-5)^2 - (-5) - 30$
$= -125 + 150 + 5 - 30 = 0$
So, factor is $(x+5)\rightarrow$ **P**
(c) $\boldsymbol{x^3 - 7x - 6}$
Test $x = -1$:
$(-1)^3 - 7(-1) - 6 = -1 + 7 - 6 = 0$
So, factor is $(x+1)\rightarrow$ **R**
(d) $\boldsymbol{x^3 - 2x^2 - 5x + 6}$
Test $x = 1$:
$1 - 2 - 5 + 6 = 0$
So, factor is $(x-1)\rightarrow$ **S**

27. Correct Answers:
a) $(x + 3)^2 - (x - 3)^2 + 4$
$= (x^2 + 6x + 9) - (x^2 - 6x + 9) + 4 =$
$12x + 4\rightarrow$ **Linear polynomial** $\rightarrow$ **Q**
b) $x(x - 2) + 2(x - 1)$
$= x^2 - 2x + 2x - 2 = x^2 - 2\rightarrow$ **Quadratic polynomial** $\rightarrow$ **R**
c) $(\sqrt{3}x)^2 - 3x^2$
$= 3x^2 - 3x^2 = 0\rightarrow$ **Zero polynomial** $\rightarrow$ **P**
d) $x^0y^0 - xy + xy$
$= 1\rightarrow$ **Constant polynomial** $\rightarrow$ **S**
Correct matching:
a–Q, b–R, c–P, d–S

COMPREHENSION TYPE

Passage – I
Using **Remainder Theorem**:
If a polynomial $p(x)$is divided by $x - 4$, the remainder is $p(4)$.
Given polynomials

$$p_1(x) = ax^3 + 3x^2 - 3$$
$$R_1 = p_1(4) = a(64) + 3(16) - 3$$
$$= 64a + 48 - 3$$
$$= 64a + 45$$
$$p_2(x) = 2x^3 - 5x + a$$
$$R_2 = p_2(4) = 2(64) - 5(4) + a$$
$$= 128 - 20 + a = 108 + a$$

28. Correct answer: B) 1
If $\boldsymbol{R_1 = R_2}$
$64a + 45 = 108 + a$
$63a = 63 \Rightarrow a = 1$

29. Correct answer: A) $-\frac{153}{65}$
If $\boldsymbol{R_1 + R_2 = 0}$
$(64a + 45) + (108 + a) = 0$
$65a + 153 = 0 \Rightarrow a = -\frac{153}{65}$

30. Correct answer: C) $\frac{18}{127}$
If $\boldsymbol{2R_1 - R_2 = 0}$
$2(64a + 45) - (108 + a) = 0$
$128a + 90 - 108 - a = 0$
$127a - 18 = 0 \Rightarrow a = \frac{18}{127}$

Passage – II

31. Correct answer: B
Factors of $\boldsymbol{4a^2 - 4\sqrt{3}a + 3}$
$4a^2 - 4\sqrt{3}a + 3 = (2a - \sqrt{3})^2$
A factor is $2a - \sqrt{3}$

32. Correct answer: A
Factors of $\boldsymbol{x^2 - 2x - 15}$
$x^2 - 2x - 15 = (x - 5)(x + 3)$
A factor is $x - 5$

33. Correct answer: C

Value of $\frac{5.01^3-0.01^3}{5.01^2+5.01\cdot0.01+0.01^2}$

Use identity $\frac{a^3-b^3}{a^2+ab+b^2}=a-b$

Here $a=5.01,\ b=0.01$

$=5.01-0.01=5$

INTEGER TYPE

34. Correct answer: $a=7, 18$

Given $x^2-25x+126=0$

Factor:$x^2-25x+126=(x-7)(x-18)$

One factor is $x-a\Rightarrow a=7$**or** 18

35. Correct answer: 6

$$p(x)=2x^2-3x+4$$

At $x=2$:

$p(2)=2(4)-3(2)+4=8-6+4=6$

36. Correct answer: 9

Remainder when $x^2+3x+5$is divided by $x-1$

Remainder $=f(1)$:

$f(1)=1+3+5=9$

37. Correct answer: 2

If $x+1$ is a factor of x^2+ax+1

Then $f(-1)=0$:

$(-1)^2-a+1=0$

$2-a=0\Rightarrow a=2$

PREVIOUS YEAR QUESTIONS

1. Correct answer: A. $p=r$

If $x-2$ and $x-\frac{1}{2}$ are factors, then $x=2$ and $x=\frac{1}{2}$ are roots.

So,

$p(2)^2+5(2)+r=0\Rightarrow 4p+10+r=0 \quad (1)$

$p\left(\frac{1}{2}\right)^2+5\left(\frac{1}{2}\right)+r=0\Rightarrow\frac{p}{4}+\frac{5}{2}+r=0$

Multiply by 4:$p+10+4r=0 \quad (2)$

From (1): $r=-4p-10$

Substitute in (2):$p+10+4(-4p-10)=0\Rightarrow -15p-30=0\Rightarrow p=-2$

Then $r=-2$

So,$p=r$

Correct Option: A. $p=r$

2. Correct answer: A. 364

Given:$x^2+\frac{1}{x^2}=51$

Then,

$\left(x-\frac{1}{x}\right)^2=x^2+\frac{1}{x^2}-2=51-2=49$

$x-\frac{1}{x}=7$

Now,$x^3-\frac{1}{x^3}=\left(x-\frac{1}{x}\right)\left(x^2+\frac{1}{x^2}+1\right)$

$=7(51+1)=7\times52=364$

3. Correct answer: D. $8y^3$

$(x-y)^3=x^3-3x^2y+3xy^2-y^3$

Subtract:$(x+y)^3-(x-y)^3=6x^2y+2y^3$

Also,$6y(x^2-y^2)=6x^2y-6y^3$

Now,$(6x^2y+2y^3)-(6x^2y-6y^3)=8y^3$

4. Correct answer: B. $3abc(a-b)(b-c)(c-a)$

Use the identity: $x^3(y-z)^3+y^3(z-x)^3+z^3(x-y)^3=3xyz(x-y)(y-z)(z-x)$

Comparing with the given expression, we get:

$a^3(b-c)^3+b^3(c-a)^3+c^3(a-b)^3=3abc(a-b)(b-c)(c-a)$

5. Correct answer: B. a^6-b^6

$(a^2-ab+b^2)(a^2+ab+b^2)=a^4+a^2b^2+b^4$

Now multiply:$(a^2-b^2)(a^4+a^2b^2+b^4)$

Using identity:$(x-y)(x^2+xy+y^2)=x^3-y^3$

Let $x=a^2,\ y=b^2$

$=a^6-b^6$

6. Correct answer: B. $x-2$

Factorize the polynomial:x^4-3x^2+2

Let $x^2=y$, then:

$y^2-3y+2=(y-1)(y-2)$

Substitute back:$(x^2-1)(x^2-2)$

Further factor:$(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$

So, the factors are:$x-1,\ x+1,\ x-\sqrt{2},\ x+\sqrt{2}$

7. Correct answer: D. −1

By Remainder Theorem, if the polynomial is divisible by $x+2$, then

$f(-2)=0$

$f(x)=2x^4+3x^3+2px^2+3x+6$

Substitute $x=-2$:

$2(-2)^4+3(-2)^3+2p(-2)^2+3(-2)+6=0$

$2(16)-24+8p-6+6=0$

$32-24-6+6+8p=0$

$8+8p=0\Rightarrow p=-1$

8. Correct answer: C. 9

Given: $x - \frac{1}{x} = \sqrt{7}$

Square both sides:

$\left(x - \frac{1}{x}\right)^2 = 7$

$x^2 + \frac{1}{x^2} - 2 = 7$

$x^2 + \frac{1}{x^2} = 9$

9. Correct answer: B. 1

Use the identity:$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

First find $x^2 + y^2 + z^2$:

$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx)$

$= 1^2 - 2(-1) = 3$

Now substitute:

$x^3 + y^3 + z^3 - 3(-1) = 1(3 - (-1))$

$x^3 + y^3 + z^3 + 3 = 4$

$x^3 + y^3 + z^3 = 1$

10. Correct answer: B. $\left(2x + \sqrt{5}\right)(x - 3\sqrt{5})$

Given:$2x^2 - 5\sqrt{5}\,x - 15$

Multiply $2 \times (-15) = -30$

Split the middle term:$2x^2 - 6\sqrt{5}\,x + \sqrt{5}\,x - 15$

Group terms:$2x(x - 3\sqrt{5}) + \sqrt{5}(x - 3\sqrt{5})$

Factor common binomial:$\left(x - 3\sqrt{5}\right)(2x + \sqrt{5})$

3. Coordinate Geometry

Solutions

Worksheet – 1

1. **Correct answer: C) 4**
   Distance between (2,3) and (6,3)
   Same y-coordinate ⇒ distance = |6 − 2| = **4**
2. **Correct answer: C) 39**
   A(0,0), B(36,15)
   AB = √[(36−0)² + (15−0)²] = √(1296 + 225) = √1521 = **39**
3. **Correct answer: B) 5**
   Distance between (1,2) and (4,6)
   = √[(4−1)² + (6−2)²] = √(9 + 16) = √25 = 5
4. **Correct answer: B) 5**
   Distance of (3,4) from origin
   = √(3² + 4²) = √25 = **5**
5. **Correct answer: B)**
   Distance between (−2,5) and (2,5)
   Same y-coordinate ⇒ |2 − (−2)| = **4**
   Ans: Correct distance = **4**
6. **Correct answer: C) 5**
   Distance between (0,2) and (0,7)
   Same x-coordinate ⇒ |7 − 2| = **5**
7. **Correct answer: C)**
   Distance between (−1,−2) and (−4,−6)
   = √[(−4+1)² + (−6+2)²] = √(9 + 16) = √25 = **5**
   Ans: B) 4 **Correct distance = 5**
8. **Correct answer: C) 2 or −4**
   Given A(5,3), B(11,−5) and ∠APB = 90°.
   Using the right-angle condition (dot product = 0), solving gives
   λ = 2 or −4.
9. **Correct answer: D) Square**
   Points: (2,5), (0,3), (2,1), (4,3)
   All four sides are equal and diagonals are equal and bisect each other at right angles.
   Hence, the figure formed is a Square.
10. **Correct answer: D)8, −2**
    Distance between two points = 5
    One point = (3,2), ordinate of other point = 2
    So distance is horizontal: |x − 3| = 5
    ⇒ x = 3 ± 5 ⇒ **x = 8 or −2**

Worksheet-2

1. **Correct answer: B) 1: 1**
   The ratio in which (1, −3) divides the line segment joining (4, 3) and (−2, −9)
   Let
   $\boldsymbol{A(4,3)}$, $\boldsymbol{B(-2,-9)}$, and $\boldsymbol{P(1,-3)}$
   Assume $\boldsymbol{P}$divides $\boldsymbol{AB}$in the ratio $\boldsymbol{m:n}$.
   Using section formula,
   $$\boldsymbol{1 = \frac{m(-2)+n(4)}{m+n} \Rightarrow -2m+4n = m+n \Rightarrow 3m = 3n \Rightarrow m = n}$$
   So, ratio $= \boldsymbol{1:1}$
2. **Correct answer: A) 2: 1**
   X-axis divides the line segment joining (−6, 4) and (3, −2) in the ratio
   Let the dividing point be $\boldsymbol{P(x,0)}$(since it lies on X-axis).
   Using y-coordinate section formula,
   $$\boldsymbol{0 = \frac{m(-2)+n(4)}{m+n} \Rightarrow -2m+4n = 0 \Rightarrow m = 2n}$$
   So, ratio $= \boldsymbol{2:1}$
3. **Correct answer: B) 2 : 1**
   Y-axis divides the line segment joining (6, 5) and (−3, −10) in the ratio
   Let the dividing point be $\boldsymbol{P(0,y)}$(since it lies on Y-axis).
   Using x-coordinate section formula,
   $$\boldsymbol{0 = \frac{m(-3)+n(6)}{m+n} \Rightarrow -3m+6n = 0 \Rightarrow m = 2n}$$
   So, ratio $= \boldsymbol{2:1}$
4. **Correct answer:** $\left(\frac{13}{5}, \frac{27}{5}\right)$
   Using **section formula**
   Let $A(x_1, y_1) = (5,7)$, $B(x_2, y_2) = (-1,3)$
   Ratio $m:n = 2:3$
   $$x = \frac{mx_2 + nx_1}{m+n} = \frac{2(-1)+3(5)}{5} = \frac{13}{5}$$
   $$y = \frac{my_2 + ny_1}{m+n} = \frac{2(3)+3(7)}{5} = \frac{27}{5}$$

5. Correct answer: (C) (3,3) and (−9,27)

Internal Division:

$$x = \frac{2(6)+3(1)}{5} = 3, y = \frac{2(-3)+3(7)}{5} = 3$$

External Division:

$$x = \frac{2(6)-3(1)}{2-3} = -9, y = \frac{2(-3)-3(7)}{2-3} = 27$$

6. Correct answer: B $\left(-\frac{7}{3}, \frac{10}{3}\right)$and $\left(\frac{1}{3}, \frac{14}{3}\right)$

First point (ratio 1:2):

$$x = \frac{1(3)+2(-5)}{3} = -\frac{7}{3}, y = \frac{1(6)+2(2)}{3} = \frac{10}{3}$$

Second point (ratio 2:1):

$$x = \frac{2(3)+1(-5)}{3} = \frac{1}{3}, y = \frac{2(6)+1(2)}{3} = \frac{14}{3}$$

7. Correct answer: D (1,1)

Step 1: Find x and y

$$\frac{2x+1}{2} = \frac{3}{2} \Rightarrow x = 1$$

$$\frac{2y+3}{2} = \frac{5}{2} \Rightarrow y = 1$$

Step 2: Required midpoint

Points are $(0, 2)$and $(2, 0)$

$$\text{Midpoint} = \left(\frac{0+2}{2}, \frac{2+0}{2}\right) = (1,1)$$

8. Correct answer: 1: 1

Using section formula:

$$2 = \frac{m(-2)+n(4)}{m+n}, 3 = \frac{m(-7)+n(8)}{m+n}$$

Solving gives:

$$m: n = 1: 1$$

9. Correct answer: C 4 : 3 externally

The ratio in which point (−3, 5) divides the line segment joining (5, 1) and (3, 2) is

Solution:

Let $A(5,1)$, $B(3,2)$

Suppose $P(-3,5)$divides ABin the ratio $m: n$

Using section formula for x-coordinate:

$$-3 = \frac{m(3)+n(5)}{m+n}$$

$$-3m - 3n = 3m + 5n \Rightarrow 6m + 8n = 0$$

$$\Rightarrow 3m + 4n = 0 \Rightarrow m: n = 4: -3$$

Negative sign ⇒ external division

10. Correct answer: C $\boxed{\frac{10}{3}}$

If $P, Q$are the points of trisection of the line segment joining $A(1,-2)$and $B(-5,6)$, then find PQ.

Solution:

Distance AB:

$$AB = \sqrt{(-5-1)^2 + (6+2)^2} = \sqrt{36+64} = \sqrt{100} = 10$$

In trisection, the segment is divided into three equal parts:

$$PQ = \frac{AB}{3} = \frac{10}{3}$$

PRACTICE EXERCISE

1. Correct answer: B) 3

Distance of point $\boldsymbol{P(2,3)}$from X-axis

Distance from X-axis = | y |

$$= |\ 3\ | = 3$$

2. Correct answer: B) 8

Distance between points $\boldsymbol{A(0,6)}$and $\boldsymbol{B(0,-2)}$

$$= |\ 6 - (-2)\ | = 8$$

3. Correct answer: C) 10

Distance of point $\boldsymbol{P(-6,8)}$from origin

$$= \sqrt{(-6)^2 + 8^2} = \sqrt{36+64} = \sqrt{100} = 10$$

4. Correct answer: B) $5\sqrt{2}$

Distance between points $(\mathbf{0}, \mathbf{5})$and $(\mathbf{-5}, \mathbf{0})$

$$= \sqrt{(0+5)^2 + (5-0)^2} = \sqrt{25+25} = \sqrt{50} = 5\sqrt{2}$$

5. Correct answer: B) (1, −10)

External division:

$P(-5,4)$, $Q(-2,-3)$, ratio $1: 2$

Using external division formula:

$$R = \left(\frac{1x_2 - 2x_1}{1-2}, \frac{1y_2 - 2y_1}{1-2}\right) = (1, -10)$$

6. Correct answer: C) $\frac{t}{1-t}$

Given

$$x = x_1 + t(x_2 - x_1), y = y_1 + t(y_2 - y_1)$$

Ratio = $t: (1-t)$

$$= \frac{t}{1-t}$$

7. **Correct answer: C) 4:3 externally**

   Point $(-3, 5)$divides line joining $(5, 1)$and $(3, 2)$

   Using section formula, ratio comes externally as 4:3

8. **Correct answer: B) 3:7**

   8. X-axis divides line joining $(2, -3)$and $(5, 7)$

   At X-axis, $y = 0$

   Ratio:

   $$= 3:7$$

9. **Correct answer: D) 3:4**

   Y-axis divides line joining $(3, 5)$and $(-4, 7)$

   At Y-axis, $x = 0$

   Ratio:

   $$= 3:4$$

10. **Correct answer: B) (17, 15)**

    Point dividing $(5, 7)$and $(-1, 3)$in the ratio $2:-3$

    Since the ratio is external, use external section formula:

    $$x = \frac{mx_2 - nx_1}{m-n}, y = \frac{my_2 - ny_1}{m-n}$$

    Where

    $$m = 2, n = 3$$

    $$(x_1, y_1) = (5,7), (x_2, y_2) = (-1,3)$$

    $$x = \frac{2(-1) - 3(5)}{2-3} = \frac{-2-15}{-1} = 17$$

    $$y = \frac{2(3) - 3(7)}{2-3} = \frac{6-21}{-1} = 15$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct answer: A) 5, B) $\sqrt{25}$**

    Distance between points $(1, 2)$and $(4, 6)$

    $$= \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25}$$
    $$= 5$$

12. **Correct answer: A, B, C**

    Points $(0, 5)$and $(0, -5)$

    - Both have $x = 0 \Rightarrow$ lie on **Y-axis**
    - Distance from origin: $|5| = |-5| = 5 \Rightarrow$ equidistant
    - Distance between them:

      $$= |5 - (-5)| = 10$$

13. **Correct answer: A, B, C**

    Distance between $(a, 3)$and $(5, 3)$is 4

    Same y-coordinate ⇒ horizontal distance

    $$|a - 5| = 4 \Rightarrow a = 9 \text{ or } a = 1$$

14. **Correct answer: A, B, C**

    Points $(2, 0)$and $(-2, 0)$

    - $y = 0 \Rightarrow$ lie on **X-axis**
    - Distance from origin:

      $|2| = |-2| = 2 \Rightarrow$ equidistant
    - Distance between points:

      $$= |2 - (-2)| = 4$$

15. **Correct answer: B, C**

    Distance between points on a vertical line depends on

    On vertical line, $x$is same ⇒ distance depends on

    $$|y_2 - y_1|$$

ASSERTION AND REASON TYPE

16. **Correct answer: (a)**

    Assertion (A): The point $(0, 0)$lies on both the x-axis and y-axis.
    True (origin lies on both axes)

    Reason (R):

    - On y-axis, $x = 0$
    - On x-axis, $y = 0$
      True and correctly explains A

17. **Correct answer: (A)**

    Assertion (A): The point $(3, -4)$lies in the fourth quadrant.
    True (x positive, y negative)

    Reason (R):
    A point with positive x-coordinate and negative y-coordinate lies in the fourth quadrant.
    True and correct explanation

18. **Correct answer: (a)**
    Assertion (A): If a point lies on the x-axis, then its ordinate is zero.
    True

Reason (R):
The ordinate of a point represents its perpendicular distance from the x-axis.
True and explains why ordinate is zero on x-axis

19. **Correct answer: (D)**
Assertion–Reason
Assertion (A): The coordinates of a point are written in the order (y, x).
False – coordinates are written as (x, y).
Reason (R):
The first coordinate represents the distance from the **y-axis** and the second coordinate represents the distance from the **x-axis**.
True

MATRIX MATCH TYPE

20. **Correct answer:**
Correct Matching
Column-I → Column-II
a) Mid-point of $A(-4,6)$and $B(8,-2)$

$$\left(\frac{-4+8}{2}, \frac{6+(-2)}{2}\right) = (2,2)$$

➞ a → P (2,2)
b) A line intersects the x-axis at P and y-axis at Q.
Given midpoint of PQ = (3, −2) and $\boldsymbol{Q} = (\mathbf{0}, -\mathbf{6})$.
Using midpoint formula:

$$\left(\frac{\boldsymbol{x_P}+\mathbf{0}}{\mathbf{2}}, \frac{\mathbf{0}+(-\mathbf{6})}{\mathbf{2}}\right) = (\mathbf{3}, -\mathbf{2})$$

From x-coordinate:

$$\frac{\boldsymbol{x_P}}{\mathbf{2}} = \mathbf{3} \Rightarrow \boldsymbol{x_P} = \mathbf{6}$$

So $\boldsymbol{P} = (\mathbf{6}, \mathbf{0})$.
Among options, matching x-intercept point is (6,2) in given set format.
b → R (6, 2)
c)
Point on y-axis ⇒ x = 0
Let point = (0, y)
Distance from (4, 2) is $\sqrt{\mathbf{20}}$:

$$\sqrt{(\mathbf{0}-\mathbf{4})^2 + (\boldsymbol{y}-\mathbf{2})^2} = \sqrt{\mathbf{20}}$$
$$\mathbf{16} + (\boldsymbol{y}-\mathbf{2})^2 = \mathbf{20}$$
$$(\boldsymbol{y}-\mathbf{2})^2 = \mathbf{4} \Rightarrow \boldsymbol{y} = \mathbf{4} \text{ or } \mathbf{0}$$

From options: (0,4)
c → Q (0, 4)
d) Fourth vertex of rectangle
Given: $(1,-2), (5,-2), (5,3)$
Missing vertex = $(1, 3)$
➞ d → S (1,3)

21. **Correct answer:**
Quadrants Match
- **First Quadrant** → x>0,y>0 → P
- **Second Quadrant** → $x < 0$,y>0 →Q
- **Third Quadrant** → x < 0, y < 0 → R
- **Fourth Quadrant** → x > 0, y< 0 → S

COMPREHENSION TYPE

Passage – I
Given points:
$A(4,3)$and $B(-4,3)$

22. **Correct answer:**
Line joining points A and B is parallel to
Both points have same y-coordinate (3) ⇒ line is horizontal

23. **Correct answer: C) 8 units**
Distance between points A and B

$$= |\ 4-(-4)\ | = 8$$

24. **Correct answer: A) (0, 3)**
Coordinates of the midpoint of AB

$$\left(\frac{4+(-4)}{2}, \frac{3+3}{2}\right) = (0,3)$$

Passage – II
Given:
$P(-2,5)$(second quadrant)
$Q(2,-5)$(fourth quadrant)

25. **Correct answer: B) 5 units**
Distance of point P from the x-axis
Distance from x-axis = $|\ y\ |$

$$= |\ 5\ | = 5$$

26. **Correct answer: C) 2 units**
Distance of point Q from the y-axis
Distance from y-axis = $|\ x\ |$

$$= |\ 2\ | = 2$$

27. **Correct answer: D) (2, −5)**
The point which lies in the fourth quadrant
Fourth quadrant ⇒ $x > 0, y < 0$

INTEGER TYPE

28. **Correct answer:**
A point lying on the **y-axis** has its x-coordinate equal to **0**.
29. **Correct answer:**
The distance of the point **(−5, 7)** from the x-axis is **7 units**.
(Distance from x-axis = |y|)
30. **Correct answer:**
The point (−3, −4) lies in the III (third) quadrant.
31. **Correct answer:**
The coordinates of the origin are **(0, 0)**.

Previous Year questions

1. **Correct answer: B) 5**
Distance of point P(4, −5) from the X-axis
Distance from X-axis = |y| = |−5| = 5
2. **Correct answer: C) 10**
Distance between A(0, 3) and B(0, −7)
Same x-coordinate ⇒ vertical distance = |3 − (−7)| = |10| = 10
3. **Correct answer: A) 5**
Distance of point P(3, 4) from the origin
Distance = $\sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$
4. **Correct answer: A) 5**
Distance between points (−4, 0) and (0, 3)
Distance = $\sqrt{[(0 + 4)^2 + (3 - 0)^2]}$
$= \sqrt{(16 + 9)} = \sqrt{25} = 5$
5. **Correct answer: B 1**
Given:
$P(-3,4)$, $Q(-4,5)$
Abscissa of P= −3
Abscissa of Q= −4
$$(-3) - (-4) = 1$$
6. **Correct answer: B** $(-2, -4)$
Given vertices of square $ABCD$:
$$A(5,3),\ B(-2,3),\ D(5,-4)$$
In a square, opposite sides are parallel and equal.
So,
x-coordinate of $C = x_B = -2$
y-coordinate of $C = y_D = -4$
$$C = (-2, -4)$$
7. **Correct answer: D**
Fill in the blanks
• Point of intersection of axes = **Origin** → **P**
• Signs of abscissa and ordinate are different in **II and IV quadrants** → **Q, R**
• Line $x = 4$is parallel to **y-axis** → **S**
• Distance from origin = | 4 |= 4units → **T**
8. **Correct answer: C —** $x - y = 0$
Short solution (text format):
From the graph, the shaded triangular region is bounded by these straight lines:
   - **x = 0** (y-axis)
   - **y = 0** (x-axis)
   - The slant line passing through $(0, 1)$and $(2, 0)$

Equation of the slant line:
$$x + 2y = 2$$
So, boundary lines are:
   - $x = 0$
   - $y = 0$
   - $x + 2y = 2$

Among the given options, $x - y = 0$(i.e., $y = x$) is **not** a boundary line.
9. **Correct answer:**
(i) Sum of abscissa of P and ordinate of T
   - Point **P** has abscissa $= 4$
   - Point **T** has ordinate $= -5$

$$4 + (-5) = -1$$
(ii) Perpendicular distance of S from the x-axis
Distance from x-axis = | y |
   - Point **S** has ordinate $= -4$

Distance $= 4$ units
(iii) Point having ordinate = −2 and abscissa = 6
That point is **R**.
10. **Correct answer: Option D — II and IV quadrants**
Short solution (text format):
Abscissa = x-coordinate
Ordinate = y-coordinate
Different signs means:
   - one is positive and the other is negative.

This happens in:
   - II quadrant $(-, +)$
   - IV quadrant $(+, -)$

4. Linear Equations in One and Two Variables

Solutions

Worksheet – 1

1. **Correct answer: (A) $\frac{8}{3}$**
   Given: $4t - \frac{5}{3} = 9$
   Add $\frac{5}{3}$ to both sides:
   $$4t = 9 + \frac{5}{3} = \frac{32}{3}$$
   Divide by 4: $t = \frac{32}{3} \div 4 = \frac{8}{3}$
2. **Correct answer: (D) $\frac{13}{6}$**
   Given: $V + \frac{4}{3} = 3\frac{1}{2} = \frac{7}{2}$
   Subtract $\frac{4}{3}$: $V = \frac{7}{2} - \frac{4}{3}$
   LCM = 6
   $V = \frac{21 - 8}{6} = \frac{13}{6}$
3. **Correct Answer: (D) 1**
   Given: $15(L - 1) + 4(L + 3) = 2(7 + L)$
   Expand: $5L - 15 + 4L + 12 = 14 + 2L$
   Simplify: $19L - 3 = 14 + 2L$
   $17L = 17 \Rightarrow L = 1$
4. **Correct Answer: (C) 3**
   Given: $8(C - 3) - (6 - 2C) = 2(C + 2) - 5(5 - C)$
   Expand: $8C - 24 - 6 + 2C = 2C + 4 - 25 + 5C$
   Simplify:
   $10C - 30 = 7C - 21$
   $3C = 9 \Rightarrow C = 3$
5. **Correct Answer: (B) $\frac{9}{2}$**
   Given: $\frac{x}{2} - \frac{1}{4} = \frac{x}{3} + \frac{1}{2}$
   Multiply both sides by LCM 12:
   $6x - 3 = 4x + 6$
   $2x = 9 \Rightarrow x = \frac{9}{2}$
6. **Correct Answer: A) −9**
   $$\frac{x - 4}{7} - \frac{x + 4}{5} = \frac{x + 3}{7}$$
   Multiply by 35:
   $5(x - 4) - 7(x + 4) = 5(x + 3)$
   $5x - 20 - 7x - 28 = 5x + 15$
   $-2x - 48 = 5x + 15 \Rightarrow -7x = 63 \Rightarrow x = -9$
7. **Correct Answer: B) 2**
   $$\frac{5x + 2}{2x + 3} = \frac{12}{7}$$
   Cross-multiply:
   $7(5x + 2) = 12(2x + 3)$
   $35x + 14 = 24x + 36$
   $11x = 22 \Rightarrow x = 2$
8. **Correct Answer: A) $\frac{3}{7}$**
   $$\frac{3t + 1}{16} - \frac{2t - 3}{7} = \frac{t + 3}{8} + \frac{3t - 1}{14}$$
   Multiply by LCM 112:
   $7(3t + 1) - 16(2t - 3)$
   $= 14(t + 3) + 8(3t - 1)$
   $21t + 7 - 32t + 48 = 14t + 42 + 24t - 8$
   $-11t + 55 = 38t + 34$
   $-49t = -21 \Rightarrow t = \frac{3}{7}$
9. **Correct Answer: A) 19**
   Total chocolates = 45, left = 7
   Given away = $45 - 7 = 38$
   Each student gets 2 ⇒ students = $38 \div 2 = 19$
10. **Correct Answer: C) 45 and 120**
    Ratio = 3: 8⇒ numbers = $3x, 8x$
    Sum = $11x = 165 \Rightarrow x = 15$
    Numbers = 45and 120

Worksheet -2

1. **Correct Answer: C) by − ax**
   Statement: "The product of x and a subtracted from the product of b and y."
   - Product of x and a = **ax**
   - Product of b and y = **by**
   - "ax subtracted from by" ⇒ **by − ax**
2. **Correct Answer: A) 13**
   Let the number be **n**.
   Given: $17n + 4 = 225$
   $\Rightarrow 17n = 221$
   $\Rightarrow n = 13$
3. **Correct Answer: B) 56**
   Let the number be x.
   $$\frac{x}{2} + 18 = 46 \Rightarrow \frac{x}{2} = 28 \Rightarrow x = 56$$

4. **Correct Answer: C) 35**
Let boys = x, girls = $5x$.
Total children = $6x$(must be a multiple of 6).
Among options, **35** is not divisible by 6.

5. **Correct Answer: B) 8**
Number of jumps $= \frac{731.2}{91.4} = 8$

6. **Correct Answer: B) 12, 17**
Let numbers be x and $x + 5$.
$x + (x + 5) = 29 \Rightarrow 2x = 24 \Rightarrow x = 12$
Numbers: 12,17

7. **Correct Answer: C)** $\boxed{\frac{3}{2}}$
Given$\frac{4p+9q}{p} = \frac{5q}{p-q}$
Cross-multiply:$(4p + 9q)(p - q) = 5pq$
Expand:
$4p^2 - 4pq + 9pq - 9q^2 = 5pq$
$4p^2 + 5pq - 9q^2 = 5pq$
Subtract $5pq$ from both sides:
$4p^2 - 9q^2 = 0$
$(2p - 3q)(2p + 3q) = 0$
Since $p, q > 0$:
$$2p = 3q \Rightarrow \frac{p}{q} = \frac{3}{2}$$

8. **Correct Answer: B)** $\frac{x}{y}$
$$\frac{x - \frac{1}{y}}{y - \frac{1}{x}} = \frac{\frac{xy-1}{y}}{\frac{xy-1}{x}} = \frac{x}{y}$$

9. **Correct Answer: C) 11**
Given: $10x + 3y = 75$.
Check options for $x + y$.
If $x + y = 8$: $x = 6, y = 2 \rightarrow 10(6) + 3(2) = 60 + 6 = 66$
If $x + y = 10$: $x = 6, y = 4 \rightarrow 10(6) + 3(4) = 60 + 12 = 72$
If $x + y = 11$: $x = 6, y = 5 \rightarrow 10(6) + 3(5) = 60 + 15 = 75$

10. **Correct Answer: B) 3**
Point $(1, 4)$lies on both lines.
From $2x + by = 6$:
$2(1) + b(4) = 6 \Rightarrow 4b = 4 \Rightarrow b = 1$
From $3y = 8 + ax$:
$3(4) = 8 + a(1) \Rightarrow a = 4$
$a - b = 4 - 1 = 3$

11. **Correct Answer: A) 1**
Given $(2, 1)$is a solution of $2x + ky = 5$.
Substitute $x = 2$, $y = 1$:
$2(2) + k(1) = 5 \Rightarrow 4 + k = 5 \Rightarrow k = 1$

12. **Correct Answer: B) (4,0)**
Check each option in $3x + 2y = 12$.
(A) (2,2): $6 + 4 = 10 \neq 12$
(B) (4,0): $12 + 0 = 12$✓
(C) (0,4): $0 + 8 = 8 \neq 12$
(D) (3,3): $9 + 6 = 15 \neq 12$

13. **Correct Answer: (B) Two**
A straight line is completely determined by two distinct points.
So, only two solution points are enough to draw the graph of a linear equation.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: (A) 12**
$$\frac{z}{2} + \frac{z}{3} - \frac{z}{6} = 8$$
LCM of 2, 3, 6 = 6
$$\frac{3z + 2z - z}{6} = 8 \Rightarrow \frac{4z}{6} = 8 \Rightarrow \frac{2z}{3} = 8 \Rightarrow z = 12$$

2. **Correct Answer: (A)** $\frac{127}{12}$
$9\frac{1}{4} = y - 1\frac{1}{3}$
Convert to improper fractions:
$\frac{37}{4} = y - \frac{4}{3}$

Add $\frac{4}{3}$to both sides:
$$y = \frac{37}{4} + \frac{4}{3}$$
LCM = 12
$$y = \frac{111 + 16}{12} = \frac{127}{12}$$

3. **Correct Answer: (B) One**
The power of a variable in a linear equation is **1**.

4. **Correct Answer: (B)** $\frac{x}{2} + 10 = 15$
Statement: Half of a number added to 10 is 15
Let the number be x.$\frac{x}{2} + 10 = 15$

5. **Correct Answer: D)16**
Point $(2, 3)$lies on both lines.
From $x + ay = 11$:
$2 + 3a = 11 \Rightarrow a = 3$

From $2x + 3y = b$:
$2(2) + 3(3) = 4 + 9 = 13 \Rightarrow b = 13$
$a + b = 3 + 13 = 16$

6. **Correct Answer: C)3**

$$\frac{x}{3}+\frac{1}{6}=\frac{x}{2}-\frac{1}{3}$$

LCM = 6:

$$2x+1=3x-2 \Rightarrow x=3$$

7. **Correct Answer: (A) 1**

Given $2x+y=7$, for $x=3$:

$$2(3)+y=7 \Rightarrow 6+y=7 \Rightarrow y=1$$

8. **Correct Answer: (A) −5**

$$\frac{3x+1}{5}=\frac{2x-4}{5}$$

Multiply both sides by 5:

$$3x+1=2x-4 \Rightarrow x=-5$$

9. **Correct Answer: (C) −3**

If $\frac{x+7}{3x+16}=\frac{4}{7}$

$7(x+7)=4(3x+16)$

$7x+49=12x+64$

10. **Correct Answer: (D) (1, 5)**

Which of the following is NOT a solution of $x+y=5$?

Check options:

(2,3): $2+3=5$

(4,1): $4+1=5$

(5,0): $5+0=5$

(1,5): $1+5=6$

11. **Correct Answer: (C) $\frac{9}{23}$**

If $\frac{8P-5}{7P+1}=\frac{-2}{4}$

Solution:

$\frac{-2}{4}=-\frac{1}{2}$

$2(8P-5)=-(7P+1)$

$16P-10=-7P-1$

$23P=9 \Rightarrow P=\frac{9}{23}$

12. **Correct Answer: (B) $2x-y=0$**

Which equation represents a line passing through the origin?

A line through origin satisfies $x=0, y=0$.

Only $2x-y=0$gives $0=0$.

13. **Correct Answer: (C) Straight line**

The graph of the equation $3x+4y=12$is a:

It is a linear equation in two variables.

14. **Correct Answer: (B) $y=3$**

Which equation represents a line parallel to the x-axis?

A line parallel to x-axis is of the form $y=$ constant.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

15. **Correct Answer: A) 27 and B) 29**

Given: Sum of two consecutive odd numbers = 56

Let numbers be $x$and $x+2$

$$x+(x+2)=56 \Rightarrow 2x+2=56 \Rightarrow x=27$$

Numbers are 27 and 29.

16. **Correct Answer: A) 3, B) $\frac{15}{5}$**

Let the number be x

$$\frac{x}{3}+5=2x \Rightarrow \frac{x+15}{3}=2x \Rightarrow x+15=6x$$
$$\Rightarrow 5x=15 \Rightarrow x=3$$

17. **Correct Answer: A) 39 and B) 104**

Ratio = 8: 3

Let numbers be $8k$and $3k$

$$8k+3k=143 \Rightarrow 11k=143 \Rightarrow k=13$$

Numbers are 104 and 39.

18. **Correct Answer: A) 78, B) 76, C) 80**

Let three consecutive even numbers be $x-2, x, x+2$

$$(x-2)+x+(x+2)=234 \Rightarrow 3x=234$$
$$\Rightarrow x=78$$

Numbers are 76, 78, 80.

19. **Correct Answer: A) 44, B) 55, C) 66**

Ratio = 4: 5: 6

Let numbers be $4k, 5k, 6k$

Largest + Smallest = Third + 55

$$6k+4k=5k+55 \Rightarrow 10k=5k+55$$
$$\Rightarrow 5k=55 \Rightarrow k=11$$

Numbers are 44, 55, 66.

20. **Correct Answer: C) 3**

Given: $3<x<6$and $2<y<4$

We need the **greatest value of** $\frac{x}{y}$.

To maximize $\frac{x}{y}$:

- Take **maximum** $x\rightarrow$ close to 6
- Take **minimum** $y\rightarrow$ close to 2

$$\frac{x}{y} \approx \frac{6}{2}=3$$

Since bounds are open, the value approaches **3**.

21. Correct Answer: A, B, C and D

Given system of linear equations:

$3x + 4y = 25$ (1)

$5x - 6y = -9$ (2)

Multiply (1) by 3 and (2) by 2:

$9x + 12y = 75$

$10x - 12y = -18$

Add:$19x = 57 \Rightarrow x = 3$

Substitute in (1):$3(3) + 4y = 25 \Rightarrow 9 + 4y = 25 \Rightarrow 4y = 16 \Rightarrow y = 4$

Check options:

A) $19x = 57$

B) $4y = 16$

C) $x = 3, y = 4$

D) $4y = 25 - 9$ (same as $4y = 16$)

ASSERTION AND REASON TYPE

22. Correct Answer: A

Assertion (A): The equation $3x - 7 = 5$is a linear equation in one variable.

True – it has only one variable $x$and its highest power is 1.

Reason (R): It contains only one variable and the highest power of that variable is one.

True – this is exactly the definition of a linear equation in one variable.

Conclusion: Reason correctly explains the Assertion.

23. Correct Answer: D

Assertion (A): If $a = 0$and $b = 0$in $ax + by + c = 0$, the equation is linear.

False – if both $a$and $b$are zero, the equation becomes $c = 0$, which is **not** a linear equation in two variables.

Reason (R): At least one of $a$or $b$must be non-zero for a linear equation in two variables.

True – this is a necessary condition.

Conclusion: Assertion is false, but Reason is true.

24. Correct Answer: A

Assertion (A): The points $(0, 3)$and $(5, 0)$lie on the line $3x + 5y = 15$.

Check by substitution:

For $(0, 3)$: $3(0) + 5(3) = 15$

For $(5, 0)$: $3(5) + 5(0) = 15$

So, **Assertion is true**.

Reason (R): Both points satisfy the equation when substituted.

True, and this is exactly why the points lie on the line.

Conclusion: Reason correctly explains the Assertion.

25. Correct Answer: B

Assertion (A): The graph of the equation $3x + y = 0$is a line passing through the origin.

True – when $x = 0$, $y = 0$, so the line passes through the origin.

Reason (R): An equation of the form $ax + by + c = 0$is called a linear equation in $x$and y.

True – this is the definition of a linear equation.

However, this reason does not explain why the line passes through the origin (that depends on $c = 0$).

Conclusion: Both are true, but Reason is not the correct explanation.

MATRIX MATCHING TYPE

26. Correct Answer: Q) −2

(a) $\frac{2-9z}{17-4z} = \frac{4}{5}$

Cross-multiply:

$5(2 - 9z) = 4(17 - 4z)$

$10 - 45z = 68 - 16z$

$-29z = 58 \Rightarrow z = -2$

b) $\frac{3(y-5)}{4} - 4y = 3 - \frac{(y-3)}{2}$

Multiply both sides by 4:

$3(y - 5) - 16y = 12 - 2(y - 3)$

$3y - 15 - 16y = 12 - 2y + 6$

$-13y - 15 = 18 - 2y$

$-11y = 33 \Rightarrow y = -3$

Correct answer: P) −3

(c) $\frac{3x+5}{4x+2} = \frac{3x+4}{4x+7}$

Cross-multiply:

$(3x + 5)(4x + 7) = (3x + 4)(4x + 2)$

$12x^2 + 41x + 35 = 12x^2 + 22x + 8$

$19x = -27 \Rightarrow x = -\frac{27}{19}$

T) −27/19

(d) $\frac{8x-3}{3x} = 2$

$8x - 3 = 6x$

$2x = 3 \Rightarrow x = \frac{3}{2}$

27. Correct Answer: (A)

$\frac{15}{2}y + 10 = -5$

$\frac{15}{2}y = -15 \Rightarrow y = -2$

Match: R) −2

(b) $\frac{7}{2}x + 2 = 14 + 3x$

$\frac{7}{2}x - 3x = 12$

$\frac{1}{2}x = 12 \Rightarrow x = 24$

Match: T) 24

(c) Let smaller part = x, greater part = $2x$

$x + 2x = 96 \Rightarrow 3x = 96 \Rightarrow x = 32$

Greater part $= 2x = 64$

Match: P) 64

(d) $\frac{4x}{5} + 9 = 2x - 3$

$\frac{4x}{5} = 2x - 12$

$4x = 10x - 60 \Rightarrow 6x = 60 \Rightarrow x = 10$

Match: Q) 10

COMPREHENSION TYPE

Passage – I:

Given:

$\frac{5a-4}{6} = 4a + 1 - \frac{3a+10}{2}$

$\frac{4b+7}{9-3b} = \frac{1}{4}$

$\frac{15(2-c) - 5(c+6)}{1-3c} = 10$

Step 1: Solve for $\boldsymbol{a}$

$$\frac{5a-4}{6} = 4a + 1 - \frac{3a+10}{2}$$

Take LCM = 6:

$5a - 4 = 6(4a + 1) - 3(3a + 10)$

$5a - 4 = 24a + 6 - 9a - 30$

$5a - 4 = 15a - 24$

$10a = 20 \Rightarrow a = 2$

Step 2: Solve for $\boldsymbol{b}$

$\frac{4b+7}{9-3b} = \frac{1}{4}$

$4(4b + 7) = 9 - 3b$

$16b + 28 = 9 - 3b$

$19b = -19 \Rightarrow b = -1$

Step 3: Solve for $\boldsymbol{c}$

$\frac{15(2-c) - 5(c+6)}{1-3c} = 10$

Simplify numerator:

$30 - 15c - 5c - 30 = -20c$

So,

$\frac{-20c}{1-3c} = 10$

$-20c = 10(1 - 3c)$

$-20c = 10 - 30c$

$10c = 10 \Rightarrow c = 1$

Final values

$a = 2, b = -1, c = 1$

28. Correct Answer: D

$$\boldsymbol{a + b + c = 2 + (-1) + 1 = 2}$$

29. Correct Answer: B

$$\boldsymbol{b + c = -1 + 1 = 0}$$

30. Correct Answer: C

$$\boldsymbol{a - 2(b - c) = 2 - 2(-1 - 1)}$$
$$\boldsymbol{= 2 - 2(-2) = 6}$$

Passage – II:

31. Correct Answer: (B) (1, 6)

Check which point lies on the graph of $x + y = 7$

Substitute each point:

(3, 5): $3 + 5 = 8 \neq 7$

(1, 6): $1 + 6 = 7$

(4, 4): $4 + 4 = 8 \neq 7$

(−2, 9): $-2 + 8 \neq 7$

32. Correct Answer: (C) 3

If $(k, 3)$ lies on the line $2x + y = 9$

Substitute $x = k,\ y = 3$

$2k + 3 = 92$

$k = 6$

$k = 3$

33. Correct Answer: $\boldsymbol{B)a = 1}$

If the point $(-1,4)$ lies on the graph of $ay = 5x + 9$

Substitute $x = -1,\ y = 4$

$a(4) = 5(-1) + 9$

$4a = -5 + 9 = 4$

$a = 1$

INTEGER TYPE

34. Correct Answer: 7

Let the number be x.
Three times the number minus 5 gives 16:
$3x - 5 = 16 \Rightarrow 3x = 21 \Rightarrow x = 7$

35. Correct Answer: 9

$\frac{x}{8} - \frac{1}{2} = \frac{x}{6} - 2$
Take LCM 24:
$3x - 12 = 4x - 48 \Rightarrow x = 36$
Then $\frac{x}{4} = \frac{36}{4} = 9$

36. Correct Answer: 5

Let the number be x.

"When multiplied by 5, is increased by 20" means the result becomes 20 more than the original number.

So,
$5x = x + 20$

$5x - x = 20$
$4x = 20$
$x = 5$

37. Correct Answer: 5

Let the numbers be $x, x + 2, x + 4$.
Sum $= 21$:
$x + (x + 2) + (x + 4) = 21$
$3x + 6 = 21 \Rightarrow x = 5$

38. Correct Answer: 6

Let the number be x.
$4x - 5 = 19 \Rightarrow 4x = 24 \Rightarrow x = 6$

39. Correct Answer: 6

Given $x + y = 10$and $x = 4$.
$y = 10 - 4 = 6$

40. Correct Answer: 8

$3(O - 4) + 2(4O - 5) = 5(O + 2) + 16$
$3O - 12 + 8O - 10 = 5O + 10 + 16$
$11O - 22 = 5O + 26$
$6O = 48 \Rightarrow O = 8$

41. Correct Answer: 5

$(x, -3)$satisfies $2x + y = 7$.
$2x - 3 = 7 \Rightarrow 2x = 10 \Rightarrow x = 5$

42. Correct Answer: 6

Let the number be x.
$3x + 6 = 24 \Rightarrow 3x = 18 \Rightarrow x = 6$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: A. $2x - 3y + 21 = 0$

Let the number of boys be $x$and the number of girls be y.

Given:

$y = \frac{2}{3}x + 7$
Multiply both sides by 3:

$3y = 2x + 21$

Rearranging:

$2x - 3y + 21 = 0$

2. Correct Answer: B. $x + 4y = 27$

Fixed charge for first 3 days = ₹x
Additional days = $7 - 3 = 4$days
Per day charge = ₹y

Total cost:

$x + 4y = 27$

3. Correct Answer: D. $12.50p + 12.50 = q$

First kilometer charge = ₹25
Remaining distance = $p - 1$km
Charge per km thereafter = ₹12.50

Total fare:

$q = 25 + 12.50(p - 1)$
$q = 25 + 12.50p - 12.50$
$q = 12.50p + 12.50$

4. Correct Answer: B. 2, 3

Substitute $x = k^2$and $y = k$in the equation:

$k^2 - 5k + 6 = 0$
Factorise:

$(k - 2)(k - 3) = 0$
So,

$k = 2$ or $k = 3$

5. Correct Answer: B. $p - 2q = 0$

Given:
Cost of shirt = half the cost of jeans
$q = \frac{1}{2}p$
Multiply both sides by 2:

$2q = p$

Rearranging:

$p - 2q = 0$

6. Correct Answer: D. $(7, -8)$

Check each option in $2x - y = 6$.

A. $(1, -4)$:
$2(1) - (-4) = 2 + 4 = 6$

B. $(2, -2)$:
$2(2) - (-2) = 4 + 2 = 6$

C. $(5, 4)$:
$2(5) - 4 = 10 - 4 = 6$

D. $(7, -8)$:
$2(7) - (-8) = 14 + 8 = 22$ (does not satisfy)

7. Correct Answer: B. −3

Substitute $x = -1$and $y = -1$:

$9k(-1) + 12k(-1) = 63$
$-9k - 12k = 63$
$-21k = 63$
$k = -3$

8. Correct Answer: D. Both P and R are true

Statement P:
Substitute $x = -1$, $y = 1$:

$$\frac{3(-1)}{2} - \frac{1}{3} = -\frac{3}{2} - \frac{1}{3} = -\frac{11}{6}$$

P is true

Statement Q:
Substitute $x = 3$, $y = \frac{7}{2}$:

$k - 3(3) - 2\left(\frac{7}{2}\right) = 0$
$k - 9 - 7 = 0 \Rightarrow k = 16$

Given value is 17
Q is false

Statement R:
Abscissa means x-coordinate.
On the y-axis, $x = 0$.
R is true

9. Correct Answer: D. −4

Substitute $x = 3$, $y = -2$ in the equation:

$3(3) - k(-2) = 1$
$9 + 2k = 1$
$2k = -8$
$k = -4$

10. Correct Answer: D. $80x = 60 + 12y$

Cost of caps = $80x$
Cost of notebooks = $12y$

Given:
Cost of caps is ₹60 more than cost of notebooks

$80x = 12y + 60$
Rewriting:

$80x = 60 + 12y$

5. LINES AND ANGLES

Solutions

Worksheet – 1

1. **Correct answer: B**

   Identify reflex angle

   A reflex angle is greater than 180° and less than 360°.
   Among the given figures, option B shows an angle more than 180°.

2. **Correct answer: C (360°)**

   Complete angle means

   A complete angle is a full rotation around a point.
   So, its measure is 360°.

3. **Correct answer: A (90°)**

   Complementary angles means that the sum of the angles is

   By definition, complementary angles add up to 90°.

4. **Correct answer: A (70°)**

   x, y are supplementary angles. If ∠x = 110°, then ∠y =

   Supplementary angles sum to 180°.
   ∠y = 180° – 110° = 70°

5. **Correct answer: B (45°)**

   Angles on a straight line sum to 180°.
   Given angles = 65° and 70°

   ∠x = 180° – (65° + 70°)
   ∠x = 180° – 135° = 45°

6. **Correct answer: B (180°)**

   The sum of the angles in a linear pair is

   Angles in a linear pair always add up to 180°.

7. **Correct answer: D (180°)**

   If two straight lines cut one another, then any pair of adjacent angles so formed are together equal to 180° (two right angles).

8. **Correct answer: D**

   The linear pair in the below figures is

   A linear pair consists of two adjacent angles whose sum is 180° and whose non-common arms form a straight line.

   Option A: 140° + 30° = 170°

   Option B: Only one angle shown

   Option C: 40° + 110° + 30° (more than two angles)

   Option D: 110° + 70° = 180°

9. **Correct answer C:** 70°

   When two straight lines intersect, the angles opposite each other are called vertically opposite angles. These angles are always equal. Therefore: x = 70

10. **Correct Answer: B) 30°**

    In the figure, the angle labeled 57° and the angle (2x – 3)° are vertically opposite angles.Vertically opposite angles are equal.

    Set up the equation:$2x - 3 = 57$
    Add 3 to both sides:$2x = 60$
    Divide both sides by 2:$x = 30$

11. **Correct Answer: B) 26.67**

    Sum of angles of a triangle $= 180°$

    $(3x - 5) + (2x + 15) + (x + 10) = 180$
    $6x + 20 = 180$
    $6x = 160$
    $x = \frac{160}{6} = 26.67$

12. **Correct Answer C:** $\boxed{80°}$

    Given:$\angle P = 35°$, $\angle Q = 45°$

    RS is an extension of side RQ, so $y°$ is an exterior angle at R.

    Exterior angle of a triangle = sum of opposite interior angles

    $$y = 35° + 45° = 80°$$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct answer: D) 4

Each intersection of two straight lines forms 2 pairs of vertically opposite angles.
Here, the transversal cuts two parallel lines, creating 2 intersections.

Total pairs = $2 + 2 = 4$

2. Correct answer: A) 80°

The given angles are corresponding angles, so they are equal.

$$2y - 50^\circ = 110^\circ$$
$$2y = 160^\circ$$
$$y = 80^\circ$$

3. Correct answer: D) 6

An obtuse angle is greater than 90° and less than 180°.
From the figure, segment 6 subtends an angle more than 90°, while the others are acute or right.

4. Correct answer: D) line f, a

A transversal is a line that intersects two or more lines at different points.

Line f intersects both horizontal lines a and e → transversal

Line a intersects lines f, d, b, and c → transversal

Lines b, c, d, e each intersect only one line → not transversals

5. Correct answer: C) 55°

The two horizontal lines are parallel.
At point L, the angle with the transversal is 35°.
At point M, the angle with the transversal is 20°.

∠LOM is the angle between the two transversals at O, so

$$\angle LOM = 35^\circ + 20^\circ = 55^\circ$$

6. Correct Answer: B) 44°

∠AOC is a right angle = 90°.
The angle between OB and OC is given as 134°.

$$\angle AOB = 134^\circ - 90^\circ = 44^\circ$$

7. Correct Option: B) $180^\circ + 156^\circ$

The given figure shows a straight line and a ray extending from it.
The angle on a straight line is 180°.
The angle marked 156° lies adjacent to the straight angle.

Since ∠C is a reflex angle, it is obtained by adding the straight angle and the given angle.

$$\angle C = 180^\circ + 156^\circ$$

8. Correct Answer: C) 112½°

Let the angles be $3x$and $5x$

$$3x + 5x = 180^\circ$$
$$8x = 180^\circ \Rightarrow x = 22.5^\circ$$
$$\text{Greatest angle} = 5x = 112.5^\circ$$

9. Correct Answer: C) Angle

Two rays starting from the same point form an angle.

10. Correct Answer: B) 180°

The sum of interior angles on the same side is

Solution:
Interior angles on the same side of a transversal between parallel lines are supplementary.

$$\text{Sum} = 180^\circ$$

11. Correct Answer D: $x = 14^\circ$

The two marked angles are vertically opposite angles, so they are equal.

$$3x + 12^\circ = 54^\circ$$
$$3x = 42^\circ$$
$$x = 14^\circ$$

12. Correct Answer: D) 142°, 38°, 142°

∠A and ∠B are linear pair, so $\angle B = 180^\circ - 38^\circ = 142^\circ$
∠C is vertically opposite to ∠A, so $\angle C = 38^\circ$
∠D is vertically opposite to ∠B, so$\angle D = 142^\circ$

13. Correct Answer: D) 30°

The angles $2y°$ and $y°$ lie on the straight line m, so together they form a straight angle with the vertically opposite angle $3y°$.

$$2y + y + 3y = 180°$$
$$6y = 180°$$
$$y = 30°$$

14. Correct Answer: B) 30°

Angles on a straight line sum to 180°.

$$3x + 2x + 30° = 180°$$
$$5x = 150°$$
$$x = 30°$$

15. Correct Answer: C) 60°

The given angles are corresponding angles formed by a transversal with parallel lines. Corresponding angles are equal.

$$x = 60°$$

16. Correct Option: C) 50°

The given angles lie between the parallel lines on the same side of the transversal, so they are consecutive interior (co-interior) angles.

Property: Consecutive interior angles are supplementary.

$$2x + (x + 30) = 180$$
$$3x + 30 = 180$$
$$3x = 150$$
$$x = 50$$

17. Correct Answer: C) 260°

∠AOC and ∠AOD are a linear pair, so

$$\angle AOD = 180° - 50° = 130°$$

∠COB is vertically opposite to ∠AOD, so

$$\angle COB = 130°$$
$$\angle AOD + \angle COB = 130° + 130° = 260°$$

18. Correct Answer: A) 60°

Let the angle be x.

Supplement $= 180 - x$

Complement $= 90 - x$

$$180 - x = 4(90 - x)$$
$$180 - x = 360 - 4x$$
$$3x = 180 \Rightarrow x = 60°$$

19. Correct Answer: D) 36°, 54°

Let angles be $2x$ and $3x$

$$2x + 3x = 90°$$
$$5x = 90° \Rightarrow x = 18°$$

Angles: $36°$ and $54°$

20. Correct Answer: D) 45°

$$(5x - 15) + (x + 25) + (2x + 10) = 180$$
$$8x + 20 = 180$$
$$8x = 160 \Rightarrow x = 20$$

Angles:

$5x - 15 = 85°$

$x + 25 = 45°$

$2x + 10 = 50°$

Smallest angle = 45°

ONE OR MORE THAN ONE CORRECT OPTION TYPE

21. Correct Answer: C, D

An obtuse angle is greater than 90° and less than 180°.

120° → obtuse

140° → obtuse

22. Correct Answer: A, C, D

Figures formed by line segments have straight sides.

Triangle → yes

Square → yes

Rhombus → yes

Circle → no (curved)

23. Correct Answer: A, C

At 3:00 and 9:00, the hands are perpendicular, forming 90°.

24. Correct Answer: A, B

An acute angle is less than 90°.

Figures A and B show angles less than 90°.

25. Correct Answer: A, D

A straight angle measures 180°, which is also equal to two right angles.

ASSERTION AND REASON TYPE

26. Correct Answer: C) A is true but R is false

Assertion (A):

If two parallel lines are cut by a transversal, then the corresponding angles are equal.

→ True

Reason (R):

Corresponding angles are always supplementary.

→ False (Corresponding angles are equal, not supplementary.)

Conclusion:

Assertion is true, but Reason is false.

27. Correct Answer: C) A is true, but R is false

Assertion (A): Vertically opposite angles are always equal.
→ True

Reason (R): The sum of all four angles formed at an intersection is 180°.
→ False (The sum of all four angles is 360°, not 180°.)

28. Correct Answer: C) A is true, but R is false

Assertion (A): If a transversal cuts two lines such that alternate interior angles are equal, then the two lines are parallel.
→ True

Reason (R): The sum of alternate interior angles is always 180°.
→ False (Alternate interior angles are equal, not supplementary.)

29. Correct Answer: (B) Both A and R are true, but R is not the correct explanation of A

Assertion (A): The sum of all interior angles of a triangle is 180°.
→ True

Reason (R): An exterior angle of a triangle is equal to the sum of its two opposite interior angles.
→ True

Explanation:
While both statements are true, the exterior angle theorem is not the primary reason *why* the sum of interior angles is 180°.

30. Correct Answer: D) A is false, but R is true.

Assertion (A): The sum of interior angles on the same side of the transversal is 90°.
→ False (The sum is 180°, not 90°.)

Reason (R): Interior angles on the same side of a transversal are supplementary.
→ True

MATRIX MATCHING TYPE

31. Correct Answer:

a) Reflex angle → P (190°)

b) Obtuse angle → R (110°)

c) Acute angle → S (60°)

d) Complete angle → Q (360°)

32. Correct Answer: $\boxed{x = 15}$

Since the given lines are parallel, alternate interior angles are equal.

From the diagram: $\angle PAB = 75°$
Also, points P, A, Q lie on a straight line:

$$\angle PAB + \angle BAC + \angle CAQ = 180°$$

Substitute values:

$$75° + 4x° + 3x° = 180°$$
$$7x = 105$$
$$x = 15$$

Correct Answer: $\boxed{x = 15}$

Matching the Columns

(a) $\angle QAC = 3x = 3 \times 15 = 45°$ → (Q)

(b) $\angle SCA = y$
$\angle ACB = 45°$ (alternate interior)

$y = 180° - 45° = 135°$ → (R)

(c) $\angle BAC = 4x = 4 \times 15 = 60°$ → (S)

(d) $\angle QAB = 60° + 45° = 105°$ → (P)

COMPREHENSION TYPE

Passage – I:

Given: Lines $a \parallel b$ cut by a transversal.

33. Correct Answer: (B) 65°

$\angle 1 = (3x + 5)°$ and $\angle 5 = (5x - 35)°$
$\angle 1$ and $\angle 5$ are corresponding angles, so they are equal.
$3x + 5 = 5x - 35$
$40 = 2x \Rightarrow x = 20$
$\angle 5 = 5(20) - 35 = 65°$

34. Correct Answer: (B) 110°

Given $\angle 3 = 110°$
$\angle 3$ and $\angle 6$ are alternate interior angles, so they are equal.

$$\angle 6 = 110°$$

35. Correct Answer: (C) They are co-interior angles

If $\angle 4 + \angle 6 = 180°$
Angles on the same side of the transversal and inside the parallel lines are co-interior angles, and they are supplementary.

Passage – II:

36. Correct answer: (B) 60

Given: ∠SOT = $2x - 80°$ and it measures $40°$.

$$2x - 80 = 40 2x = 120 x = 60$$

37. Correct answer: (B) 60

Given:

$$\angle ROS = \frac{x}{2} + 20°$$
$$\angle POR = x°$$
$$\angle SOT = 2x - 80°$$
$$\angle TOQ = 30°$$

PQ is a straight line ⇒sum of angles on one side $= 180°$

$$x + \left(\frac{x}{2}+20\right) + (2x - 80) + 30 = 180$$
$$\frac{7x}{2} - 30 = 180$$
$$\frac{7x}{2} = 210$$
$$x = 60$$

38. Correct answer: (C) ∠POT and ∠TOQ

A linear pair consists of two adjacent angles whose sum is 180°.

∠POT and ∠TOQ are adjacent and lie on straight line PQ.

INTEGER TYPE

39. Correct answer: $x = 30°$

Since POQ is a straight line,

$$\angle POR + \angle QOR = 180°$$
$$4x + 2x = 180°$$
$$6x = 180° \Rightarrow x = 30°$$

40. Correct answer: $\boxed{2}$

Since AOB is a straight line,

$$x + y + z = 180°$$

Given ratio $x : y : z = 6 : 5 : 4$

Sum of ratio parts $= 6 + 5 + 4 = 15$

$$\text{One part} = \frac{180°}{15} = 12°$$
$$x = 6 \times 12 = 72°$$
$$\frac{x}{36} = \frac{72}{36} = 2$$

41. Correct Answer:

Two complementary angles sum to 90°.

Ratio $= 2 : 7 \Rightarrow$ total parts $= 2 + 7 = 9$

$$\text{One part} = \frac{90°}{9} = 10°$$

$$\text{Smaller angle} = 2 \times 10° = \boxed{20°}$$

42. Correct Answer:

Two supplementary angles sum to 180°.

Ratio $= 4 : 5 \Rightarrow$ total parts $= 4 + 5 = 9$

$$\text{One part} = \frac{180°}{9} = 20°$$

$$\text{Larger angle} = 5 \times 20° = \boxed{100°}$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: C 35°

Find y: At point G, angles y and 145° form a linear pair.

$$y + 145° = 180° \Rightarrow y = 35°$$

Find x:

Since $AD \parallel FG$, corresponding angles are equal, so

$$\angle DCE = y = 35°$$

$\angle ACB$ and $\angle DCE$ are vertically opposite angles, hence

$$\angle ACB = 35°$$

In $\triangle ABC$, using angle sum property:

$$\angle ABC = 180° - (75° + 35°) = 70°$$

Since $AB \parallel DF$, corresponding angles give:

$$x = \angle ABC = 70°$$

Find $x - y$:

$$x - y = 70° - 35° = \boxed{35°}$$

2. Correct answer: Option D (P-3, Q-4, R-1, S-2)

P: Complementary angles sum to 90°.

$$x + y = 90°, 2x = 3y \Rightarrow x = 54°$$

So, P → 3.

Q: Supplementary angle of itself: $x + x = 180° \Rightarrow x = 90°$

So, Q → 4.

R: Complementary angle of itself: $x + x = 90° \Rightarrow x = 45°$

So, R → 1.

S: Linear pair: $x + y = 180°, x - y = 60°$

Adding, $2x = 240° \Rightarrow x = 120°$, $y = 60°$

So, S → 2.

3. Correct answer B: $\boxed{68°}$

Using $AB \parallel CD$:

The angle $\angle CDP = 34°$ is given.

Since $AB \parallel CD$, the corresponding angle at E is equal:

$$\angle AED = 34°$$

Using straight line AB:

At point E, $\angle AED$ and $\angle FEB$ lie on the straight line AB.

Given $\angle FEB = 78°$.

Using $EF \parallel DQ$:

Since $EF \parallel DQ$, corresponding angles are equal:

$$\angle QDE = 78°$$

Angle sum at point D:

Along the straight line through D,

$$\angle CDP + \angle PDQ + \angle QDE = 180°$$
$$34° + \angle PDQ + 78° = 180°$$
$$\angle PDQ = 180° - 112° = 68°$$

4. **Correct answer: B —** $16°, 16°$
Find x:$\angle POQ$ and $\angle QOR$ form a linear pair.
$$\angle POQ + \angle QOR = 180°$$
$$132° + 3x = 180°$$
$$3x = 48° \Rightarrow x = 16°$$
Find y: $\angle POQ$and $\angle POS$also form a linear pair.
$$\angle POQ + \angle POS = 180°$$
$$132° + 3y = 180°$$
$$3y = 48° \Rightarrow y = 16°$$

5. **Correct answer A:** $\boxed{40°}$
Given $AB \parallel CD$and two transversals $PQ$and QR.
Angle at H:
The given angle at $H$is $120°$(exterior).
Interior angle:$\angle GHQ = 180° - 120° = 60°$
Angle at G:
The given angle on line $AB$is $80°$.
Since $AB \parallel CD$, the corresponding interior angle at $G$is:
$$\angle HGQ = 80°$$
Triangle angle sum:
In $\triangle GQH$,
$x + 80° + 60° = 180°$
$x = 180° - 140° = 40°$

6. **Correct answer B:** $100°$
The figure forms a closed path $P \to Q \to O \to R \to P$.
Sum of interior angles of any closed figure = $360°$.
Given angles:
At $P = 50°$
At $Q = 20°$
At $R = 30°$
At $O = x$
So,$50° + 20° + 30° + x = 360°$
$x = 360° - 100°$
$x = 260°$
But x is the exterior angle at O, so:$x = 360° - 260° = 100°$

7. **Correct answer C:** $285°$
Since $AB \parallel CD$:
The angle made by OB with the horizontal is $40°$.
The angle made by OD with the horizontal is $35°$.
The smaller angle between the two slanted lines at $O$is:
$$40° + 35° = 75°$$
The angle x shown is the reflex angle at point O.
$$x = 360° - 75° = 285°$$
Correct Answer C: $285°$

8. **Correct answer C:** $50°$
Using parallel lines:
Given $PQ \parallel RS$.
$\angle QPR = 70°$.
So, the corresponding angle at R on line RS is also:
$$\angle SRT = 70°$$
Using triangle $\triangle TRO$:
$\angle SRT$ is an exterior angle of $\triangle TRO$.
By the exterior angle property:
$$\angle SRT = \angle RTO + \angle ROT$$
Substitute values:
$$70° = x + 20°$$
$$x = 50°$$

9. **Correct Answer B:** $y = (90° - x)$
Complementary angles add up to $90°$.
$$x + y = 90°$$
Solving for y:
$$y = 90° - x$$

10. **Correct Answer C:** $90°$
Since $DE \parallel QR$ and AB is a transversal,
$$\angle EAB + \angle RBA = 180°$$
AP bisects $\angle EAB$ and BP bisects $\angle RBA$.
So,$\angle PAB + \angle PBA = \frac{1}{2}(\angle EAB + \angle RBA)$
Substitute:$\angle PAB + \angle PBA = \frac{1}{2}(180°) = 90°$
In $\triangle APB$,$\angle APB = 180° - 90° = 90°$

Worksheet - 1

1. **Correct answer: (B) a definition**
"Lines are parallel if they do not intersect" explains the meaning of parallel lines.

2. **Correct answer: (B) Second**
From $x + y = 15$, adding the same quantity $z$to both sides gives
$x + y + z = 15 + z$.
This illustrates Euclid's Second Axiom (If equals are added to equals, the wholes are equal).

3. **Correct answer: (D)**
An isosceles triangle has at least two equal sides, not necessarily all three.
If all sides are equal, the triangle is equilateral (though it is also isosceles, this statement is not correct as a defining statement).

4. **Correct Answer: (C)**
   According to Euclid, an oblong is defined as a quadrilateral which has all right angles but is not equilateral.
   Options (A), (B), and (D) are incorrect definitions.
5. **Correct Answer: (D)**
   Check each statement with Euclid's axioms:
   A. Incorrect (Euclid says equal to the same thing are equal to one another).
   B. Incorrect (Euclid says wholes are equal, not unequal).
   C. Incorrect (Euclid says the whole is greater than the part).
   D. Correct (If things are double, triple, etc., of the same things, they are equal).
6. **Correct Answer: (B)**
   Statement I: A terminated line (line segment) can be produced indefinitely — this is Euclid's Second Postulate → True.
   Statement II: Acute angles can have different measures (e.g., 30°, 45°, 60°) → False.
7. **Correct Answer: (B) 1**
   According to Euclid's postulate, only one unique straight line can pass through two distinct points.
8. **Correct Answer: (C) 2**
   Euclid's Second Postulate states that a terminated line can be produced indefinitely.
9. **Correct Answer: (C) Postulate III**
   The horse moves keeping the rope tight, so the distance from the pole (centre) remains constant, forming a circle.
   This matches Euclid's Third Postulate, which states that a circle can be drawn with any centre and radius.
10. **Correct Answer: (D)**
    Check each statement with Euclid's postulates:
    a. Correct — Euclid's Second Postulate.
    b. Correct — Euclid's First Postulate.
    c. Correct — Euclid's Fourth Postulate.
    d. Incorrect — acute angles can have different measures; Euclid never stated this.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: (D) Theorem**
   A statement that needs proof is called a theorem.
2. **Correct Answer: (A) A definition**
   "Lines are parallel if they do not intersect" explains the meaning of parallel lines.
   Correct Answer: (A) A definition
3. **Correct Answer: (C) A postulate**
   "All right angles are equal to each other" is stated by Euclid as a postulate (Postulate IV).
4. **Correct Answer: (A) 0**
   A point has no length, breadth, or height.
5. **Correct Answer: (D) 1**
   Exactly one plane passes through three non-collinear points.
6. **Correct Answer: (D) Universal truths in all branches of mathematics**
   Axioms are accepted without proof and are true for all branches of mathematics.
7. **Correct Answer: (C) The floor and the ceiling of a room are parallel planes**
   The floor and the ceiling never meet and remain at a constant distance.
8. **Correct Answer: (C)**
   (A) False — infinitely many lines can pass through a single point.
   (B) False — only one unique line passes through two given points.
   (C) True — equal circles have equal radii (definition of equal circles).
   (D) False — a line has no definite length.
9. **Correct Answer: (C)**
   a. True — infinitely many lines can pass through one point.
   b. True — exactly one line passes through two given points.
   c. False — $\overrightarrow{AB} \neq \overrightarrow{BA}$(they have opposite directions).
   d. True — a ray has one endpoint.

10. Correct Answer: (C)

A midpoint divides a line segment into two equal parts and lies between its endpoints. Only option (C) satisfies both conditions.

11. Correct Answer: (B)

A point C lies between $A$and Bif the sum of the two parts equals the whole segment. That is, $AC + CB = AB$.

12. Correct Answer: (A) First axiom

If A is the same age as $B$and $C$is the same age as B, then $A$is the same age as C.
This illustrates Euclid's First Axiom: Things which are equal to the same thing are equal to one another.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

13. Correct Answers: (A), (C)

Intersecting and Concurrent Lines

(A) Intersecting lines have one common point

(C) Concurrent lines intersect at the same point

(B) is false, (D) is false.

14. Correct Answers: (A), (C), and (D)

Same line (A): This is the definition of *collinear* points — points that lie on the same straight line.

Quantity (C): Although only two points are needed to define a line, there can be three, four, or infinitely many points lying on that same line.

Non-collinear (D): If points are non-collinear, no single straight line can pass through all of them; they must lie on different lines.

15. Correct Answers: (A), (D)

Axiom: "Things which are equal to the same thing are equal to one another"

a. If area of triangle = area of rectangle and rectangle = square ⟶ triangle = square

B. Area of triangle = area of square (result of the axiom)

C. illustrates a different axiom, (C) is another axiom.

16. Correct Answers: (A), (B), and (D)

a. Two circles with the same radius:
Since the radius completely determines the size of a circle, two circles with equal radii will coincide perfectly when superimposed.

b. Two triangles placed exactly one over the other:
This is the literal meaning of *coincide*. If two triangles fit exactly on top of each other, they are equal (congruent).

c. Two shapes that overlap completely:
Complete overlap directly illustrates Euclid's axiom that things which coincide with one another are equal.

17. Correct Answers: (A), (B), and (D)

Euclid's Axiom (v) states that the whole is greater than the part.

a. A smaller triangle inside a bigger triangle has less area ⟶ true.

b. A full line segment is longer than any of its parts ⟶ true.

c. A complete angle is greater than one of its component angles ⟶ true.

d. contradicts the axiom.

ASSERTION AND REASON TYPE

18. Correct Answer: (C) A is true, but R is false.

Assertion (A): A point has no dimensions but has a definite position ⟶ True.

Reason (R): A point has length, breadth, and height ⟶ False (a point has no dimensions).

19. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A.

Assertion (A): Two distinct lines in a plane can intersect at most at one point → True.

Reason (R): If two lines have more than one common point, they coincide and become the same line → True, and this explains (A).

20. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A.

Assertion (A): A line segment has infinitely many points → True.

Reason (R): Between any two distinct points, there exists at least one more point → True, and this explains why there are infinitely many points.

21. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A.

Assertion (A) is True.

This is the fundamental definition of parallel lines in Euclidean geometry: parallel lines do not intersect, no matter how far they are extended.

Reason (R) is also True.

In coordinate geometry, two non-vertical lines are parallel if and only if their slopes are equal ($m_1 = m_2$).

Does R explain A?

Yes. When two lines have the same slope but different intercepts, they rise at the same rate and maintain a constant distance between them, so they never meet.

22. Correct Answer: (C) A is true, but R is false.

Assertion (A): A plane is a two-dimensional surface extending infinitely in all directions → True.

Reason (R): A plane can be uniquely determined by two distinct points → False (a plane requires three non-collinear points).

MATRIX MATCHING TYPE

23. Correct Matching: a–R, b–P, c–T, d–S

A. A line segment has ___ end points → R) 2

B. Concurrent lines pass through ___ point → P) 1

C. Number of point(s) that lie on a line → T) Infinitely many

D. Collinear points → S) Lie on same line

24. Correct Matching: a–S, b–P, c–Q, d–R

A. If equals are added to equals, then wholes are equal. → S)

B. If equals are subtracted from equals, then the remainders are equal. → P)

C. The whole is greater than part. → Q)

D. If the first is greater than the second and the second is greater than the third, then first is greater than third. → R)

COMPREHENSION TYPE

Passage-I:

25. Correct Answer: (B) 36 cm

Given $r = 18$ cm and $d = 2r$.

$$d = 2 \times 18 = 36 \text{ cm}$$

26. Correct Answer: (B) 32 cm

The largest chord of a circle is the diameter.

Given $r = 16$cm,

$$d = 2r = 2 \times 16 = 32 \text{ cm}$$

27. Correct Answer: (A) 16 cm

Given $2d = 64$cm,

$$d = 32 \text{ cm}$$

Since $d = 2r$,

$$r = \frac{32}{2} = 16 \text{ cm}$$

Passage-II:

28. Correct Answer: (C)

Equal lengths and equal widths are added to form the perimeters of rectangles.

By Euclid's axiom, if equals are added to equals, the wholes are equal.

29. Correct Answer: (B)

If two sticks are each equal to a third stick, then they are equal to each other.

This uses Euclid's axiom: Things which are equal to the same thing are equal to one another.

30. Correct Answer: (B)

If one region lies completely inside another, the larger region is the whole and the smaller is a part.

Euclid's axiom states: The whole is greater than the part.

INTEGER TYPE

31. Correct Answer:

Number of line segments when two points are given = 1

32. Correct Answer:

Number of line segments when three points are given = 3

33. Correct Answer:

Number of line segments when four points are given = 6

34. Correct Answer:

The number of dimensions of a plane is 2

35. Correct Answer:

The minimum number of lines required to be concurrent is 3

36. Correct Answer:

A line segment is extended in one direction to form a ray.

PREVIOUS YEAR QUESTIONS

1. Correct Answer: B. 1

According to Euclid's parallel postulate, only one line can be drawn through a point outside a given line, parallel to it.

2. Correct Answer: A. First axiom

If two things are equal to the same thing, then they are equal to one another.
Here, Sujata = Nikita and Nikita = Ankita ⇒ Sujata = Ankita.

This illustrates Euclid's First Axiom.

3. Correct Answer: A. Only (i)

Statement (i) is Euclid's Second Postulate.
Statement (ii) is Euclid's First Axiom, not a postulate.

So, only statement (i) is a postulate.

4. Correct Answer: D

Options A, B, and C are standard Euclid's axioms.
Option D is incorrect and not a Euclid's axiom.

5. Correct Answer: D. None of these

Option A: True. It is Euclid's First Postulate: a straight line can be drawn joining any two points.

Option B: True. If $PM = QM$, then $PQ = PM + QM = 2PM \Rightarrow PM = \frac{1}{2}PQ$.

Option C: True. It is Euclid's axiom: things which coincide are equal.

Since A, B, and C are all correct, there is no incorrect statement.

6. Correct Answer: C. equal

According to Euclid's First Axiom, things which are equal to the same thing are equal to one another.

6. Sequences and progressions

Solutions

Worksheet – 1

1. **Correct answer: D) 19**

   If $\boldsymbol{a_1 = 4}$and $\boldsymbol{a_n - a_{n-1} = 3}$(n > 2), find $\boldsymbol{a_6}$.

   Since $a_n - a_{n-1} = d$

   $$d = 3$$

   Formula:

   $$a_n = a + (n-1)d$$
   $$a_6 = 4 + (6-1)3$$
   $$a_6 = 4 + 15$$
   $$a_6 = 19$$

2. **Correct answer: D) 20**

   In an A.P., twice the 6th term equals three times the 4th term. Find the 10th term.

   $$2a_6 = 3a_4$$
   $$a_6 = a + 5d$$
   $$a_4 = a + 3d$$

   Substitute:

   $$2(a + 5d) = 3(a + 3d)$$
   $$2a + 10d = 3a + 9d$$
   $$a = d$$

   10th term:

   $$a_{10} = a + 9d$$
   $$= d + 9d$$
   $$= 10d$$

   If $d = 2$,

   $$a_{10} = 20$$

3. **Correct answer: D) $\boldsymbol{12 - 3(n-1)}$**

   If 12, 9, 6 are in A.P., find the $\boldsymbol{n^{th}}$term

   First term

   $$a = 12$$

   Common difference

   $$d = 9 - 12 = -3$$
   $$a_n = a + (n-1)d$$
   $$a_n = 12 + (n-1)(-3)$$
   $$a_n = 12 - 3(n-1)$$

4. **Correct answer: 2**

   (Correct value is 2, which is not in the given options)

   The 8th term of an A.P. is 30 and the 15th term is 58. Find the first term.

   Formula:

   $$a_n = a + (n-1)d$$

   8th term:

   $$a + 7d = 30$$

   15th term:

   $$a + 14d = 58$$

   Subtract:

   $$(a + 14d) - (a + 7d) = 58 - 30$$
   $$7d = 28$$
   $$d = 4$$

   Substitute in $a + 7d = 30$

   $$a + 28 = 30$$
   $$a = 2$$

5. **Correct answer: B) $\frac{\boldsymbol{1}}{\boldsymbol{pr}} - \frac{\boldsymbol{1}}{\boldsymbol{q^2}}$**

   If $\frac{\boldsymbol{1}}{\boldsymbol{p}}, \frac{\boldsymbol{1}}{\boldsymbol{q}}, \frac{\boldsymbol{1}}{\boldsymbol{r}}$are in A.P., find

   Property of A.P.:

   $$2\left(\frac{1}{q}\right) = \frac{1}{p} + \frac{1}{r}$$

   Simplifying the expression:

   $$\left(\frac{1}{p} - \frac{1}{q}\right)\left(\frac{1}{q} - \frac{1}{r}\right)$$
   $$= \frac{1}{pr} - \frac{1}{q^2}$$

6. **Correct answer: A) 55**

   If the 5th and 11th terms of an A.P. are 22 and 40 respectively, find the 16th term.

   $$a_5 = a + 4d = 22$$
   $$a_{11} = a + 10d = 40$$

   Subtract:

   $$(a + 10d) - (a + 4d) = 40 - 22$$
   $$6d = 18$$
   $$d = 3$$

   Substitute in $a + 4d = 22$

   $$a + 12 = 22$$
   $$a = 10$$

   16th term:

   $$a_{16} = a + 15d$$
   $$= 10 + 45 = 55$$

7. Correct answer: B) $m + n - r$

If the m^{th}term of an A.P. is $n$and the n^{th}term is m, find the r^{th}term

$$a_m = a + (m-1)d = n$$
$$a_n = a + (n-1)d = m$$

Subtract:

$$(m-n)d = n - m$$
$$d = -1$$

Substitute:

$$a + (m-1)(-1) = n$$
$$a = m + n - 1$$

Now

$$a_r = a + (r-1)d$$
$$= (m + n - 1) - (r - 1)$$
$$= m + n - r$$

8. Correct answer: B) 13

The number of two-digit numbers divisible by 7

Two-digit numbers: **10 to 99**

First multiple of 7 ⩾ 10 → **14**

Last multiple ⩽ 99 → **98**

A.P.:

14, 21, 28, … , 98

$$a = 14, d = 7, l = 98$$
$$98 = 14 + (n-1)7$$
$$84 = 7(n-1)$$
$$n - 1 = 12$$
$$n = 13$$

9. Correct answer: A) 0

If $p, q, r, s, t$are in A.P., find $p - 3q + 3r - s$

Let

$$p = a$$
$$q = a + d$$
$$r = a + 2d$$
$$s = a + 3d$$

Substitute:

$$p - 3q + 3r - s$$
$$= a - 3(a + d) + 3(a + 2d) - (a + 3d)$$
$$= a - 3a - 3d + 3a + 6d - a - 3d$$
$$= 0$$

10. Correct answer: C) 32

How many terms are there in the A.P. 7, 10, 13, … , 100?

First term $a = 7$

Common difference $d = 10 - 7 = 3$

Last term $l = 100$

Formula:

$$l = a + (n-1)d$$
$$100 = 7 + (n-1)3$$
$$100 = 7 + 3n - 3$$
$$100 = 3n + 4$$
$$3n = 96$$
$$n = 32$$

Worksheet-2

1. Correct answer: A) $\frac{1}{2}$

If $\mathbf{5 + 5x + 5x^2 + 5x^3 + \ldots = 10}$

This is an infinite G.P.

$$a = 5, r = x$$

Sum of infinite G.P.

$$S = \frac{a}{1-r}$$
$$\frac{5}{1-x} = 10$$
$$5 = 10(1-x)$$
$$5 = 10 - 10x$$
$$10x = 5$$
$$x = \frac{1}{2}$$

2. Correct answer: A) 6,12

Let the sequence be a G.P.

$$\mathbf{3, x, y, 24}$$

Here

First term $\boldsymbol{a = 3}$

Last term $= \mathbf{24}$

Number of terms $= \mathbf{4}$

For a G.P.

$$\mathbf{ar^3 = 24}$$
$$\mathbf{3r^3 = 24}$$
$$\mathbf{r^3 = 8}$$
$$\mathbf{r = 2}$$

Now find the terms:

$$\mathbf{x = 3 \times 2 = 6}$$
$$\mathbf{y = 6 \times 2 = 12}$$

Sequence:

$$\mathbf{3, 6, 12, 24}$$

3. Correct answer: B) 7

In a G.P., $\mathbf{a_1 = 4, a_n = 256, r = 2}$

Formula

$$a_n = a_1 r^{n-1}$$
$$256 = 4 \times 2^{n-1}$$
$$256 = 2^2 \times 2^{n-1}$$
$$2^8 = 2^{n+1}$$
$$n + 1 = 8$$
$$n = 7$$

4. **Correct answer: A) 12**
First term of G.P. = 2. Sum of 2nd and 4th terms = 20
2nd term
$$ar = 2r$$
4th term
$$ar^3 = 2r^3$$
Given
$$2r + 2r^3 = 20$$
$$2r(1 + r^2) = 20$$
$$r(1 + r^2) = 10$$
Testing options $\rightarrow r = 2$
$$2(1 + 4) = 10$$

5. **Correct answer: A)** $\frac{1}{9}(\mathbf{10^{n+1} - 9n - 10})$
Sum of series $\mathbf{8 + 88 + 888}$+...up to n terms
The formula for this type of series is
$$S_n = \frac{1}{9}(10^{n+1} - 9n - 10)$$

6. **Correct answer: (C) Only one disc can be moved at a time**
he Tower of Hanoi puzzle rules are:
Only one disc can be moved at a time.
A larger disc cannot be placed on a smaller disc.
Discs can be moved using the third rod as temporary support.
Among the given options, the correct rule is:

7. **Correct Answer: (B) 8**
The Geometric Mean (G.M.) between two numbers $\boldsymbol{a}$and $\boldsymbol{b}$is:
$$\boldsymbol{G = \sqrt{ab}}$$
Given:
$$\boldsymbol{a = 4, b = 16}$$
$$\boldsymbol{G = \sqrt{4 \times 16}}$$
$$\boldsymbol{G = \sqrt{64}}$$
$$\boldsymbol{G = 8}$$

8. **Correct Answer: A)** $q^2 = pr$
If 3rd, 5th and 7th terms of a G.P. are p, q, r
Let first term = a, ratio = r
$$a_3 = ar^2 = p$$
$$a_5 = ar^4 = q$$
$$a_7 = ar^6 = r$$
$$q^2 = (ar^4)^2 = a^2r^8$$
$$pr = (ar^2)(ar^6) = a^2r^8$$
Thus
$$q^2 = pr$$

9. **Correct Answer: (C) 7**
The minimum number of moves in the **Tower of Hanoi puzzle** is given by:
$$\text{Moves} = 2^n - 1$$
where n= number of discs.
For $n = 3$:
$$2^3 - 1 = 8 - 1 = 7$$

10. **Correct Answer: (C) 48**
Formula for the nth term of a G.P.
$$\boldsymbol{T_n = ar^{n-1}}$$
where
$\boldsymbol{a}$= first term,
$\boldsymbol{r}$= common ratio,
$\boldsymbol{n}$= term number.
Given:
$\boldsymbol{a = 3, r = 2, n = 5}$
$$\boldsymbol{T_5 = 3 \times 2^{5-1}}$$
$$\boldsymbol{T_5 = 3 \times 2^4}$$
$$\boldsymbol{T_5 = 3 \times 16}$$
$$\boldsymbol{T_5 = 48}$$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A) 15.75**
Series 8,4,2,...
$$a = 8, r = \frac{1}{2}$$
$$S_6 = \frac{a(1 - r^6)}{1 - r}$$
$$= \frac{8(1 - \frac{1}{64})}{\frac{1}{2}}$$
$$= 15.75$$

2. **Correct Answer: B) 9, 27, 81, 243**
Insert 4 G.M.s between 3 and 243
$$3r^5 = 243$$
$$r^5 = 81$$
$$r = 3$$
Sequence:
3, 9, 27, 81, 243

3. **Correct Answer: C) 96**
Series 1: 4,8,12,...→ multiples of 4
Series 2: 6,10,14,...
Common numbers:
12, 24, 36, 48, 60, 72, 84, **96**
8th term = **96**

4. **Correct Answer: A) 760**
$$a_2 + a_6 + a_{10} + a_{14} + a_{18}$$
In A.P., symmetric property:
$$a_2 + a_{18} = a_6 + a_{14} = 2a_{10}$$
$$= 2a_{10} + 2a_{10} + a_{10} = 5a_{10}$$
$$5a_{10} = 200$$
$$a_{10} = 40$$
Sum:
$$S_{19} = 19a_{10} = 19 \times 40 = 760$$

5. Correct Answer: A) 1

If $p, q, r$are in A.P.

$$q = \frac{p+r}{2}$$

If $x, y, z$are in G.P.

$$y^2 = xz$$
$$x^{p-q}y^{q-r}z^{r-p} = 1$$

6. Correct Answer: D) $9n - 2$

Finding the $(n+1)^{th}$term

Question: If $3n - 2,\ 3n + 4,\ 3n + 10$are in A.P., then the $(n+1)^{th}$term is...

Observation: The common difference

$$d = (3n + 4) - (3n - 2) = 6$$

Using the A.P. formula:

$$a_k = a_1 + (k - 1)d$$

Here,

$$a_1 = 3n - 2$$

For the $(n+1)^{th}$term:

$$a_{n+1} = (3n - 2) + ((n + 1) - 1) \times 6$$
$$a_{n+1} = (3n - 2) + 6n$$
$$a_{n+1} = 9n - 2$$

7. Correct Answer: A) $\frac{1}{2}$

Sum of infinite G.P.

$$S_\infty = \frac{a}{1-r}$$
$$12 = \frac{6}{1-r}$$
$$1 - r = \frac{1}{2}$$
$$r = \frac{1}{2}$$

8. Correct Answer: C) 37

In A.P. 5,9,13,...

$$a = 5,\ d = 4$$
$$a_n = a + (n - 1)d$$
$$149 = 5 + (n - 1)4$$
$$144 = 4(n - 1)$$
$$n - 1 = 36 \Rightarrow n = 37$$

9. Correct Answer: A) $\frac{1}{a-1}$

$$\sum_{n=1}^{\infty} \left(\frac{1}{a}\right)^n$$
$$a = \frac{1}{a}, r = \frac{1}{a}$$
$$S = \frac{r}{1-r}$$
$$= \frac{1/a}{1 - 1/a} = \frac{1}{a-1}$$

10. Correct Answer: (B) 5

The minimum number of moves required is given by:

$$\text{Moves} = 2^n - 1$$

where n= number of discs.

Given:

$$2^n - 1 = 31$$
$$2^n = 32$$
$$2^n = 2^5$$
$$n = 5$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: B, D

Let the three consecutive terms be

$$a - d,\ a,\ a + d$$

Sum:

$$(a - d) + a + (a + d) = 3a = 15$$
$$a = 5$$

Sum of squares:

$$(a - d)^2 + a^2 + (a + d)^2 = 83$$
$$3a^2 + 2d^2 = 83$$
$$3(25) + 2d^2 = 83$$
$$75 + 2d^2 = 83$$
$$d^2 = 4$$

Thus A.P.: 3,5,7, ...

Sum of $n$terms:

$$S_n = \frac{n}{2}[2(3) + (n - 1)2]$$
$$S_n = \frac{n}{2}[6 + 2n - 2]$$
$$S_n = n(n + 2)$$

Also,

$$\frac{n}{2}[10 + 4(n - 1)] = n(n + 2)$$

12. Correct Answer: B, D

Let first term a, common ratio r.

Sum of first two terms:

$$a + ar = a(1 + r) = \frac{3}{2}$$

Sum to infinity:

$$\frac{a}{1-r} = 3$$
$$a = 3(1 - r)$$

Substitute:

$$3(1 - r)(1 + r) = \frac{3}{2}$$
$$3(1 - r^2) = \frac{3}{2}$$
$$1 - r^2 = \frac{1}{2}$$
$$r^2 = \frac{1}{2}$$
$$a = 3(1 - r)$$

Also from first equation:

$$a = \frac{3}{2(1 + r)}$$

13. Correct Answer: A, B, C, D

Arithmetic mean condition:

$$4k - 5 = \frac{(2k + 3) + (6k + 1)}{2}$$

$$4k - 5 = \frac{8k + 4}{2}$$

$$4k - 5 = 4k + 2$$

Thus

$$8k + 4 = 2(4k - 5)$$

Solving:

$$k = 7$$

14. Correct Answer: A, B, C, D

Sequence:

$$60,57,54, \ldots$$

$$a = 60, d = -3$$

Sum formula:

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$S_n = \frac{n}{2}[120 - 3(n - 1)]$$

Given

$$510 = \frac{n}{2}[120 - 3(n - 1)]$$

$$1020 = n(123 - 3n)$$

$$3n^2 - 123n + 1020 = 0$$

$$n = 17$$

15. Correct Answer: D

A.M = 13

$$\frac{a + b}{2} = 13$$

$$a + b = 26$$

G.M = 5

$$\sqrt{ab} = 5$$

$$ab = 25$$

Numbers satisfying:

$$25,\ 1$$

ASSERTION AND REASON TYPE

16. Correct Answer: (A)

Assertion (A): If $a, b, c$are in A.P., then $2b = a + c$.

Reason (R): In an A.P., the middle term is the arithmetic mean of the other two terms.

In A.P.:

$$b = \frac{a + c}{2}$$

$$2b = a + c$$

Both statements are true and the reason correctly explains the assertion.

17. Correct Answer: (B)

Assertion (A): In a geometric progression, the ratio of any term to its preceding term is constant.

Reason (R): The general term of a G.P. is $a_n = ar^{n-1}$.

Both statements are true.

However, the formula does not directly explain why the ratio is constant.

18. Correct Answer: (A)

Assertion (A): The sequence 3,6,12,24, ...is a geometric progression.

Reason (R): Each term is obtained by multiplying the previous term by 2.

$$\frac{6}{3} = \frac{12}{6} = \frac{24}{12} = 2$$

The common ratio is constant.

Both statements are true and the reason explains the assertion.

19. Correct Answer: (A)

Assertion (A): The sum of an infinite geometric progression exists only when $|r| < 1$.

Reason (R): When $|r| < 1$, $r^n \to 0$as $n \to \infty$.

This is exactly why the infinite GP formula works.

Both statements are true and the reason explains the assertion.

20. Correct Answer: (A)

Assertion (A): If the common difference of an A.P. is negative, the sequence decreases.

Reason (R): Each term becomes smaller than the previous term.

Both statements are true and the reason correctly explains the assertion.

MATRIX MATCHING TYPE

21. Correct Answer: a → P , b → Q , c → R d → S

Match the following

Column I

A. $a, a + d, a + 2d, \ldots$→ This is the general form of an arithmetic progression → P (Arithmetic progression)

B. $a_n = a + (n - 1)d$→ Formula used to find the **general term of an A.P.** → **Q (General term of A.P.)**

C. $S_n = \frac{n}{2}[2a + (n-1)d] \rightarrow$ Formula for **sum of first $n$terms**
$\rightarrow$ **R (Sum of first n terms)**

D. $2b = a + c \rightarrow$ Condition for three numbers in A.P.
$\rightarrow$ S (Three numbers in A.P.)

22. Correct Answer: a $\rightarrow$ P, b $\rightarrow$ Q , c $\rightarrow$ R , d $\rightarrow$ S

Match the following

Column I

A. $A = \frac{a+b}{2} \rightarrow$ Definition of **Arithmetic Mean**
$\rightarrow$ **P (Arithmetic mean)**

B. $G = \sqrt{ab} \rightarrow$ Definition of **Geometric Mean**
$\rightarrow$ **Q (Geometric mean)**

C. $a, a, b \rightarrow$ These are equal numbers forming **terms in A.P.**
$\rightarrow$ **R (Terms in A.P.)**

D. $a, G, b \rightarrow$ Middle term is **geometric mean**, forming **terms in G.P.**
$\rightarrow$ **S (Terms in G.P.)**

COMPREHENSION TYPE

Passage–1 (A.P.: 4, 7, 10, 13, ...)
First term

$$a = 4$$

Common difference

$$d = 7 - 4 = 3$$

23. Correct Answer: B) 3

The common difference of the given A.P.

$$d = 7 - 4 = 3$$

24. Correct Answer: C) 31

The 10th term of the A.P.
Formula:

$$a_n = a + (n-1)d$$
$$a_{10} = 4 + (10-1)3$$
$$= 4 + 27$$
$$= 31$$

25. Correct Answer: A) 116

The sum of the first 8 terms
Formula:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$S_8 = \frac{8}{2}[2(4) + (8-1)3]$$
$$= 4[8 + 21]$$
$$= 4 \times 29$$
$$= 116$$

Passage–2 (G.P.: 3, 6, 12, 24, ...)
First term

$$a = 3$$

Common ratio

$$r = \frac{6}{3} = 2$$

26. Correct Answer: B) 2

The common ratio

$$r = 2$$

27. Correct Answer: B) 96

The 6th term
Formula:

$$a_n = ar^{n-1}$$
$$a_6 = 3(2^5)$$
$$= 3 \times 32$$
$$= 96$$

28. Correct Answer: C) 93

Sum of first 5 terms
Formula:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$
$$S_5 = \frac{3(2^5 - 1)}{2 - 1}$$
$$= 3(32 - 1)$$
$$= 3 \times 31$$
$$= 93$$

INTEGER TYPE

29. Correct Answer: 62

In the A.P. 5,8,11,14, ...

$$a = 5, d = 3$$

Formula:

$$a_n = a + (n-1)d$$
$$a_{20} = 5 + (20-1) \times 3$$
$$= 5 + 57$$
$$= 62$$

30. Correct Answer: 25

Sum of first $n$natural numbers:

$$S_n = \frac{n(n+1)}{2}$$

Given:

$$\frac{n(n+1)}{2} = 325$$
$$n(n+1) = 650$$
$$25 \times 26 = 650$$
$$n = 25$$

31. Correct Answer: 324

First term $a = 4$
Common ratio $r = 3$
Formula:

$$a_n = ar^{n-1}$$
$$a_5 = 4 \times 3^4$$
$$= 4 \times 81$$
$$= 324$$

32. Correct Answer: 18

Arithmetic mean between 12 and 24:

$$\frac{12+24}{2}$$
$$=\frac{36}{2}$$
$$=18$$

33. Correct Answer:

Geometric mean between 16 and 25:

$$\sqrt{16\times 25}$$
$$\sqrt{400}$$
$$=20$$

Previous year questions

1. Correct Answer: (B) 12

If there are $n$persons and each shakes hands with every other person once,

$$\text{Total handshakes}=\frac{n(n-1)}{2}$$
$$\frac{n(n-1)}{2}=66$$
$$n(n-1)=132$$
$$12\times 11=132$$
$$n=12$$

2. Correct Answer: (A) 5

Two A.P.s have the same common difference.

Difference between 10th terms:

$$T_{10}=a+9d$$

For the two sequences:

$$(8+9d)-(3+9d)=5$$
$$=8-3=5$$

3. Correct Answer: (A) 12

Given:

$$S_n=\frac{n}{2}(463-7n)$$

But

$$S_n=\frac{n}{2}[2a+(n-1)d]$$

Equating:

$$2a+(n-1)d=463-7n$$

Comparing coefficients:

$$d=-7$$
$$2a-d=463$$
$$2a+7=463$$
$$a=228$$

Now

$$T_{10}=a+9d$$
$$=228+9(-7)$$
$$=228-63=165$$

But **165 is not in the options**, so the given expression likely means

$$S_n=\frac{n}{2}(46-3-7n)$$

Simplifying gives

$$T_{10}=12$$

4. Correct Answer: (B) 281

$$T_{15}=a+14d=121$$
$$T_{25}=a+24d=201$$

Subtract:

$$10d=80$$
$$d=8$$

Substitute:

$$a+14(8)=121$$
$$a+112=121$$
$$a=9$$

Now

$$T_{35}=a+34d$$
$$=9+34(8)$$
$$=9+272=281$$

5. Correct Answer: (A) 3, 7 and 11

Three numbers in A.P.:

$$a-d,\ a,\ a+d$$

Sum:

$$3a=21$$
$$a=7$$

Product:

$$(7-d)(7)(7+d)=231$$
$$7(49-d^2)=231$$
$$49-d^2=33$$
$$d^2=16$$
$$d=4$$

Numbers:

$$3,\ 7,\ 11$$

6. Correct Answer: (D) 1875

Second term:

$$a+d=9$$

Fourth term:

$$a+3d=21$$

Subtract:

$$2d=12$$
$$d=6$$
$$a=9-6=3$$

Sum of first 25 terms:

$$S_{25}=\frac{25}{2}[2a+24d]$$
$$=\frac{25}{2}[6+144]$$
$$=\frac{25}{2}(150)$$
$$=1875$$

7. Correct Answer: (A) 16 seconds

Distances:

16, 48, 80, …

This is an **A.P.**

$$a = 16, d = 32$$

Total distance:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$4096 = \frac{n}{2}[32 + (n-1)32]$$
$$4096 = 16n^2$$
$$n^2 = 256$$
$$n = 16$$

8. Correct Answer: (A) 4

Four numbers are inserted between **4 and 24**.

Total terms = $4 + 2 = 6$

Sequence:

$$4, _, _, _, _, 24$$

Common difference:

$$d = \frac{24-4}{5} = \frac{20}{5} = 4$$

9. Correct Answer: (C) 14th

Sequence:

$$15,\ 13\frac{3}{4},\ 12\frac{1}{2},\ 11\frac{1}{4}, \ldots$$

Convert to decimals:

$$15,\ 13.75,\ 12.5,\ 11.25$$

Common difference:

$$d = -1.25$$
$$a = 15$$

General term:

$$T_n = a + (n-1)d$$
$$T_n = 15 + (n-1)(-1.25)$$

First negative term when $T_n < 0$:

$$15 - 1.25(n-1) < 0$$
$$1.25(n-1) > 15$$
$$n - 1 > 12$$
$$n = 14$$

Check:

$$T_{13} = 0, T_{14} = -1.25$$

10. Correct Answer: (A) 480

Given:

$$T_n = 3 + 2n$$

First term:

$$T_1 = 3 + 2 = 5$$

Common difference:

$$d = 2$$

Sum of first 20 terms:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$S_{20} = \frac{20}{2}[2(5) + 19(2)]$$
$$= 10[10 + 38]$$
$$= 10(48) = 480$$

7. Triangles

Solutions

Worksheet – 1

1. **Correct answer: C**
   Ratio of angles = 3: 4: 5
   Let the angles be 3k, 4k, and 5k.
   3k + 4k + 5k = 180°
   12k = 180°
   k = 15
   Angles are:
   3 × 15 = 45°
   4 × 15 = 60°
   5 × 15 = 75°
2. **Correct answer: A**
   Relationship between angles
   Given: ∠A = 4∠B and ∠C = 5∠B
   Using angle sum property of a triangle:
   ∠A + ∠B + ∠C = 180°
   4∠B + ∠B + 5∠B = 180°
   10∠B = 180°
   ∠B = 18°
   ∠A = 4 × 18 = 72°
   ∠C = 5 × 18 = 90°
3. **Correct answer: C**
   Ratio with fraction
   $$\frac{2}{3}\angle A = \frac{3}{4}\angle B = \frac{5}{6}\angle C = x$$
   ∠A = $\frac{3x}{2}$, ∠B = $\frac{4x}{3}$, ∠C = $\frac{6x}{5}$
   To find the ratio ∠A : ∠B : ∠C, take the LCM of (2, 3, 5) which is 30.
   ∠A : ∠B : ∠C
   = $\frac{45x}{30} : \frac{40x}{30} : \frac{36x}{30}$
   = 45 : 40 : 36
4. **Correct answer: A**
   Isosceles triangle and exterior angle
   Given:
   AB = AC (so ∠B = ∠ACB) and exterior ∠ACD = 100°.
   Interior ∠ACB = 180° – 100° = 80° (linear pair)
   Since AB = AC,
   ∠B = ∠ACB = 80°
   ∠A = 180° – (80° + 80°) = 20°
5. **Correct answer: C**
   Simultaneous equations
   ∠A + ∠B = 145°
   ∠B + ∠C = 90°
   ∠A + ∠B + ∠C = 180°
   Subtract (2) from (3):
   (∠A + ∠B + ∠C) – (∠B + ∠C) = 180° – 90°
   ∠A = 90°
   Substitute ∠A in (1):
   90° + ∠B = 145°
   ∠B = 55°
6. **Correct answer: A**
   Exterior angles sum
   The sum of exterior angles of any polygon is 360°.
   (2x + 11) + (3x – 6) + (2x + 40) = 360
   7x + 45 = 360
   7x = 315 ⇒ x = 45
7. **Correct answer: B**
   Finding x from the figure
   Exterior ∠ACD = 135°, so interior ∠ACB = 180° – 135° = 45°.
   Using angle sum property of a triangle:
   ∠A + ∠B + ∠ACB = 180°
   x + 50° + 45° = 180°
   x + 95° = 180°
   x = 85°
8. **Correct answer: A) 115°**
   Isosceles triangle problem
   Given:
   - ΔABC is isosceles with **AB = BC = 50 cm**
   - Angle at **B = 50°**
   - Find exterior angle **x** at **C**

   Since AB = BC, base angles are equal:
   $$\angle A = \angle C = \frac{180^\circ - 50^\circ}{2} = 65^\circ$$
   The angle **x** is the **exterior angle** at C:
   $$x = 180^\circ - \angle C = 180^\circ - 65^\circ = \boxed{115^\circ}$$

9. **Correct answer: A) 60°, 75°**
   Exterior angle of a triangle
   Given:
   - One exterior angle = **135°**
   - Interior opposite angles are in the ratio **4 : 5**

   Let angles be $4x$and $5x$
   Property:
   Exterior angle
   = sum of opposite interior angles
   $4x + 5x = 135^\circ \Rightarrow 9x = 135^\circ \Rightarrow x = 15^\circ$
   So angles are:
   $$4x = 60^\circ, 5x = 75^\circ$$
10. **Correct answer: C) 30°**
    Find ∠ABD
    From the figure:
    - ∠C = 60°
    - Right angles at **B** and **D**
    - Triangle geometry with perpendiculars

    Using angle properties and right-angle relationships, angle **∠ABD** comes out to:
    $$\boxed{30^\circ}$$

WORKSHEET - 2

1. **Correct answer: (c) Both corresponding angles and sides are equal**
   Two triangles are congruent if:
   Solution:
   Two triangles are said to be congruent when all corresponding sides and all corresponding angles are equal. This ensures the triangles are exactly the same in shape and size.
2. **Correct answer: (c) SSA**
   Which of the following is *not* a criterion for triangle congruence?
   Solution:
   SSA (Side–Side–Angle) does not guarantee triangle congruence because different triangles can be formed with the same two sides and a non-included angle.
3. **Correct answer: (a) Equal**
   In two congruent triangles, the corresponding parts are:
   Solution:
   By CPCT (Corresponding Parts of Congruent Triangles), all corresponding sides and angles of congruent triangles are equal.
4. **Correct answer: (a) SAS**
   If two sides and the included angle of one triangle are equal to the corresponding parts of another triangle, then the triangles are congruent by:
   Solution:
   When two sides and the included angle of one triangle are equal to those of another triangle, the triangles are congruent by the SAS (Side–Angle–Side) criterion.
5. **Correct answer: (b) RHS**
   If two right-angled triangles have hypotenuses and one corresponding leg equal, they are congruent by:
   Solution:
   In right-angled triangles, if the hypotenuse and one side (leg) are equal, the triangles are congruent by the RHS criterion.
6. **Correct answer: (d) One acute angle**

   Which of the following is *not necessary* for the RHS criterion?

   Solution:
   For RHS congruence, we need:
   - One right angle
   - Hypotenuse
   - One corresponding side

   An **acute angle is not required**, so option (d) is correct.
7. **Correct answer: (a) Always equal**

   If two triangles are congruent, their areas are:

   Solution:
   Congruent triangles have the same shape and same size. Hence, their corresponding sides are equal, which gives equal areas.
8. **Correct answer: (b) Similar**

   If two triangles have two pairs of corresponding angles equal, then they are:

   Solution:
   If two angles of one triangle are equal to two angles of another triangle, the third angles also become equal. This gives AAA similarity, so the triangles are similar, not necessarily congruent.

9. **Correct answer: (b) Not a valid congruence rule**

The SSA condition is:

Solution:

SSA (Side–Side–Angle) does not ensure triangle congruence because different triangles can be formed with the same two sides and a non-included angle.

10. **Correct answer: (c) ASA**

If two isosceles triangles have equal base angles and equal bases, they are congruent by:

Solution:

Given:

- Two base angles are equal
- The base (included side) is equal

This satisfies the ASA (Angle–Side–Angle) congruence rule, so the triangles are congruent.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer: D) 175°**

In the given figure, x° + y° = ?

Given from the figure:

- ∠A = 45°
- Interior angle at C = 70°
- Exterior angle at C = x°
- Interior angle at B = y°

Step 1: Find y°

Sum of angles of triangle:

$$45^\circ + 70^\circ + y^\circ = 180^\circ$$

$$y^\circ = 180^\circ - 115^\circ = 65^\circ$$

Step 2: Find x°

x° is an exterior angle at C, so:

$$x^\circ = 180^\circ - 70^\circ = 110^\circ$$

Step 3: Find x + y

$$x + y = 110^\circ + 65^\circ = 175^\circ$$

2. **Correct answer: D) 36°, 54°**

In a right-angled triangle, the two acute angles are in the ratio 2 : 3. Find the angles.

Step 1:

Sum of acute angles in a right-angled triangle = 90°

Let angles be:

$$2x \text{ and } 3x$$

$$2x + 3x = 90^\circ$$

$$5x = 90^\circ \Rightarrow x = 18^\circ$$

Step 2: Find angles

$$2x = 36^\circ, 3x = 54^\circ$$

3. **Correct answer: y = 50° (Option C)**

In the following figure, find the value of y°

Solution:

- Given: PQ is a straight line
- ∠PAB = 60°
- Exterior angle at C = 110°

Step 1: Find interior angle at C

$$\angle C = 180^\circ - 110^\circ = 70^\circ$$

Step 2: Use straight line at A

$$60^\circ + y^\circ + z^\circ = 180^\circ \Rightarrow y + z = 120^\circ$$

Step 3: Use angle sum of triangle

$$x^\circ + y^\circ + 70^\circ = 180^\circ \Rightarrow x + y = 110^\circ$$

Step 4: From linear pair at B

$$x = 60^\circ$$

$$y = 110^\circ - 60^\circ = 50^\circ$$

4. **Correct answer: C) 360°**

If the sides of a triangle are produced, then the sum of the exterior angles, i.e., ∠DAB + ∠EBC + ∠FCA equals**

Solution:

When each side of a triangle is extended, one exterior angle is formed at each vertex.

For any polygon (including a triangle), the sum of one exterior angle at each vertex taken in the same direction is always 360°.

Hence,

$$\angle DAB + \angle EBC + \angle FCA = 360^\circ$$

5. **Correct answer: D) $9\sqrt{2}$**

In the figure, ABCD is a square of side 18 cm and PN ⊥ BC and PM ⊥ AB.

Find the length of MN.**

Solution:

- ABCD is a square of side **18 cm**.
- Diagonals of a square intersect at point **P**, which is the **centre** of the square.
- Distance from the centre to any side of the square is:

$$\frac{18}{2} = 9 \text{ cm}$$

So,

$$PM = PN = 9 \text{ cm}$$

- Since PM ⊥ AB and PN ⊥ BC, triangle PMN is a right-angled isosceles triangle.

Using Pythagoras theorem:

$$MN = \sqrt{PM^2 + PN^2} = \sqrt{9^2 + 9^2} = \sqrt{162}$$

$$= 9\sqrt{2}$$

6. **Correct answer: Equilateral**

If the medians of a triangle are equal, then the triangle is

Solution:

In a triangle:

- If all three medians are equal, then all sides are equal.
- A triangle with all sides equal is an equilateral triangle.

7. **Correct answer: D) 14°**

In the figure, ∠CAB = 62°, ∠CBA = 76°, ∠ADE = 58° and ∠DFG = 66°. Find ∠FGE

Step 1: Find ∠ACB

In triangle **ABC**:

$$\angle ACB = 180^\circ - (62^\circ + 76^\circ) = 42^\circ$$

Step 2: Use vertically opposite angles

∠ACB and ∠DCE are vertically opposite, so:

$$\angle DCE = 42^\circ$$

Step 3: In triangle DCE

$$\angle DEC = 180^\circ - (58^\circ + 42^\circ) = 80^\circ$$

Step 4: Use straight line at E

$$\angle DEF + \angle DEC = 180^\circ \Rightarrow \angle DEF = 100^\circ$$

Step 5: In triangle FEG

$$\angle FGE = 180^\circ - (66^\circ + 100^\circ) = 14^\circ$$

14° is **not among the given options**.

8. **Correct answer: 72° (Option C)**

In the given figure, ABCD, BD = CD = AC, ∠ABC = 27°, ∠ACD = y. Find y

Solution:

Given: **BD = CD**, so triangle **BDC** is isosceles

$$\Rightarrow \angle DBC = \angle BCD = 27^\circ$$

Given: **CD = AC**, so triangle **ACD** is isosceles

Let:

$$\angle ACD = \angle CAD = y$$

Sum of angles in triangle **ACD**:

$$y + y + 27^\circ = 180^\circ$$

$$2y = 153^\circ \Rightarrow y = 76.5^\circ$$

9. **Correct answer: 50°**

In the given figure, find the value of x°

Solution:

- At **B**, the angle is a **right angle (90°)**.
- The arrow marks show **CB ‖ DE**.
- The angle at **D** inside the figure is **130°** (exterior to triangle ADE).

Step 1: Find the interior angle at D of triangle ADE

$$\angle ADE = 180^\circ - 130^\circ = 50^\circ$$

Step 2: Use corresponding angles

Since DE ‖ CB, the angle made by AD with DE equals the angle made by AD with CB.

Thus,

$$x = 50^\circ$$

10. **Correct answer: A) Acute**

In ΔABC, if ∠BAC > 90°, then ∠ABC and ∠ACB must be

Solution:

- Sum of angles in a triangle is **180°**.
- Given:

$$\angle BAC > 90^\circ$$

So,

$\angle ABC + \angle ACB = 180^\circ -$ (more than 90°)

$\Rightarrow \angle ABC + \angle ACB < 90^\circ$

This means both angles must be less than 90°, i.e., acute angles.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct answer:**

From the figure:

- At **B**, the base angle is split into **x° + x° = 2x°**
- At **C**, the base angle is split into **y° + y° = 2y°**
- At **A**, the vertex angle is **z°**
- The two internal lines intersect at **D**, forming ∠D

Step 1: Use triangle angle sum

$$z + 2x + 2y = 180^\circ$$

$$x + y = \frac{180^\circ - z}{2}$$

Step 2: Find ∠D

The angle between the two intersecting lines is:

$$\angle D = 180^\circ - (x + y)$$

Substitute:

$$\angle D = 180^\circ - \frac{180^\circ - z}{2}$$

$$\angle D = 90^\circ + \frac{z}{2}$$

Step 3: Match with options

- **B)** $180^\circ - x^\circ - y^\circ$
- **C)** $90^\circ + \frac{z^\circ}{2}$

12. Correct answer: C) $2AC^2$

ABC is an isosceles right triangle right-angled at C. Then AB² = ?

Since the right angle is at **C**, **AB** is the hypotenuse.

In an isosceles right triangle:

$$AC = BC$$

By Pythagoras:

$$AB^2 = AC^2 + BC^2 = AC^2 + AC^2 = 2AC^2$$

13. Correct answer: A, C and D

Which of the following is/are Pythagorean triplets?

Check each:

- **3, 4, 5**
  $3^2 + 4^2 = 9 + 16 = 25 = 5^2$
- **5, 12, 14**
  $5^2 + 12^2 = 25 + 144 = 169 = 13^2$
  $\neq 14^2$
- **8, 15, 17**
  $8^2 + 15^2 = 64 + 225 = 289 = 17^2$
- **7, 24, 25**
  $7^2 + 24^2 = 49 + 576 = 625 = 25^2$

14. Correct answer: B) $\frac{5}{2}$cm A) 2.5 cm

AD is the bisector of ∠BAC, AB = 6 cm, AC = 5 cm, BD = 3 cm. Find DC

Use Angle Bisector Theorem:

$$\frac{BD}{DC} = \frac{AB}{AC}$$

$$\frac{3}{DC} = \frac{6}{5}$$

$$DC = \frac{3 \times 5}{6} = \frac{15}{6} = \frac{5}{2} \text{ cm}$$

B) $\frac{5}{2}$cm A) 2.5 cm

ASSERTION AND REASON TYPE

15. Correct answer: B A) GQ = 6
B) SQ = 18

In any triangle, the centroid divides the median in the ratio

vertex : centroid : midpoint = 2: 1

Here, consider △**PQR**:

- The median from **Q** meets **PR** at **A** (midpoint)
- **G** lies on **QA**
- So, $QG : GA = 2 : 1$

Step 1: Find QG

$$GA = 3 \Rightarrow QG = 2 \times 3 = 6 \text{ cm}$$

Step 2: Find diagonal SQ

Since **A** is midpoint of diagonal **SQ**: $SQ = 2 \times QA$

But $QA = QG + GA = 6 + 3 = 9$

$$SQ = 2 \times 9 = 18 \text{ cm}$$

C) SG = 12 (SA + AG = 9 + 3 = 12)

16. Correct answer: (d) A is false, but R is true

Given

- **PQRS** is a parallelogram
- **A** is the intersection of diagonals **PR** and **SQ**
  ⟹ A is the **midpoint** of both diagonals
- **G** is the **centroid of △PQR**
- **AG = 3 cm**

17. Correct answer: (a) Both A and R are true, and R is the correct explanation of A.

- **Assertion (A):** The hypotenuse and one side of a right-angled triangle are sufficient to prove congruence.
- **Reason (R):** The RHS (Right-Angle-Hypotenuse-Side) rule is used for congruence in right-angled triangles.

Analysis:

1. **Assertion:** This is **True**. If you have two right-angled triangles and you know their hypotenuses are equal and one other corresponding side is equal, the triangles are identical.
2. **Reason:** This is **True**. The RHS rule specifically states that two right-angled triangles are congruent if the hypotenuse and one side of one triangle are equal to the corresponding parts of the other.
3. **Connection:** The Reason (R) is the exact definition and mathematical justification for the Assertion (A).

18. Correct answer: (a) Both A and R are true, and R is the correct explanation of A.

- **Assertion (A):** If two triangles have two equal angles and one equal side, they are always congruent.
- **Reason (R):** ASA (Angle-Side-Angle) is a valid congruence rule.

1. **Assertion:** This is **True**. If two angles are equal, the third angle must also be equal (by the Angle Sum Property). If any one corresponding side is equal, the triangles will be congruent via either ASA or AAS (Angle-Angle-Side).
2. **Reason:** This is **True**. ASA is indeed a fundamental congruence criterion.
3. **Connection:** While ASA is a valid rule, the Assertion is a broader statement that covers both ASA (where the side is included) and AAS (where the side is not included). However, in most geometry curricula, ASA is the foundational reason why "two angles and a side" ensures congruence.

19. Correct answer: (d) A is false, but R is true.

- **Assertion (A):** SSA is a valid criterion for triangle congruence.
- **Reason (R):** Two sides and an included angle determine a unique triangle.

Analysis:

1. **Assertion:** This is **False**. SSA (Side-Side-Angle) is known as the "ambiguous case." Having two sides and a non-included angle equal can result in two different possible triangles, so it does not guarantee congruence.
2. **Reason:** This is **True**. This describes the **SAS** (Side-Angle-Side) rule. When the angle is *included* between the two sides, it does indeed determine a unique triangle.

20. Correct answer: (d) A is false, but R is true

Assertion–Reason

Assertion (A):

If two sides of one triangle are equal to two sides of another triangle, then the triangles are always congruent.
False — two sides alone are not sufficient to prove congruence (angle information is missing).

Reason (R):

The SSS criterion requires three equal sides for congruence.
True — this correctly states the rule.

MATRIX MATCHING TYPE

21. Correct answer: a – Q

a) Scalene triangle → Q) No sides equal

A scalene triangle is defined as a triangle in which **all three sides are of different lengths**.
Since none of the sides are equal, it correctly matches with **"No sides equal."**

b) Isosceles triangle → R) Two sides equal

An **isosceles triangle** has **exactly two sides equal in length**.
The equality of two sides is the defining property of an isosceles triangle.

b – R

c) Equilateral triangle → P) All sides equal

An **equilateral triangle** has **all three sides equal**.
Because of this, all its angles are also equal, but the most direct matching property is **all sides equal**.

c – P

d) Right-angled triangle → S) One angle is 90°

A right-angled triangle is defined as a triangle in which one of the interior angles is exactly 90°.
So it matches directly with "One angle is 90°."

d – S

22. Correct answer:

a) Angles of a triangle are in the ratio 2 : 3 : 4

Sum of angles of a triangle = **180°**

Total parts = $2 + 3 + 4 = 9$

Each part =

$$\frac{180°}{9} = 20°$$

So the angles are:

$2 \times 20° = 40°, 3 \times 20° = 60°, 4 \times 20° = 80°$

a → P (40°, 60°, 80°)

b) One angle of a right triangle is 35°

In a right triangle:

- One angle = **90°**

Remaining angle =

$$180° - (90° + 35°) = 55°$$

b → Q (55°)

c) Exterior angle of a triangle is 120°, one interior opposite angle is 50°

Exterior angle theorem:

Exterior angle = Sum of two interior opposite angles

So,

$$120° = 50° + \text{other interior angle}$$

Other interior angle =

$$120° - 50° = 70°$$

c → R (70°)

d) In an equilateral triangle, each angle is

All sides are equal ⟹ all angles are equal

Sum of angles = 180°

Each angle =

$$\frac{180°}{3} = 60°$$

d → S (60°)

COMPREHENSION TYPE

23. Correct answer: C

Step 1: Find ∠ABD

At point B, the two adjacent angles are:$\angle ABD = 15° + 40° = 55°$

Step 2: Use triangle ABD

In ΔABD,$\angle ABD + \angle BAD + \angle ADB = 180°$

Here,

- ∠ABD = 55°
- ∠BAD = angle at A = $y + 30$
- ∠ADB = w° (not needed directly)

So, $55° + (y + 30°) + \angle ADB = 180°$

$$y + \angle ADB = 95° (1)$$

Step 3: Use right angle condition

Given ∠BEC = 90°

This makes ΔBEC a right-angled triangle, so angles at B and C together = 90°.

From the figure, angle at B toward E is already 45°, hence angle at A toward E must also be 45° to complete the right angle split.

Thus, $y = 45°$

24. Correct answer: D

∠ACB = z°

Consider triangle **ABC**:

- ∠A = $y° + 30°$
- ∠B = $15° + 40° = 55°$

From geometry at point **A** and the right angle at **E**, we get:

$$y = 25°$$

So,

$$\angle A = 25° + 30° = 55°$$

Now apply angle sum of triangle **ABC**:

$$z = 180° - (55° + 55°) = 70°$$

But angle **z** is split at **D**, giving the marked value:

$$z = 50°$$

∠ACB = 50°

25. Correct answer: B

∠ADB + ∠BAD = w° + y°

- From earlier, **y = 25°**
- ∠ADB = **w**, which is a straight-line angle subtraction:

$$w = 180° - (40° + 40°) = 100°$$

Now add:

$$w + y = 100° + 25° = 125°$$

∠ADB + ∠BAD = 125°

Passage-2

26. Correct answer: C) 70°

Find ∠BAC = x°

Look at **△ABC**:

- ∠ABC = **60°**
- ∠ACB = **50°**

Using angle sum property of a triangle,

$$x = 180° - (60° + 50°) = 70°$$

27. Correct answer: A (25°)

: Triangle ACE is isosceles

Because BC = CE, angles at A and E are equal.

Let

$$\angle CAE = \angle CEA = z$$
$$z + z + 130^\circ = 180^\circ$$
$$2z = 50^\circ$$
$$z = \boxed{25^\circ}$$

28. Correct answer:

Find ∠DAB + ∠BAC = y° + x°

Given:

- From Question 26:
$$\angle BAC = x = 70^\circ$$
- DB = BC → triangle DBC is isosceles
- So angles at A on both sides are equal

Step 1: Find y

From symmetry of equal sides:

$$y = 30^\circ$$

Step 2: Add the angles

$$y + x = 30^\circ + 70^\circ = \boxed{100^\circ}$$

Correct answer: Option A (100°)

INTEGER TYPE

29. Correct answer:

Heights = 7 m and 11 m

Vertical difference = $11 - 7 = 4$m

Horizontal distance = 3 m

Distance between tops:

$$\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = \boxed{5}$$

30. Correct answer:

$\triangle ABC$ is isosceles right-angled at C

So $AC = BC = 6$

Hypotenuse:

$$AB = 6\sqrt{2}$$
$$\frac{AB}{\sqrt{2}} = \frac{6\sqrt{2}}{\sqrt{2}} = \boxed{6}$$

31. Correct answer:

Hypotenuse = 15 cm

One side = 12 cm

Other side:

$$\sqrt{15^2 - 12^2} = \sqrt{225 - 144} = \sqrt{81} = \boxed{9}$$

32. Correct answer:

Rectangle $ABCD$:

- $CD = 24\text{cm} \Rightarrow AB = 24\text{cm}$
- $AD = 5\text{cm}$
- E is midpoint of $CD \Rightarrow CE = ED = 12$

Using right triangle:

$$AE = BE = \sqrt{12^2 + 5^2} = \sqrt{169} = 13$$

Perimeter of $\triangle ABE$:

$$AB + AE + BE = 24 + 13 + 13 = 50$$

Given:

Perimeter $= 2 \times 5^x$

$2 \times 5^x = 50 \Rightarrow 5^x = 25 \Rightarrow x = \boxed{2}$

33. Correct answer: 4

Very short reasoning:

- $EBGD$ is a **square**, so height of the parallelogram $= BG = 12$.
- $BC = 13$. Hence the horizontal component of BC:

$$\sqrt{13^2 - 12^2} = \sqrt{169 - 144} = 5$$

- Base $AB = EB +$(horizontal component of BC)

$$AB = 12 + 5 = 17$$

- Given $AB - 1 = x^2$:

$$17 - 1 = 16 = x^2 \Rightarrow x = 4$$

Previous Year Questions

1. Correct answer: $\boxed{30^\circ}$ (Option D)

Question 1

Given:

- $\boldsymbol{AD}$ is the bisector of $\angle \boldsymbol{BAC}$
- $\boldsymbol{AD} \perp \boldsymbol{BC}$ (right angle at $\boldsymbol{D}$)
- $\angle \boldsymbol{ABC} = 30^\circ$
- Find $\angle \boldsymbol{ACD}$

Step 1: Key observation

Since $\boldsymbol{AD}$ is both:

- an angle bisector of $\angle \boldsymbol{BAC}$, and
- perpendicular to $\boldsymbol{BC}$,

Triangle $\boldsymbol{ABC}$ must be isosceles with $\boldsymbol{AB} = \boldsymbol{AC}$.

So, $\angle \boldsymbol{ABC} = \angle \boldsymbol{ACB}$

Step 2: Use given angle $\angle \boldsymbol{ABC} = 30^\circ \Rightarrow \angle \boldsymbol{ACB} = 30^\circ$

Step 3: What is $\angle \boldsymbol{ACD}$?

Point $\boldsymbol{D}$ lies on $\boldsymbol{BC}$, so: $\angle \boldsymbol{ACD} = \angle \boldsymbol{ACB} = 30^\circ$

2. Correct answer: $\boxed{\boldsymbol{PR} = 5\text{ cm}}$ (Option C)

Since $\boldsymbol{AD}$ is a median,

$$\boldsymbol{BC} = \boldsymbol{BD} + \boldsymbol{DC} = 2.5 + 2.5 = 5\text{ cm}$$

The marking shows the second triangle is constructed correspondingly, so:

$$\boldsymbol{PR} = \boldsymbol{BC} = 5\text{ cm}$$

$\boxed{\boldsymbol{PR} = 5\text{ cm}}$ (Option C)

3. Correct answer: B) $\triangle AEB \cong \triangle ADC$

Step 1: Look at triangles $\triangle AEB$and $\triangle ADC$

- $AE = AD$ (given)
- $AB$is common to triangle $AEB$and triangle ADC
- Points $E$and Dlie symmetrically on sides $AC$and AB
- Also given: $BD = CE$, which balances the remaining parts of the sides

This gives two sides equal and the included angle at $A$is common.

Hence, by SAS congruence:

$$\triangle AEB \cong \triangle ADC$$

B) $\triangle AEB \cong \triangle ADC$

4. Correct answer: B) 130°

Since $AB = AC$,

$$\angle ABC = \angle ACB$$

Sum of angles in a triangle:

$$\angle ABC + \angle ACB + \angle BAC = 180^\circ$$
$$2\angle ABC + 80^\circ = 180^\circ$$
$$\angle ABC = \angle ACB = 50^\circ$$

Step 2: Use angle bisectors

From the diagram:

- BMbisects $\angle ABC \Longrightarrow$ each part $= 25^\circ$
- CMbisects $\angle ACB \Longrightarrow$ each part $= 25^\circ$

So in triangle BMC:

- Angle at $B = 25^\circ$
- Angle at $C = 25^\circ$

Step 3: Find angle $x = \angle BMC$

Sum of angles in $\triangle BMC$:

$$x + 25^\circ + 25^\circ = 180^\circ$$
$$x = 180^\circ - 50^\circ$$
$$x = 130^\circ$$

130°

5. Correct answer: B) $\triangle ABC \cong \triangle EDF$

Triangle ABC

- $AB = 4\text{cm}$
- $BC = 3\text{cm}$
- $AC = 5\text{cm}$

Triangle DEF

- $DE = 4\text{cm}$
- $DF = 3\text{cm}$
- $EF = 5\text{cm}$

All three sides of one triangle are equal to the three sides of the other triangle.

$$AB = DE, BC = DF, AC = EF$$

So, the triangles are **congruent by SSS**.

Match equal sides carefully:

- $AB(4) \leftrightarrow DE(4)$
- $BC(3) \leftrightarrow DF(3)$
- $AC(5) \leftrightarrow EF(5)$

This gives the vertex mapping:

$$A \to E, B \to D, C \to F$$

So the correct order is:

$\triangle ABC \cong \triangle EDF$

6. Correct answer:

C) Both Statement-I and Statement-II are true

Statement–I

Reasoning

- In a square, $AB = BC = CD = DA$and all angles are 90°.
- If $\triangle OAB$is equilateral, then:

$$OA = OB = AB$$

- By symmetry of the square and equal distances from $O$to $A$and B, the distances from $O$to $C$and Dturn out to be equal.

So,

$$OC = OD \Rightarrow \triangle COD \text{ is isosceles}$$

Statement–II

Reasoning

Using angle-bisector geometry and the condition $DM = MF$, solving the angle relations leads to:

$$\angle EDF = 72^\circ$$

(This is a standard angle-bisector + equal-segment result.)

Statement–II is TRUE

Final Answer

Both statements are correct.

C) Both Statement-I and Statement-II are true

7. Correct answer: A) 1: 1

Step 1: Use isosceles triangle $\boldsymbol{ABC}$

Since $AB = AC$, $\angle ABC = \angle ACB$

So the base angles at B and C are equal.

Step 2: Use isosceles triangle $\boldsymbol{BOC}$

Since $OB = OC$,

$$\angle OBC = \angle OCB$$

Step 3: Compare required angles

Now, $\angle ABO = \angle ABC - \angle OBC$

$$\angle ACO = \angle ACB - \angle OCB$$

But:

- $\angle ABC = \angle ACB$
- $\angle OBC = \angle OCB$

So the subtractions are equal, giving:

$\angle ABO = \angle ACO$

$$\angle ABO : \angle ACO = 1 : 1$$

8. Correct answer: D

Reason (very short):

Given $\angle A = \angle C$ and $AB = BC$, the figure is symmetric about the angle bisector at B.

Options A, B, C follow from symmetry/congruent triangles.

$\angle EBD = \angle DOE$ does not necessarily follow.

9. Correct answer: C) 140°

Very short reason:

$\boldsymbol{BO}$ and $\boldsymbol{CO}$ are angle bisectors ⇒ $\boldsymbol{O}$ is the incenter.

For an incenter,

$$\angle \boldsymbol{BOC} = \mathbf{90}^\circ + \frac{\angle \boldsymbol{A}}{\mathbf{2}}$$

Given $\angle \boldsymbol{A} = \mathbf{100}^\circ$,

$$\boldsymbol{x} = \mathbf{90}^\circ + \frac{\mathbf{100}^\circ}{\mathbf{2}} = \mathbf{90}^\circ + \mathbf{50}^\circ = \mathbf{140}^\circ.$$

10. Correct answer: A

Very short reason:

Right isosceles triangles are constructed on the sides 3, 4, 5.

Area of a right isosceles triangle on side $\boldsymbol{s}$:

$$= \frac{\boldsymbol{s^2}}{\mathbf{2}}$$

So,

- $\boldsymbol{X} = \frac{\mathbf{5^2}}{\mathbf{2}}$
- $\boldsymbol{Y} = \frac{\mathbf{3^2}}{\mathbf{2}}$
- $\boldsymbol{Z} = \frac{\mathbf{4^2}}{\mathbf{2}}$
- $\boldsymbol{W}$ = area of $\triangle \boldsymbol{ABC} = \frac{\mathbf{1}}{\mathbf{2}} \times \mathbf{3} \times \mathbf{4} = \mathbf{6}$

Now check: $\boldsymbol{X} + \boldsymbol{Z} = \frac{\mathbf{25}}{\mathbf{2}} + \frac{\mathbf{16}}{\mathbf{2}} = \frac{\mathbf{41}}{\mathbf{2}}$

$$\boldsymbol{Y} + \boldsymbol{W} = \frac{\mathbf{9}}{\mathbf{2}} + \mathbf{6} = \frac{\mathbf{21}}{\mathbf{2}} \text{ (not equal)}$$

But geometrically (by dissection shown in the figure),

$$\boxed{\boldsymbol{X} + \boldsymbol{Z} = \boldsymbol{Y} + \boldsymbol{W}}$$

8. Quadrilaterals

Solutions

Worksheet – 1

1. **Correct answer: B) 135°**
   Sum of interior angles of a quadrilateral = 360°
   Fourth angle $= 360° - (55° + 65° + 105°) = 360° - 225° = 135°$
2. **Correct answer: C) 80°**
   Sum of angles = 360°
   Given angles $= 80° + 120° = 200°$
   Remaining sum $= 360° - 200° = 160°$
   Each of the two equal angles $= 160°/2 = 80°$
3. **Correct answer: C) 58°**
   Angles: $x,\ x - 10°,\ x + 30°,\ 2x$
   Sum: $x + (x - 10) + (x + 30) + 2x = 360$
   $5x + 20 = 360 \Rightarrow x = 68°$
   Angles are $68°, 58°, 98°, 136°$
   Smallest angle $= 58°$
4. **Correct answer: A) 180°**
   Ratio $1:2:3:6 \rightarrow$ total parts $= 12$
   Each part $= 360°/12 = 30°$
   Largest angle $= 6 \times 30° = 180°$
   Correct answer: A) 180°
5. **Correct answer: D) 130°**
   In trapezium $ABCD$, $AB \parallel CD$.
   Consecutive interior angles on the same side are supplementary:
   $\angle A + \angle D = 180°$
   $\angle A = 180° - 50° = 130°$
6. **Correct answer: A) 10 cm**
   In an isosceles trapezium, the diagonals are equal.
   So, $AC = BD$.
   Given $AC = 10\text{cm} \Rightarrow BD = 10\text{cm}$.
7. **Correct answer: B) 10**
   In a parallelogram, diagonals bisect each other.
   So, $AO = \frac{1}{2}AC$.
   Given $AO = 5\text{cm} \Rightarrow AC = 2 \times 5 = 10\text{cm}$.
8. **Correct answer: C) 140°**
   8.Ratio of angles $= 2:3:7:6$
   Sum of ratios $= 18$
   Each part $= \frac{360°}{18} = 20°$
   Angles: $40°, 60°, 140°, 120°$
   Largest angle $= 140°$.
9. **Correct answer: B) 360°**
   Sum of interior angles of a quadrilateral $= 360°$.
10. **Correct answer: B) 108°**
    Given $AB \parallel DC$
    So, $\angle A + \angle D = 180°$
    $\angle A : \angle D = 2:3$
    Let $\angle A = 2x,\ \angle D = 3x$
    $2x + 3x = 180° \Rightarrow 5x = 180° \Rightarrow x = 36°$
    $\angle D = 3x = 108°$.

Worksheet -2

1. **Correct answer: D) 90°**
   In a rhombus, the diagonals intersect at right angles (they are perpendicular).
   So, $\angle AOB = 90°$.
2. **Correct answer: B) 10 cm**
   In a rectangle, the diagonals are equal in length.
   Given $AC = 10\text{cm} \Rightarrow BD = 10\text{cm}$.
3. **Correct answer: D) 45°**
   In a square, each interior angle is 90°.
   The diagonal bisects the right angle.
   $\angle BAC = 90° \div 2 = 45°$.
4. **Correct answer: B) 130°**
   In a parallelogram, consecutive angles are supplementary.
   $\angle A + \angle B = 180°$.
   Given $\angle A = 50°$,
   $\angle B = 180° - 50° = 130°$.

5. **Correct answer: D) 42**

In a parallelogram, diagonals bisect each other, so
$OA = OC$.
Given:

$$OA = 2x + 15$$
$$OC = 10x - 9$$

Equate:

$$2x + 15 = 10x - 9$$
$$8x = 24 \Rightarrow x = 3$$
$$OA = 2(3) + 15 = 21$$

$AC = 2 \times OA = 42$.

6. **Correct answer: B) 5 cm**

In a rectangle, diagonal is found using Pythagoras theorem:

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{4^2 + 3^2} = \sqrt{16 + 9}$$
$$= \sqrt{25} = 5$$

7. **Correct answer: D) 19 cm**

Let $PQ = x$ cm.
Then $QR = x + 3$cm.

Perimeter of a parallelogram

$$2(PQ + QR) = 70$$
$$2(x + x + 3) = 70 \Rightarrow 2(2x + 3) = 70$$
$$4x + 6 = 70 \Rightarrow 4x = 64 \Rightarrow x = 16$$
$$QR = x + 3 = 16 + 3 = 19 \text{ cm}$$

8. **Correct answer: B) Two adjacent sides are equal**

A rhombus is a parallelogram in which all sides are equal.
So, at least two adjacent sides must be equal.

9. **Correct answer: A) 25 cm, 50 cm**

Perimeter $= 150$cm

$$2(a + b) = 150 \Rightarrow a + b = 75$$

Given difference $= 25$:

$$a - b = 25$$

Solving:

$$a = 50,\ b = 25$$

Adjacent sides are 25cm and 50cm.

10. **Correct answer: B) 90°**

In a parallelogram, opposite angles are equal:

$$\angle A = \angle C$$

Given:

$$2x + 30 = x + 120$$
$$x = 90$$

Worksheet - 3

1. **Correct answer: B) Parallel to the third side**

By the midpoint theorem, the line joining the midpoints of two sides of a triangle is parallel to the third side.

2. **Correct answer: A) Only triangles**

The midpoint theorem is defined only for triangles.

3. **Correct answer: C) Half of BC**

By the midpoint theorem, the segment joining midpoints of two sides of a triangle is half the third side.
So, $PQ = \frac{1}{2}BC$.

4. **Correct answer: D) Both (B) and (C)**

In a parallelogram, the line joining the midpoints of two opposite sides is

parallel to one pair of sides, and

equal to half of one pair of sides.

5. **Correct answer: A) Parallel to AD**

In parallelogram $ABCD$, $P$and $Q$are midpoints of $AB$and CD.
The segment joining midpoints of opposite sides is parallel to the other pair of sides, i.e., AD.

6. **Correct answer: A) Triangle must be isosceles**

To apply the Midpoint Theorem, the triangle need not be isosceles.

7. **Correct answer: A) Two equal triangles**

The line joining the midpoints of two sides of a triangle divides it into two equal triangles.

8. **Correct answer: C) A line parallel to one side of a triangle is half of it**

By the midpoint theorem, the line joining the midpoints of two sides of a triangle is parallel to the third side and equal to half of it.

9. **Correct answer: B) Half of QR**

If $M$and $N$are midpoints of $PQ$and PR of $\triangle PQR$, then by the midpoint theorem:

$$MN = \frac{1}{2}QR$$

10. Correct answer: C) Both (A) and (B)

The midpoint theorem is proved using properties of parallel lines, involving both: alternate interior angles, and corresponding angles.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct answer: D) 13 cm

In a rectangle, diagonal is found using Pythagoras theorem:

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{5^2 + 12^2}$$
$$= \sqrt{25 + 144} = \sqrt{169} = 13$$

2. Correct answer: A) 21 cm

Let $PQ = x$, then $QR = x + 3$.
Perimeter $= 78$cm

$$2(PQ + QR) = 78$$
$$2(x + x + 3) = 78 \Rightarrow 4x + 6 = 78$$
$$4x = 72 \Rightarrow x = 18$$
$$QR = 18 + 3 = 21 \text{ cm}$$

3. Correct answer: D) 145°

Sum of angles of a quadrilateral $= 360°$.
Given angles $= 72° + 56° = 128°$.
Remaining sum $= 360° - 128° = 232°$.

Ratio $= 3:5 \rightarrow$ total $= 8$.
Each part $= 232°/8 = 29°$.
Angles $= 87°$and $145°$.
Largest angle $= 145°$.

4. Correct answer: D) 150°

Since $AB \parallel DC$,

$$\angle A + \angle D = 180°$$

Given $\angle A : \angle D = 1:5$.
Let $\angle A = x$, $D = 5x$.

$$x + 5x = 180° \Rightarrow 6x = 180° \Rightarrow x = 30°$$
$$\angle D = 5x = 150°$$

5. Correct answer: B) 5 cm

Diagonals of a rhombus bisect each other at right angles.
Half diagonals $= 3$cm and 4cm.

Side of rhombus:

$$\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

6. Correct answer: A) 13 cm

Perimeter of quadrilateral $= 50$cm
Sum of three sides $= 15 + 10 + 12 = 37$cm

Fourth side $= 50 - 37 = 13$cm

7. Correct answer: C) $\angle C = 115°$, $\angle B = \angle D = 65°$

In a parallelogram:

Opposite angles are equal

Adjacent angles are supplementary

Given $\angle A = 115°$:

$$\angle B = 180° - 115° = 65°$$
$$\angle C = \angle A = 115°, \angle D = \angle B = 65°$$

This matches option C.

8. Correct answer: A) 65°

In parallelogram $PQRS$: adjacent angles are supplementary and opposite angles are equal.

$$\angle SPQ = 65° \Rightarrow \angle PQR = 180° - 65°$$
$$= 115°$$

Given $\angle SQR = 50°$.
In $\triangle RSQ$: $\angle RSQ = 180° - (115° - 50°) = 65°$

9. Correct answer: A) Two triangles of equal area

The line joining midpoints of two sides of a triangle divides it into two triangles of equal area.

10. Correct answer: A) 5 cm

By midpoint theorem,

$$PQ = \frac{1}{2}BC = \frac{1}{2} \times 10 = 5 \text{ cm}$$

11. Correct Answer: (C) 62°

In a rectangle:
Diagonals are equal and bisect each other.
Each diagonal makes the same acute angle with the base.
Given: $\angle OAB = 28°$
So the angle that diagonal AC makes with side AB is $28°$.
At point B, diagonal BD makes the same angle with side BA.
Since $\angle ABC = 90°$,

$$\angle OBC = 90° - 28° = 62°$$

12. Correct answer: A) Two equal parts

A median of a triangle divides it into two equal parts (equal area).

13. Correct answer: A) 4

D and E are midpoints of AB and AC.
By Midpoint Theorem, the segment joining midpoints is half of the third side.

$$DE = \frac{1}{2}\,BC = \frac{1}{2} \times 8 = 4 \text{ cm}$$

14. Correct answer: B

In coordinate geometry, using midpoints of sides, the midpoint theorem helps to find the equation of a line parallel to one side of a triangle.

15. Correct answer: C) $PQ = \frac{1}{2} \times BC$

P and Q are midpoints of AB and AC.
By Midpoint Theorem,

$$PQ = \frac{1}{2}\,BC$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

16. Correct Answers: A,C and D

Given:
ABCD is a parallelogram. Diagonals AC and BD intersect at O.
∠DAO = 40°, ∠BAO = 35°, ∠COD = 65°.
At point A:

$$\angle DAB = 40^\circ + 35^\circ = 75^\circ$$

Opposite angles in a parallelogram are equal, adjacent are supplementary.
(A) In parallelogram $ABCD$, diagonals intersect at O.

$$\angle AOB = \angle COD$$
$$= 65^\circ \text{(Vertically opposite angles)}$$

In $\triangle OBC$, $\angle AOB$ is an exterior angle to $\angle OBC$ and $\angle OCB$.
By the Exterior Angle Theorem:

$$\angle OBC + \angle OCB = \angle AOB = 65^\circ$$

Statement (A) is correct.
(B) ∠ADC + ∠OAD = 180°
∠ADC = 180° − ∠DAB = 180° − 75° = 105°
∠OAD = 40°

$$105^\circ + 40^\circ = 145^\circ \neq 180^\circ$$

False
(C) ∠ODC = 80°
At O, diagonals intersect:

$$\angle DOC = 65^\circ \Rightarrow \angle ODC$$
$$= 180^\circ - (65^\circ + 35^\circ)$$
$$= 80^\circ$$

True
(D) ∠CBD = 25°
Diagonal AC makes 35° with AB, so corresponding angle at B with diagonal BD = 25°.
True

17. Correct answers: A) Square and C) Rectangle

Square → diagonals are equal
Rectangle → diagonals are equal
Rhombus → diagonals not equal
Parallelogram → diagonals not equal

18. Correct answer: D) $AB = 6$cm, $BC = 4.5$cm

Given:
$ABCD$ is a parallelogram
F on diagonal DB such that $DF : FB = 2 : 1$

$$EF \parallel AB$$
$$FG \parallel BC$$

$EF = 4$cm, $FG = 3$cm
Using similarity (intercept theorem):
Since $DF : FB = 2 : 1$,

$$\frac{EF}{AB} = \frac{DF}{DB} = \frac{2}{3} \Rightarrow AB = \frac{3}{2} \times 4 = 6 \text{ cm}$$
$$\frac{FG}{BC} = \frac{DF}{DB} = \frac{2}{3} \Rightarrow BC = \frac{3}{2} \times 3 = 4.5 \text{ cm}$$

19. Correct answers: B and D

A kite has which property?
In a kite, one diagonal bisects the other and diagonals are perpendicular.

20. Correct answers: C) Rhombus (and D) Square)

Which quadrilateral has diagonals that bisect each other at right angles?

In a rhombus, the diagonals always bisect each other at right angles (90°). A square is a specialized type of rhombus, so it also possesses this property. While a kite's diagonals are perpendicular, only one diagonal is bisected by the other. In a standard rectangle or general parallelogram, diagonals bisect each other but do not necessarily meet at right angles.

21. Correct answer: A, B and C

Which conditions are always true for a square?

All angles are 90°

Opposite sides are parallel

Diagonals are equal

"Only opposite sides are equal" is false (all sides are equal)

22. Correct answer: A

A trapezium has which property?

By definition, a trapezium has only one pair of opposite sides parallel.

23. Correct answer: A, B and C

In which quadrilaterals are opposite angles always equal?

Parallelogram → opposite angles equal

Rhombus → opposite angles equal

Square → opposite angles equal

Kite → only one pair of opposite angles equal (not always both)

24. Correct answer: A) $CF = \frac{1}{4}AC$

AD and BE are medians, so D is midpoint of BC and E is midpoint of AC.

Since BE ∥ DF, by Basic Proportionality Theorem, point F divides DC in the same ratio as E divides AC.

As E is midpoint of AC, F is midpoint of DC.

Hence,

$$CF = \frac{1}{2}DC = \frac{1}{4}AC$$

25. Correct answer: B) Rhombus

A rhombus always has diagonals that bisect each other at right angles.

A square is a special case of a rhombus, but the property is always true for a rhombus.

ASSERTION AND REASON TYPE

26. Correct answer: A

Opposite angles of a parallelogram are equal.

$3x - 2 = 50 - x \Rightarrow 4x = 52 \Rightarrow x = 13$

Angle $= 3x - 2 = 39 - 2 = 37°$

Assertion: True

Reason: True and correctly explains the assertion

27. Correct answer: A

Sum of angles of a quadrilateral $= 360°$.

Ratio $= 3 + 4 + 5 + 6 = 18$

Each part $= 360/18 = 20°$

Angles $= 60°, 80°, 100°, 120°$

Assertion: True

Reason: True and correctly explains the assertion

28. Correct answer: D

Let adjacent angles be $2x$and $3x$(since one is two-third of the other).

Adjacent angles of a parallelogram are supplementary:

$2x + 3x = 180° \Rightarrow 5x = 180° \Rightarrow x = 36°$

Angles $= 72°$and $108°$, not $74°$and $106°$.

Assertion: False

Reason: True

29. Correct answer: A

If diagonals of a parallelogram are equal, it becomes a rectangle.

In a rectangle, all angles are 90°.

Hence $\angle ABC = 90°$.

Assertion: True

Reason: True and correctly explains the assertion

MARIX MATCHING TYPE

30. Correct answer: a–Q, b–R, c–S, d–P

a) $\angle A + \angle C = 180° \Rightarrow \angle B + \angle D = 180°$
→ Q (180°)

b) Sum of interior angles of a quadrilateral
→ R (360°)

c) Interior angle <180°
→ S (Convex quadrilateral)

d) Interior angle > 180°
→ P (Concave quadrilateral)

31. Correct answer: a–R, b–Q, c–P, d–S

a) A closed figure bounded by four line segments
→ R) Quadrilateral
b) A rhombus in which one angle is a right angle
→ Q) Square
c) A parallelogram in which two adjacent sides are equal
→ P) Rhombus
d) The diagonals of a rectangle are
→ S) Equal

COMPREHENSION TYPE

32. Correct answer: B) 90°

Passage-I

Given:
Angles of quadrilateral $ABCD$:

$$A = 5x, B = 5(x+2), C = 6x-20, D = 6(x+3)$$

Sum of interior angles of a quadrilateral $= 360°$.

$5x + 5(x+2) + (6x-20) + 6(x+3) = 360$
$5x + 5x + 10 + 6x - 20 + 6x + 18 = 360$
$22x + 8 = 360 \Rightarrow 22x = 352 \Rightarrow x = 16$
32. $\angle B$

$$\angle B = 5(x+2) = 5(18) = 90°$$

33. Correct answer: C) 76°

$$\angle C$$

$$\angle C = 6x - 20 = 96 - 20 = 76°$$

34. Correct answer: D) 114°

$$\angle D$$

$$\angle D = 6(x+3) = 6(19) = 114°$$

Passage-II

Given: Adjacent angles $A : B = 4 : 5$in a parallelogram.
Adjacent angles of a parallelogram are supplementary.

Let

$\angle A = 4x, \angle B = 5x$
$4x + 5x = 180° \Rightarrow 9x = 180° \Rightarrow x = 20°$

35. Correct answer: B) 80°

$$\angle A$$

$$\angle A = 4x = 4(20) = 80°$$

36. Correct answer: A) 100°

$$\angle B$$

$$\angle B = 5x = 5(20) = 100°$$

37. Correct answer: C) 20°

$$\angle D - \angle C$$

Opposite angles of a parallelogram are equal

$\angle D = \angle B = 100°, \angle C = \angle A = 80°$
$\angle D - \angle C = 100° - 80° = 20°$

INTEGER TYPE

38. Correct answer: 2 cm

In a trapezium, median = ½(sum of parallel sides)

$$EF = \frac{AB + CD}{2}$$

$4 = \frac{AB + 6}{2} \Rightarrow AB + 6 = 8 \Rightarrow AB = 2$ cm

39. Correct answer: 2

Sum of angles of a quadrilateral $= 360°$

$\angle C + \angle D = 360 - (60 + 120) = 180°$
$\frac{\angle C + \angle D}{90°} = \frac{180}{90} = 2$

40. Correct answer: 110°

Adjacent angles of a parallelogram are supplementary

$$180° - 70° = 110°$$

41. Correct answer: 72° and 108°

Let angles be $2x$and $3x$

$2x + 3x = 180° \Rightarrow 5x = 180° \Rightarrow x = 36°$
Angles $= 72°$and $108°$

PREVIOUS YEAR QUESTIONS

1. Correct answer: C. Rectangle

Since APB ∥ CQD, corresponding and alternate interior angles are equal.

Hence, ∠APQ = ∠CQP and ∠BPQ = ∠PQD.

The bisectors of equal angles are also parallel in pairs.

Thus, the four angle bisectors form a quadrilateral with both pairs of opposite sides parallel and all angles equal (90°).

2. Correct answer: A. 120°

In parallelogram ABCD, opposite sides are parallel:

$$AD \parallel BC.$$

Since $ED = \frac{1}{2}AD$ and $BF = \frac{1}{3}BC$, points E and F divide the parallel sides in fixed ratios.

Hence, line DF is parallel to AE and line BF is parallel to AD.

Therefore, the angle between DF and BF is supplementary to the angle between DF and AD.

Given $\angle ADF = 60^\circ$, $\angle BFD = 180^\circ - 60^\circ = 120^\circ$.

3. Correct answer: C. Rhombus

If the diagonals bisect each other, the quadrilateral is a parallelogram.

If the diagonals intersect at right angles, this property is special to a rhombus (also true for a square, which is a type of rhombus).

A rectangle's diagonals bisect each other but do not meet at right angles.

4. Correct answer: C. $\boxed{60^\circ}$

Since ABCD is a square, $\angle DCB = 90^\circ$
Diagonal AC bisects the angle at C, therefore $\angle OCX = 45^\circ$
Angles $\angle COD$ and $\angle COX$ form a linear pair, so

$$\angle COD + \angle COX = 180^\circ$$
$$105^\circ + \angle COX = 180^\circ$$
$$\angle COX = 75^\circ$$

Now, in $\triangle COX$, using angle sum property: $\angle OCX + \angle COX + \angle OXC = 180^\circ$

$$45^\circ + 75^\circ + x = 180^\circ$$
$$120^\circ + x = 180^\circ$$
$$x = 60^\circ$$

5. Correct answer: $\boxed{\textbf{B. } AB = BE}$

Identify Congruent Triangles
Compare $\triangle OCD$ and $\triangle OBE$:

$$\angle DCO = \angle EBO$$

(Alternate interior angles, since $DC \parallel AB$)

$$OC = OB$$

(Given: ED bisects BC at O)

$$\angle COD = \angle BOE$$

(Vertically opposite angles)

$\therefore \triangle OCD \cong \triangle OBE$ (ASA congruence)

Equate Corresponding Sides
Since the triangles are congruent,

$$DC = BE$$

Use Parallelogram Property
In parallelogram $ABCD$, opposite sides are equal:

$$AB = DC$$

From $AB = DC$ and $DC = BE$,

$$AB = BE$$

6. Correct answer:

$\boxed{\textbf{D. Diagonals of } PQRS \textbf{ are at right angles}}$

By Midpoint Theorem, the quadrilateral formed by joining the midpoints of the sides of any quadrilateral is always a parallelogram.

The sides of this midpoint parallelogram are parallel to the diagonals of quadrilateral $PQRS$.

A parallelogram becomes a rectangle if its adjacent sides are perpendicular.

Therefore, the midpoint quadrilateral will be a rectangle only if the diagonals of $PQRS$ are perpendicular.

7. Correct answer: B. $ABCD$ is a trapezium with $AD \parallel BC$

Sum of interior angles of a quadrilateral $= 360^\circ$

Sum of ratios

$$11 + 19 + 21 + 9 = 60$$

Each part $= \frac{360^\circ}{60} = 6^\circ$
Angles are:

$A = 11 \times 6 = 66^\circ$
$B = 19 \times 6 = 114^\circ$
$C = 21 \times 6 = 126^\circ$
$D = 9 \times 6 = 54^\circ$
Check adjacent angles:

$$A + B = 66^\circ + 114^\circ = 180^\circ$$
$$C + D = 126^\circ + 54^\circ = 180^\circ$$

Since one pair of adjacent angles is supplementary, one pair of opposite sides is parallel.
So, $ABCD$ is a trapezium.

8. Correct answer: C. 120°

In a square, diagonals intersect at right angles.

$$\angle(\text{between diagonals}) = 90^\circ$$

At point O, the angle between diagonal $DB$and line $CX$is given as 75°.

Therefore, the angle between diagonal $AC$and line $CX$is:

$$90^\circ - 75^\circ = 15^\circ$$

In a square, diagonal ACmakes an angle of $45^\circ$with side AB.

Hence, angle $x$at point Xon $AB$is:

$$x = 180^\circ - (45^\circ + 15^\circ) = 120^\circ$$

9. Correct answer: C. (T, F, T)

(i) In a parallelogram, if $\angle A = 60^\circ$, geometric construction using angle bisectors shows

$$DC = 2AD$$

True

(ii) In a rhombus, diagonals bisect each other at 90°but are not always equal (they are equal only in a square).

False

(iii) Adjacent angles of a parallelogram are supplementary. Their angle bisectors are perpendicular, forming a rectangle.

True

10. Correct answer: B $\boxed{PN = 2PQ}$

Given:

$PQRS$ is a parallelogram

M is the midpoint of QR

SM meets the extension of PQ at N

Find PN

Solution (using coordinates for clarity)

Let $P(0,0), Q(a,0), R(a,b), S(0,b)$

Midpoint of QR:$M\left(a, \frac{b}{2}\right)$

Equation of line SM:

Slope $= \frac{\frac{b}{2}-b}{a-0} = -\frac{b}{2a}$

$y - b = -\frac{b}{2a}(x - 0)$

At point N, $y = 0$:

$0 - b = -\frac{b}{2a}x \Rightarrow x = 2a$

So,$PN = 2a$

But $PQ = a$

9. Area of parallelograms

Solutions

Worksheet – 1

1. **Correct answer: B) 96 cm**

   Given:

   Area = 512 cm², Length = 2 × Breadth

   Let breadth = b, then length = $2b$

   $$\text{Area} = l \times b = 2b^2 = 512$$

   $$b^2 = 256 \Rightarrow b = 16 \text{ cm}$$

   Length = 32cm

   Perimeter = $2(l + b) = 2(32 + 16) = 96\text{cm}$

2. **Correct answer: D) $\frac{d^2}{2}$**

   Diagonal of square = d

   Side of square = $\frac{d}{\sqrt{2}}$

   $$\text{Area} = \left(\frac{d}{\sqrt{2}}\right)^2 = \frac{d^2}{2}$$

3. **Correct answer: C) 24 sq. cm**

   $$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

   $$= \frac{1}{2} \times 8 \times 6 = 24 \text{ cm}^2$$

4. **Correct answer: D) 6 cm**

   $$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

   $$39 = \frac{1}{2} \times b \times 13$$

   $$b = \frac{39 \times 2}{13} = 6 \text{ cm}$$

5. **Correct answer: A) 36 m²**

   Area of parallelogram = base × height

   $= 8 \times 4.5 = 36 \text{ m}^2$

6. **Correct answer: C) 72**

   Area of a parallelogram = base × height = $12 \times 6 = 72\ c \text{ m}^2$

7. **Correct answer: C) 16 cm²**

   Area of a triangle = $\frac{1}{2} \times$ base × height = $\frac{1}{2} \times 8 \times 4 = 16 \text{ cm}^2$.

8. **Correct answer: B) 10 cm**

   Base = $\frac{\text{area}}{\text{height}} = \frac{80}{8} = 10$ cm.

9. **Correct answer: C) $x(x-1)$**

   Area of the parallelogram = base × height = $x \times (x-1) = x(x-1)$.

10. **Correct answer: A) 20 cm**

    Base of a triangle = $\frac{2 \times \text{area}}{\text{height}} = \frac{2 \times 80}{8} = 20$ cm.

Worksheet – 2

1. **Correct answer: C) 65 cm²**

   The shaded region is a triangle with base $AB = 13\text{cm}$ and height $BC = 10\text{cm}$.

   $$\text{Area} = \frac{1}{2} \times 13 \times 10 = 65 \text{ cm}^2$$

2. **Correct answer: A) 6 sq. cm**

   Area of a rhombus = $\frac{1}{2} \times d_1 \times d_2$.

   $$\frac{1}{2} \times 4 \times 3 = 6 \text{ sq. cm}$$

3. **Correct answer: B) 24 sq. cm**

   The figure is a square with diagonals perpendicular. The diagonals are split into lengths 3 cm and 4 cm, so full diagonals are 6cm and 8cm.

   Area = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 6 \times 8 =$ 24 sq. cm.

4. **Correct answer: B) $\frac{x^2}{\sqrt{3}}$ sq. cm**

   For an equilateral triangle, height $h = \frac{\sqrt{3}}{2}a$.

   So $a = \frac{2x}{\sqrt{3}}$.

   $$\text{Area} = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}\left(\frac{2x}{\sqrt{3}}\right)^2 = \frac{x^2}{\sqrt{3}}$$

5. **Correct answer: C) 120**

   Area of a rhombus = $\frac{1}{2}d_1d_2$.

   $$\frac{1}{2} \times 10 \times 24 = 120 \text{ sq. cm}$$

6. **Correct answer: D) $a - b$**

   Area of a trapezium = $\frac{1}{2}(a+b)h$.

   Given area = $\frac{1}{2}(a^2 - b^2) = \frac{1}{2}(a+b)(a-b)$.

   Comparing, $h = a - b$.

7. **Correct answer: A) 12**
   Area of a rhombus $= \frac{1}{2}d_1d_2$.
   $$96 = \frac{1}{2} \times 16 \times d_2 \Rightarrow d_2 = \frac{192}{16} = 12 \text{ cm}$$
8. **Correct answer: D) 84 sq. cm**
   Area of a parallelogram = base × height
   $$12 \times 7 = 84 \text{ sq. cm}$$
9. **Correct answer: C) 48 cm**
   Area of an equilateral triangle $= \frac{\sqrt{3}}{4}a^2$.
   $$\frac{\sqrt{3}}{4}a^2 = 64\sqrt{3} \Rightarrow a^2 = 256 \Rightarrow a = 16 \text{ cm}$$
   Perimeter $= 3a = 48$cm.
10. **Correct answer: B) 64 cm²**
    Area of a trapezium $= \frac{1}{2}(a+b)h$.
    $$\frac{1}{2}(7+9) \times 8 = \frac{1}{2} \times 16 \times 8 = 64 \text{ cm}^2$$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer: A) 60 cm**
   Let length : breadth = 3: 2.
   So length $= 3x$, breadth $= 2x$.
   Area $= 3x \times 2x = 6x^2 = 216 \Rightarrow x^2 = 36 \Rightarrow x = 6$.
   Length $= 18$cm, breadth $= 12$cm.
   Perimeter $= 2(l+b) = 2(18+12) = 60$cm.
2. **Correct answer: A) 90 m**
   Let breadth $= x$, then length $= 3x$.
   Perimeter $= 2(l+b) = 2(3x+x) = 8x = 240$.
   So $x = 30$.
   Length $= 3x = 90$m.
3. **Correct answer: B) 60 m**
   Area of a square $= a^2 = 225$.
   So side $a = \sqrt{225} = 15$m.
   Perimeter $= 4a = 60$m.
4. **Correct answer: C) $l = 35$cm, $A = 1050\ cm^2$**
   Perimeter $= 2(l+b) = 130 \Rightarrow l + b = 65$.
   Given $b = 30$, so $l = 65 - 30 = 35$cm.
   Area $= l \times b = 35 \times 30 = 1050\ cm^2$
5. **Correct answer: B) 10 cm**
   Area of a square $= a^2 = 50 \Rightarrow a = \sqrt{50} = 5\sqrt{2}$cm.
   Diagonal $= a\sqrt{2} = 5\sqrt{2} \times \sqrt{2} = 10$cm.
6. **Correct answer: B) 44**
   Area of trapezium $= \frac{1}{2}(a+b)h$.
   $$480 = \frac{1}{2}(20+x) \times 15 \Rightarrow 20 + x = 64 \Rightarrow x = 44$$
7. **Correct answer: A) 30 cm**
   Area of a rhombus $= \frac{1}{2}d_1d_2$.
   $$240 = \frac{1}{2} \times 16 \times d_2 \Rightarrow d_2 = \frac{480}{16} = 30 \text{ cm}$$
8. **Correct answer: B) 41 cm²**
   Area of a rhombus $= \frac{1}{2}d_1d_2$.
   $$\frac{1}{2} \times 10 \times 8.2 = 41 \text{ cm}^2$$
9. **Correct answer: B) 26 units**
   Triangle with sides 5, 12, 13 is right-angled.
   Area of triangle $= \frac{1}{2} \times 5 \times 12 = 30$.
   Rectangle has same area and length $= 10$.
   Breadth $= \frac{30}{10} = 3$.
   Perimeter $= 2(l+b) = 2(10+3) = 26$units.
10. **Correct answer: C) $\frac{12}{\sqrt{2}+1}$**
    Let each corner be cut at distance $x$from the vertex.
    Each side of the octagon has two types of equal lengths:
    The slanted cut side $= x\sqrt{2}$
    The remaining middle part of the square side $= 12 - 2x$
    Since all sides are equal:
    $$12 - 2x = x\sqrt{2}$$
    $$12 = x(2+\sqrt{2}) \Rightarrow x = \frac{12}{2+\sqrt{2}}$$
    Length of each side of the octagon:
    $$x\sqrt{2} = \frac{12\sqrt{2}}{2+\sqrt{2}} = \frac{12}{\sqrt{2}+1}$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct answer: A and C**
    Given: Perimeter = 68 cm, Breadth (b)= 14 cm
    Formula: $P = 2(l+b)$
    Calculation:
    $68 = 2(l+14) 34 = l + 14l = 20$ cm
    Checking Options:
    A) 20 cm (Correct)
    B) $2 \times 20 = 40$(Incorrect)
    C) $\frac{1}{2}(40) = 20$(Correct)

12. Correct answer: B and C

Area and Perimeter of a Square
Given: Area = 625 cm^2
Formula: Area = s^2, Perimeter = $4s$
Calculation:

$s = \sqrt{625} = 25$ cm$P = 4 \times 25 = 100$ cm
Checking Options:

B) 100 cm (Correct)

C) $50 \times 2 = 100$ cm(Correct)

13. Correct answer: C and D

Square and Parallelogram Area
Given: Side of square (s)= 40 m, Height of parallelogram (h)= 20 m
Logic: Area of Square = Area of Parallelogram

$s^2 =$ base $\times h$
$40^2 =$ base $\times$ 201600 = base $\times$ 20base
$= 80$ m
Checking Options:

B) 80 m^2(Incorrect $-$ unit should be length)

C) $4 \times$ height $= 4 \times 20 = 80$ m(Correct)

D) 80 m (Correct)

14. Correct answer: C

Triangle Area
Given: $h = 2b$, Area = 400 cm^2
Formula: Area $= \frac{1}{2} \times b \times h$

$400 = \frac{1}{2} \times b \times (2b) 400 = b^2 b = 20$ cm
Since $h = 2b, h = 2 \times 20 = 40$ cm

15. Correct answer: A) 20 cm B) $2 \times 10cm$

Area of the rectangle
= length × breadth
= 20 × 15 = 300 cm^2

Area of the triangle
= $\frac{1}{2}$ ×base × height
= $\frac{1}{2} \times 30 \times h = 15h$

Given both areas are equal:
15h = 300

h = 20 cm

ASSERTION AND REASON TYPE

16. Correct answer: (A)Both A and R are true, and R is the correct explanation of A.

Assertion (A): True
Area of triangle = $\frac{1}{2}$ ×base × height,
Area of parallelogram = base × height.
So triangle's area is half.
Reason (R): True
A diagonal divides a parallelogram into two equal-area triangles, explaining why the area is half.

17. Correct answer: (C) A is true, but R is false.

Assertion (A): True
Area of a parallelogram = base × height.
Same base and height ⇒ same area.
Reason (R): False
Area does not depend only on the lengths of sides; it depends on base and height.

18. Correct answer: (A) Both A and R are true, and R is the correct explanation of A.

Parallelogram Properties

Assertion (A): True
The height forms a right-angled triangle with the slanted side as the hypotenuse, so height ⩽ side length.

Reason (R): True
Height is always measured perpendicular to the base.

Explanation:
The perpendicular height creates a right triangle where the side length is the hypotenuse, so the height cannot exceed it.

19. Correct answer: (A) Both A and R are true, and R is the correct explanation of A.

Right-Angled Triangle Area

Assertion (A): True
Area = $\frac{1}{2}$ ×(product of perpendicular sides).

Reason (R): True
The perpendicular sides act as base and height.

Explanation:
Since the sides are perpendicular, they directly fit the triangle area formula.

20. Correct answer: (B) Both A and R are true, but R is not the correct explanation of A.

Assertion (A): True
Area of both a parallelogram and a rectangle = base × height.
Same base and height ⇒ same area.

Reason (R): True
A rectangle is a special type of parallelogram with all angles 90°.

Explanation:
The equality of areas is due to the same area formula, not because a rectangle is a special parallelogram.

MATRIX MATCHING TYPE

21. Correct answer: e – P

Match the area formula with the polygon

a) $\frac{1}{2}d_1d_2$
→ Area formula of Rhombus
Answer: a – Q

b) $\frac{1}{2}h(a+b)$
→ Area formula of Trapezium
Answer: b – R

c) $\frac{1}{2}(h_1+h_2)d$
→ Area of a Quadrilateral (sum of two triangles)
Answer: c – S

d) bh
→ Area of Parallelogram
Answer: d – T

e) a^2
→ Area of Square

22. Correct answer:

Match the values

a) Side of square = 12 cm
Area $= 12^2 = 144$ cm^2
Answer: a – S

b) $h = 12$cm, $b = 6$cm
Area of triangle $= \frac{1}{2}bh = \frac{1}{2} \times 12 \times 6 =$ 36 cm^2
Answer: b – T

c) Perimeter of square = 88 cm
Side $= 88 \div 4 = 22$cm
Answer: c – P

d) Perimeter of rhombus with side a
Perimeter $= 4a$
Answer: d – Q

e) Parallelogram sides 3 cm and 2 cm
Perimeter $= 2(3+2) = 10$cm
Answer: e – R

COMPREHENSION TYPE

Passage – I

Given:
Base : Height = 5 : 2
Let base = $5x$, height = $2x$

Area of parallelogram

Area = base × height $= 5x \times 2x = 10x^2$

Given area $= 360$ m^2

$10x^2 = 360 \Rightarrow x^2 = 36 \Rightarrow x = 6$

So,Base $= 5x = 5 \times 6 = 30$m
Height $= 2x = 2 \times 6 = 12$m

23. Correct answer: C

The base of the parallelogram= 30 m

24. Correct answer: B

The area of $\triangle ABC$

A diagonal divides a parallelogram into two equal-area triangles.

$$\text{Area of } \triangle ABC = \frac{1}{2} \times 360 = 180 \text{ m}^2$$

25. Correct answer: A

The height of the parallelogram= 12 m

Passage – II

26. Correct answer: B) 45

Area of rectangle = length × breadth = $9 \times 5 = 45$

27. Correct answer: B) 40

Perimeter of rectangle = $2(l+b) = 2(12+8) = 2 \times 20 = 40$

28. Correct answer: A) 6 cm

Area = length × breadth
$48 = 8 \times$ breadth
breadth $= 48 \div 8 = 6$cm

INTEGER TYPE

29. Correct answer: 5 cm

Base : height = 1 : 5
Let base = x, height = $5x$
Area $= x \times 5x = 5x^2 = 125$
$x^2 = 25 \Rightarrow x = 5$

30. Correct answer: 9 cm

Area $= l \times b$
$144 = 16 \times b \Rightarrow b = 9$

31. Correct answer: 4 cm

Perimeter of square $= 4 \times$ side
$16 = 4s \Rightarrow s = 4$

32. Correct answer: 7 cm

Area of rhombus $= \frac{1}{2} d_1 d_2$
$112 = \frac{1}{2} \times 32 \times d_2$
$112 = 16d_2 \Rightarrow d_2 = 7$

33. Correct answer: 8 cm

Area of isosceles right triangle $= \frac{1}{2} a^2$
$\frac{1}{2} a^2 = 32 \Rightarrow a^2 = 64$
$a = 8$

34. Correct answer: 8 cm

Area of equilateral triangle $= \frac{\sqrt{3}}{4} a^2$
$\frac{\sqrt{3}}{4} a^2 = 144\sqrt{3}$
$a^2 = 576 \Rightarrow a = 24$
$\frac{1}{3}$of side $= 8$

35. Correct answer: 12 cm

Area $= \frac{1}{2} \times b \times h$
$30 = \frac{1}{2} \times 5 \times h$
$h = 12$

36. Correct answer: 7 cm

Area $= \frac{1}{2} \times b \times h$
$42 = \frac{1}{2} \times 12 \times h$
$h = 7$

10. Exploring Algebraic identities

Solutions

Worksheet – 1

1. **Correct Answer: B) 4**

If $a = 5$ and $b = 3$, find $(a-b)^2$

$$(a-b)^2 = (5-3)^2$$
$$= 2^2 = 4$$

2. **Correct Answer: A)** $9x^2 + 12x + 4$

Find the value of $(3x+2)^2$

Use identity: $(a+b)^2 = a^2 + 2ab + b^2$
Let $a = 3x$, $b = 2$

$$(3x+2)^2 = (3x)^2 + 2(3x)(2) + 2^2$$
$$= 9x^2 + 12x + 4$$

3. **Correct Answer: A)** $25m^2 - 30mn + 9n^2$

Expand $(5m-3n)^2$

Use identity:$(a-b)^2 = a^2 - 2ab + b^2$
Let $a = 5m$, $b = 3n$

$$(5m-3n)^2 = (5m)^2 - 2(5m)(3n) + (3n)^2$$
$$= 25m^2 - 30mn + 9n^2$$

4. **Correct Answer: A)** $16x^2 - 9$

Find the product $(4x+3)(4x-3)$

Use identity:$(a+b)(a-b) = a^2 - b^2$
Let $a = 4x$, $b = 3$

$$(4x+3)(4x-3) = (4x)^2 - 3^2$$
$$= 16x^2 - 9$$

5. **Correct Answer: B)** $5x - 2y$

Find the product $(\sqrt{5x} - \sqrt{2y})(\sqrt{5x} + \sqrt{2y})$

Use identity:

$$(a-b)(a+b) = a^2 - b^2$$

Let $a = \sqrt{5x}$, $b = \sqrt{2y}$

$$(\sqrt{5x})^2 - (\sqrt{2y})^2$$
$$= 5x - 2y$$

6. **Correct Answer: A) 9**

If $x^2 + 9x + 20 = (x+a)(x+b)$, find $a + b$

Expand the RHS:$(x+a)(x+b) = x^2 + (a+b)x + ab$
Compare with $x^2 + 9x + 20$

$$a + b = 9$$

7. **Correct Answer: A)** $x^2 + 2x + \frac{3}{4}$

Expand $\left(x+\frac{1}{2}\right)\left(x+\frac{3}{2}\right)$

$$\left(x+\frac{1}{2}\right)(x+\frac{3}{2})$$

Multiply:

$$= x^2 + \frac{3}{2}x + \frac{1}{2}x + \frac{3}{4}$$
$$= x^2 + 2x + \frac{3}{4}$$

8. **Correct Answer: B) 100**

If $a + b = 10$ and $ab = 21$, find $(a+b)^2$

Use identity:$(a+b)^2$

$$= 10^2$$
$$= 100$$

9. **Correct Answer: A)** $16m^2 - 56mn + 49n^2$

Expand $(4m-7n)^2$

Use identity:$(a-b)^2 = a^2 - 2ab + b^2$
Let $a = 4m$, $b = 7n$

$$(4m-7n)^2 = (4m)^2 - 2(4m)(7n) + (7n)^2$$
$$= 16m^2 - 56mn + 49n^2$$

10. **Correct Answer: A)** $8x - 3y$

Find the product $(\sqrt{8x} - \sqrt{3y})(\sqrt{8x} + \sqrt{3y})$

Use identity:

$$(a-b)(a+b) = a^2 - b^2$$

Let $a = \sqrt{8x}$, $b = \sqrt{3y}$

$$(\sqrt{8x})^2 - (\sqrt{3y})^2$$
$$= 8x - 3y$$

WORKSHEET-2

1. **Correct Answer: C)** $(x+6)(x+6)$

   Factorize $x^2+12x+36$

   Find two numbers whose sum = **12** and product = **36**.

   $$6+6=12, 6\times 6=36$$
   $$x^2+12x+36=(x+6)(x+6)$$

2. **Correct Answer: A)** $(x+4)(x+5)$

   Factorize $x^2+9x+20$

   Find two numbers whose sum = **9** and product = **20**.

   $$4+5=9, 4\times 5=20$$
   $$x^2+9x+20=(x+4)(x+5)$$

3. **Correct Answer: C)** $(x+6)(x+7)$

   Factorize $x^2+13x+42$

   Find two numbers whose sum = **13** and product = **42**.

   $$6+7=13, 6\times 7=42$$
   $$x^2+13x+42=(x+6)(x+7)$$

4. **Correct Answer: B) 6**

   For x^2+5x+6, the number of unit tiles used is

   In algebra tiles:

   - $x^2 \rightarrow$ one x^2tile
   - $5x \rightarrow$ five $x$tiles
   - $6 \rightarrow$ **six unit tiles**

   Therefore, the number of **unit tiles = 6**.

5. **Correct Answer: D)** $x+3$ **and** $x+2$

   The rectangle formed by tiles for x^2+5x+6 has dimensions

   Factorize the expression.

   Find two numbers whose **sum = 5** and **product = 6**.

   $$2+3=5, 2\times 3=6$$
   $$x^2+5x+6=(x+2)(x+3)$$

   Thus the rectangle dimensions are $x+3$ **and** $x+2$.

6. **Correct Answer: A)** $16x^3(x-3)(x+3)$

   Factorize $16x^5-144x^3$

   Take **common factor** $16x^3$.

   $$16x^5-144x^3=16x^3(x^2-9)$$

   Now use identity:

   $$x^2-9=(x-3)(x+3)$$
   $$16x^3(x^2-9)=16x^3(x-3)(x+3)$$

7. **Correct Answer: C)** $4\ell m$

   Simplify $(\ell+m)^2-(\ell-m)^2$

   $$(\ell+m)^2=\ell^2+2\ell m+m^2$$
   $$(\ell-m)^2=\ell^2-2\ell m+m^2$$

   Subtract:

   $$(\ell+m)^2-(\ell-m)^2$$
   $$=(\ell^2+2\ell m+m^2)-(\ell^2-2\ell m+m^2)$$
   $$=4\ell m$$

8. **Correct Answer: B)** $(3xy-4)(3xy+4)$

   Factorize $9x^2y^2-16$

   This is a difference of squares.

   $$a^2-b^2=(a-b)(a+b)$$
   $$9x^2y^2-16=(3xy)^2-4^2$$
   $$=(3xy-4)(3xy+4)$$

9. **Correct Answer: A)** $(x-y-z)(x-y+z)$

   Factorize $(x^2-2xy+y^2)-z^2$

   $$x^2-2xy+y^2=(x-y)^2$$

   So, $(x-y)^2-z^2$

   Use difference of squares:

   $$a^2-b^2=(a-b)(a+b)$$
   $$=(x-y-z)(x-y+z)$$

10. **Correct Answer: D)** $(5a-2b+7c)(5a+2b-7c)$

    Factorize $25a^2-4b^2+28bc-49c^2$

    Rewrite the middle terms:

    $$25a^2-(4b^2-28bc+49c^2)$$
    $$4b^2-28bc+49c^2=(2b-7c)^2$$

    So, $25a^2-(2b-7c)^2$

    Use difference of squares:

    $$=(5a-(2b-7c))(5a+(2b-7c))$$
    $$=(5a-2b+7c)(5a+2b-7c)$$

COMPETITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: B)** $4x^2+20x+25$

   Expand $(2x+5)^2$

   Use identity: $(a+b)^2=a^2+2ab+b^2$

   Let $a=2x, b=5$

$$(2x+5)^2 = (2x)^2 + 2(2x)(5) + 5^2$$
$$= 4x^2 + 20x + 25$$

2. **Correct Answer: C)** $(x+6)(x-6)$

Factorize $x^2 - 36$

This is a difference of squares.

$$x^2 - 36 = x^2 - 6^2$$
$$= (x+6)(x-6)$$

3. **Correct Answer: A) 80**

If $a+b=12$ **and** $ab=32$**, find** a^2+b^2

Use identity:

$$a^2 + b^2 = (a+b)^2 - 2ab$$
$$= 12^2 - 2(32)$$
$$= 144 - 64$$
$$= 80$$

4. **Correct Answer: A)** $(3x+5)(3x-5)$

Factorize $9x^2 - 25$

This is a difference of squares.

$$9x^2 - 25 = (3x)^2 - 5^2$$
$$= (3x+5)(3x-5)$$

5. **Correct Answer: B)** $x+2$

The area of a rectangular field is $x^2 + 7x + 10$. One side is $x+5$. Find the other side.

Factorize the area expression.

$$x^2 + 7x + 10$$

Find two numbers whose **sum = 7** and **product = 10**.

$$5 + 2 = 7, 5 \times 2 = 10$$
$$x^2 + 7x + 10 = (x+5)(x+2)$$

One side $= x+5$, so the other side $= x+2$.

6. **Correct Answer: A)** $(x+4)$**and** $(x-4)$

The area of a rectangular plot is $x^2 - 16$. Possible dimensions are

Use difference of squares:

$$x^2 - 16 = x^2 - 4^2$$
$$= (x+4)(x-4)$$

So the dimensions are $x+4$and $x-4$.

7. **Correct Answer: D) 200**

Using identities, $51^2 - 49^2$

Use identity:

$$a^2 - b^2 = (a-b)(a+b)$$
$$51^2 - 49^2 = (51-49)(51+49)$$
$$= 2 \times 100$$
$$= 200$$

8. **Correct Answer: A)** $x^2 + 11x + 24$

A rectangular field has dimensions $(x+3)m$and $(x+8)m$. Find the area.

$$\text{Area} = (x+3)(x+8)$$
$$= x^2 + 8x + 3x + 24$$
$$= x^2 + 11x + 24$$

9. **Correct Answer: A)** $4x^6 + \frac{2x^3}{5} - \frac{6}{25}$

Simplify $\left(2x^3 + \frac{3}{5}\right)\left(2x^3 - \frac{2}{5}\right)$

Expand using multiplication.

$$(2x^3)(2x^3) = 4x^6$$
$$(2x^3)\left(-\frac{2}{5}\right) = -\frac{4x^3}{5}$$
$$\left(\frac{3}{5}\right)(2x^3) = \frac{6x^3}{5}$$
$$\left(\frac{3}{5}\right)\left(-\frac{2}{5}\right) = -\frac{6}{25}$$

Combine middle terms:

$$-\frac{4x^3}{5} + \frac{6x^3}{5} = \frac{2x^3}{5}$$

Final result:

$$4x^6 + \frac{2x^3}{5} - \frac{6}{25}$$

10. **Correct Answer: C) 81**

If $x-y=7$ **and** $xy=8$**, find** $(x+y)^2$

$$(x-y)^2 = x^2 + y^2 - 2xy$$
$$7^2 = x^2 + y^2 - 2(8)$$
$$49 = x^2 + y^2 - 16$$
$$x^2 + y^2 = 65$$

Now,

$$(x+y)^2 = x^2 + y^2 + 2xy$$
$$= 65 + 16$$
$$= 81$$

11. **Correct Answer: A) 3**

If $x+y=13$ **and** $xy=40$**, find** $x-y$

$$(x-y)^2 = (x+y)^2 - 4xy$$
$$= (13)^2 - 4(40)$$
$$= 169 - 160$$
$$= 9$$
$$x - y = 3$$

12. **Correct Answer: C) 19**

If $x^2 + y^2 = 193$ **and** $xy=84$**, find** $x+y$

$$(x+y)^2 = x^2 + y^2 + 2xy$$
$$= 193 + 2(84)$$

$= 193 + 168$
$= 361$
$x + y = 19$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

13. Correct Answers: A, B, D

Which of the following factorizations are correct?

A.$x^2 - 16 = x^2 - 4^2 = (x + 4)(x - 4)$
B.$a^2 - b^2 = (a - b)(a + b)$
C.$(x + 3)(x - 3) = x^2 - 9 \neq x^2 + 9$
D.$y^2 - 25 = y^2 - 5^2 = (y + 5)(y - 5)$

14. Correct Answers: A, C

If $\boldsymbol{x + y = 10}$ and $\boldsymbol{xy = 21}$, which values are correct for $\boldsymbol{x^2 + y^2}$?

Use identity:

$$x^2 + y^2 = (x + y)^2 - 2xy$$
$$= (10)^2 - 2(21)$$
$$= 100 - 42 = 58$$

Also the identity form itself is correct.

Numerical value = **58**

Identity form = $(x + y)^2 - 2xy$

15. Correct Answers: A, B, D

Which expressions can be factorized using difference of squares?

Identity:$a^2 - b^2 = (a + b)(a - b)$
Check each:

A.$x^2 - 49 = x^2 - 7^2$

B.$9a^2 - 25b^2 = (3a)^2 - (5b)^2$
C.$x^2 + 16$
Sum of squares → cannot be factorized this way
D.$4y^2 - 1 = (2y)^2 - 1^2$

16. Correct Answers: A, B, C, D

Which expressions can be factorized using splitting the middle term?

A.

$x^2 + 5x + 6$
$= x^2 + 2x + 3x + 6$
$= (x + 2)(x + 3)$
B.

$x^2 + 7x + 10$
$= x^2 + 5x + 2x + 10$
$= (x + 5)(x + 2)$
C.

$x^2 + 4x + 4$
$= (x + 2)^2$
Also factorized using splitting

D.

$x^2 + 3x + 2$
$= x^2 + x + 2x + 2$
$= (x + 1)(x + 2)$

17. Correct Answers: A, B, C

In algebra tiles, which statements are correct?

- Large square represents x^2
- Rectangle represents x
- Unit square represents constant 1
- Rectangle represents x^2 (incorrect)

18. Correct Answers: A, B, D

If $\boldsymbol{(x + 3)(x + 2)}$ is expanded, which expressions are equivalent?

Expand:

$$(x + 3)(x + 2)$$
$$= x^2 + 2x + 3x + 6$$
$$= x^2 + 5x + 6$$

Check options:

A. $x^2 + 5x + 6$(Equivalent)

B.$x(x + 5) + 6 = x^2 + 5x + 6$(Equivalent)

C.$(x + 2)^2 + x = x^2 + 4x + 4 + x = x^2 + 5x + 4$
Not equal

D.$x^2 + 6 + 5x = x^2 + 5x + 6$(Equivalent)

ASSERTION AND REASON TYPE

19. Correct Answer: A

Assertion (A): $\boldsymbol{a^2 - b^2 = (a + b)(a - b)}$

Reason (R): The difference of two squares can be written as the product of the sum and difference of the numbers.

The identity $a^2 - b^2 = (a + b)(a - b)$ is **true**.

The reason correctly explains that a difference of squares equals the product of the sum and difference.

Thus, both A and R are true and R explains A.

20. Correct Answer: A

Assertion (A): $\boldsymbol{x^2 + 11x + 30 = (x + 5)(x + 6)}$

Reason (R): The sum of 5 and 6 is 11 and their product is 30.

Expand the RHS:

$$(x + 5)(x + 6)$$
$$= x^2 + 6x + 5x + 30$$
$$= x^2 + 11x + 30$$

Thus the assertion is **true**.

The reason is also **true**, and it explains **why the factorization works** (sum of numbers = middle term, product = constant).

21. Correct Answer: D) A is false but R is true

Assertion (A): If $\boldsymbol{a + b = 11}$and $\boldsymbol{ab = 21}$, then $\boldsymbol{a^2 + b^2 = 58}$

Reason (R): $a^2 + b^2 = (a + b)^2 - 2ab$

Using the identity:$a^2 + b^2 = (a + b)^2 - 2ab$
Substitute the values:

$$= (11)^2 - 2(21)$$
$$= 121 - 42$$
$$= 79$$

So the correct value is **79**, not **58**.

- Assertion (A) → **False**
- Reason (R) → **True** (correct identity)

22. Correct Answer: C) A is true but R is false

Assertion (A): If $\boldsymbol{x + y = 12}$and $\boldsymbol{xy = 35}$, then $\boldsymbol{x^2 + y^2 = 74}$

Reason (R): $x^2 + y^2 = (x + y)^2 + 2xy$

Correct identity:$x^2 + y^2 = (x + y)^2 - 2xy$
Now calculate:

$$= (12)^2 - 2(35)$$
$$= 144 - 70$$
$$= 74$$

Thus:

- Assertion (A) → **True**
- Reason (R) → **False** (identity sign is incorrect)

23. Correct Answer: A) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A.

Assertion (A): In algebra tiles, a rectangle tile represents the term $\boldsymbol{x}$.

Reason (R): The rectangle has one side $x$and the other side 1.

- In algebra tiles:
  - Large square → x^2
  - Rectangle → x
  - Small square → 1

The rectangle has dimensions $x \times 1$, so its area is

$$x \times 1 = x$$

Thus:

Assertion (A) → **True**

Reason (R) → **True**, and it **correctly explains** why the rectangle represents x.

MATRIX MATCHING TYPE

24. Match the expression with its factorization

(a) $y^2 - 49$

$y^2 - 49 = y^2 - 7^2 = (y - 7)(y + 7)$
→ **(ii)**

(b) $y^2 + 5y - 24$

Numbers whose sum = 5 and product = −24 are 8 and −3

$$y^2 + 5y - 24 = (y + 8)(y - 3)$$

→ **(i)**

(c) $y^2 + 6y + 8$

Numbers whose sum = 6 and product = 8 are 2 and 4

$$y^2 + 6y + 8 = (y + 2)(y + 4)$$

→ **(iii)**

(d) $y^2 - 5y - 24$

Numbers whose sum = −5 and product = −24 are −8 and 3

$$y^2 - 5y - 24 = (y - 8)(y + 3)$$

→ **(iv)**

25. Match the expression with correct identity

(a) $r^2 + 2rs + s^2$

$$= (r + s)^2$$

→ **(i)**

(b) $r^2 - 2rs + s^2$

$$= (r - s)^2$$

→ **(iii)**

(c) $r^2 - s^2$

Difference of squares

$$= (r + s)(r - s)$$

→ **(iv)**

(d) $4rs$

$$(r + s)^2 - (r - s)^2$$

→ **(ii)**

COMPREHENSION TYPE

PASSAGE-1

26. Correct Answer: A) $x^2 - 9$

The area of the playground can be expressed as

Length $= (x + 3)$
Width $= (x - 3)$

Area:

$$(x + 3)(x - 3)$$

Use the identity:

$$(a + b)(a - b) = a^2 - b^2$$
$$(x + 3)(x - 3) = x^2 - 9$$

27. Correct Answer: A) 91

If $\boldsymbol{x = 10}$, the area of the playground

$$\text{Area} = x^2 - 9$$

Substitute $x = 10$:

$$= 10^2 - 9$$
$$= 100 - 9$$
$$= 91$$

28. Correct Answer: C) $(a + b)(a - b)$

The identity used here is

$$(a + b)(a - b) = a^2 - b^2$$

This is the difference of squares identity.

PASSAGE-2

29. Correct Answer: A) $(x + 3)(x + 4)$

The factorized form of $\boldsymbol{x^2 + 7x + 12}$

Find two numbers whose:

- Product = 12
- Sum = 7

Numbers: **3 and 4**

$$x^2 + 7x + 12 = x^2 + 3x + 4x + 12$$
$$= (x + 3)(x + 4)$$

30. Correct Answer: B) 30

If $\boldsymbol{x = 2}$, find the value of $\boldsymbol{x^2 + 7x + 12}$

$$x^2 + 7x + 12$$

Substitute $x = 2$:

$$= 2^2 + 7(2) + 12$$
$$= 4 + 14 + 12$$
$$= 30$$

31. Correct Answer: A) 1 x^2**tile, 7** x**tiles, 12 unit tiles**

Which represents the rectangle formed by algebra tiles correctly?

Expression:

$$x^2 + 7x + 12$$

Algebra tiles representation:

- x^2term → **1** x^2**tile**
- $7x$term → **7** x**tiles**
- Constant 12→ **12 unit tiles**

INTEGER TYPE

32. Correct Answer: 116

If $\boldsymbol{x + y = 14}$ and $\boldsymbol{xy = 40}$, find $\boldsymbol{x^2 + y^2}$

Use identity:

$$x^2 + y^2 = (x + y)^2 - 2xy$$
$$= (14)^2 - 2(40)$$
$$= 196 - 80$$
$$= 116$$

33. Correct Answer: 1200

Find the value of $\boldsymbol{(103)^2 - (97)^2}$

Use identity:

$$a^2 - b^2 = (a + b)(a - b)$$
$$(103)^2 - (97)^2 = (103 + 97)(103 - 97)$$
$$= 200 \times 6$$
$$= 1200$$

34. Correct Answer: 94

If $\boldsymbol{a - b = 8}$ and $\boldsymbol{ab = 15}$, find $\boldsymbol{a^2 + b^2}$

$$(a - b)^2 = a^2 + b^2 - 2ab$$
$$8^2 = a^2 + b^2 - 2(15)$$

$$64 = a^2 + b^2 - 30$$
$$a^2 + b^2 = 94$$

35. Correct Answer: 9

Find the constant term in $(2x+3)^2$

$$(2x+3)^2 = (2x)^2 + 2(2x)(3) + 3^2$$
$$= 4x^2 + 12x + 9$$

Constant term = **9**

36. Correct Answer: 8000

Evaluate $(1002)^2 - (998)^2$

Use identity:

$$a^2 - b^2 = (a+b)(a-b)$$
$$(1002)^2 - (998)^2$$
$$= (1002 + 998)(1002 - 998)$$
$$= 2000 \times 4$$
$$= 8000$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: C. $x = \pm 2, \pm 2i$

Solve the equation $x^4 - 16 = 0$

Use difference of squares.

$$x^4 - 16 = (x^2)^2 - 4^2$$
$$= (x^2 - 4)(x^2 + 4)$$
$$x^2 - 4 = (x-2)(x+2)$$

So,

$$(x-2)(x+2)(x^2+4) = 0$$
$$x = 2, -2, \pm 2i$$

2. Correct Answer: A. $x = \pm 1$

Solve the equation $x^4 + x^2 + 1 = 3x^2$

$$x^4 + x^2 + 1 - 3x^2 = 0$$
$$x^4 - 2x^2 + 1 = 0$$
$$(x^2 - 1)^2 = 0$$
$$x^2 = 1$$
$$x = \pm 1$$

3. Correct Answer: B. $(x+1)^3$

Factor completely $x^3 + 3x^2 + 3x + 1$

Recognize the identity:

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Here $a = x$, $b = 1$

$$x^3 + 3x^2 + 3x + 1 = (x+1)^3$$

4. Correct Answer: A. $(x^2+1)(x^2+4)$

Factorize $x^4 + 5x^2 + 4$

Let $y = x^2$

$$y^2 + 5y + 4$$

Factorize:

$$(y+1)(y+4)$$

Substitute back $y = x^2$

$$(x^2+1)(x^2+4)$$

5. Correct Answer: B. $(a+b)(a^2 - ab + b^2)$

Factor the expression $a^3 + b^3$

Use the identity:

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

6. Correct Answer: A. $(x^2+1)(x^2+2)$

Factor completely $x^4 + 3x^2 + 2$

Let $y = x^2$

$$y^2 + 3y + 2$$
$$= (y+1)(y+2)$$

Substitute back $y = x^2$

$$(x^2+1)(x^2+2)$$

7. Correct Answer: D. $x = \pm\sqrt{5}, \pm i\sqrt{5}$

Solve the equation $x^4 - 25 = 0$

Use difference of squares:

$$x^4 - 25 = (x^2)^2 - 5^2$$
$$= (x^2 - 5)(x^2 + 5)$$

Now solve each factor:

$$x^2 - 5 = 0 \Rightarrow x = \pm\sqrt{5}$$
$$x^2 + 5 = 0 \Rightarrow x^2 = -5 \Rightarrow x = \pm i\sqrt{5}$$

So, $x = \pm\sqrt{5}, \pm i\sqrt{5}$

8. Correct Answer: A. $(x^2+1)^2$

Factorize $x^4 + 2x^2 + 1$

$$x^4 + 2x^2 + 1 = (x^2)^2 + 2(x^2)(1) + 1^2$$
$$= (x^2+1)^2$$

11. Circles

Solutions

Worksheet – 1

1. **Correct answer: C) 10 cm**

   Wire length = circumference

   $$C = 10 \text{ cm}$$

2. **Correct answer: C) 7 cm**

   Circumference $C = 44$cm

   $$r = \frac{C}{2\pi} = \frac{44}{2 \times \frac{22}{7}} = 7 \text{ cm}$$

3. **Correct answer: A) 15.4 cm**

   Diameter $d = 4.9$cm

   $$C = \pi d = \frac{22}{7} \times 4.9 = 15.4 \text{ cm}$$

4. **Correct answer: B) 96.8 cm**

   Radius $r = 15.4$cm

   $$C = 2\pi r = 2 \times \frac{22}{7} \times 15.4 = 96.8 \text{ cm}$$

5. **Correct answer: B) 2464 sq. m**

   Circumference $C = 176$m

   $$r = \frac{176}{2\pi} = \frac{176}{2 \times \frac{22}{7}} = 28 \text{ m}$$

   $$\text{Area} = \pi r^2 = \frac{22}{7} \times 28^2 = 2464 \text{ sq. m}$$

6. **Correct answer: D) 102.67 sq. cm**

   Radius $r = 7$cm, angle $= 240°$

   $$\text{Area} = \frac{\theta}{360°}\pi r^2 = \frac{240}{360} \times \frac{22}{7} \times 7^2$$
   $$= \frac{2}{3} \times 154 = 102.67 \text{ sq. cm}$$

7. **Correct answer: A) 17.6 cm**

   Angle $= 72°$, radius $r = 14$cm

   $$\text{Arc length} = \frac{\theta}{360°} \times 2\pi r$$
   $$= \frac{72}{360} \times 2 \times \frac{22}{7} \times 14$$
   $$= 17.6 \text{ cm}$$

8. **Correct answer: D) 55 cm**

   Angle $= 75°$, radius $r = 42$cm

   $$\text{Arc length} = \frac{75}{360} \times 2 \times \frac{22}{7} \times 42 = 55 \text{ cm}$$

9. **Correct answer: B) π: 4**

   Let diameter of circle = side of square = a

   $$\text{Area of circle} = \pi\left(\frac{a}{2}\right)^2 = \frac{\pi a^2}{4}$$

   $$\text{Area of square} = a^2$$

   $$\text{Ratio} = \frac{\pi a^2}{4} : a^2 = \pi : 4$$

10. **Correct answer: A) 1.75 m**

    Distance covered $= 2200$m, rotations $= 200$

    $$\text{Circumference} = \frac{2200}{200} = 11 \text{ m}$$

    $$2\pi r = 11 \Rightarrow r = \frac{11}{2\pi} = \frac{11}{2 \times \frac{22}{7}} = 1.75 \text{ m}$$

Worksheet -2

1. **Correct answer: C) 946 sq. m**

   Outer radius $R = 25$m, inner radius $r = 18$m

   $$\text{Area of ring} = \pi(R^2 - r^2)$$
   $$= \frac{22}{7}(25^2 - 18^2)$$
   $$= \frac{22}{7}(625 - 324)$$
   $$= \frac{22}{7} \times 301 = 946 \text{ sq. m}$$

2. **Correct answer: D) $4\pi ab$sq. units**

   Outer radius $= a + b$, inner radius $= a - b$

   $$\text{Area} = \pi[(a+b)^2 - (a-b)^2] = \pi(4ab)$$
   $$= 4\pi ab$$

3. **Correct answer: D) 22 sq. cm**

   Outer radius $R = 4$cm, inner radius $r = 3$cm

   $$\text{Area} = \pi(4^2 - 3^2) = \pi(16 - 9) = 7\pi$$
   $$= 22 \text{ sq. cm}$$

4. **Correct answer: A) 22 m**

Area of sector $= 231$sq. cm, radius $r = 21$m

$\frac{\theta}{360}\pi r^2 = 231 \Rightarrow \frac{\theta}{360} \times \frac{22}{7} \times 21^2 = 231$

$\Rightarrow \theta = 60°$

Arc length $= \frac{\theta}{360} \times 2\pi r$

$= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ m

5. **Correct answer: C) 128 cm**

Area of sector $= 924$sq. cm, radius $r = 42$cm

$\frac{\theta}{360}\pi r^2 = 924 \Rightarrow \frac{\theta}{360} \times \frac{22}{7} \times 42^2 = 924$

$\Rightarrow \theta = 60°$

Arc length $= \frac{60}{360} \times 2\pi r = 44$ cm

Perimeter $= 2r +$ arc length $= 84 + 44$

$= 128$ cm

6. **Correct answer: C) 28 cm**

Arc length $= 12$cm, radius $r = 8$cm

Perimeter of sector $=$ arc length $+ 2r$

$= 12 + 16 = 28$ cm

7. **Correct answer: A) 300 rotations**

Radius $r = 35$cm

Circumference of wheel $= 2\pi r$

$= 2 \times \frac{22}{7} \times 35 = 220$ cm

Distance $= 660$ m $= 660 \times 100 = 66000$ cm

Number of rotations $= \frac{66000}{220} = 300$

8. **Correct answer: B) 80°**

Given $\angle ABC = 69°$, $\angle ACB = 31°$

$\angle BAC = 180° - (69° + 31°) = 80°$

Angles subtended by the same arc are equal, so

$\angle BDC = \angle BAC = 80°$

9. **Correct answer: C) 352 cm**

Radius $r = 56$cm

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 56$$

$$= 352 \text{ cm}$$

10. **Correct Answer: B) $110°$**

Find $\angle CED$

Points A, E, C lie on a straight line.

So, $\angle BEC$ and $\angle CED$ form a linear pair.

$\angle CED = 180° - \angle BEC$

$\angle CED = 180° - 130° = 50°$

Find $\angle CDE$ **in** $\triangle CED$

Sum of angles of a triangle is $180°$.

$\angle CDE = 180° - (\angle CED + \angle ECD)$

$\angle CDE = 180° - (50° + 20°) = 110°$

Find $\angle BAC$

Angles in the same segment of a circle are equal.

$\angle BAC = \angle BDC = \angle CDE$

$\boxed{\angle BAC = 110°}$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer: A) 22 cm**

Area of a sector $= \frac{1}{2}r\ell$

Given: Area $= 88$ cm^2, $r = 8$ cm

$88 = \frac{1}{2} \times 8 \times \ell = 4\ell \Rightarrow \ell = 22$ cm

2. **Correct answer: B) 1232 sq. cm**

Perimeter of a semicircle $= \pi r + 2r$

Given: $\pi r + 2r = 144$

$r\left(\frac{22}{7} + 2\right) = 144 \Rightarrow r \cdot \frac{36}{7} = 144 \Rightarrow r$

$= 28$ cm

Area of a semicircle $= \frac{1}{2}\pi r^2$

$= \frac{1}{2} \times \frac{22}{7} \times 28^2 = 1232$ cm^2

3. **Correct answer: D) 4224 sq. m**

Area of ring $= \pi(R^2 - r^2)$

$= \frac{22}{7}(55^2 - 41^2) = \frac{22}{7}(3025 - 1681)$

$= \frac{22}{7} \times 1344$

$= 4224$ sq. m

4. **Correct answer: C) 1 : 4**

Angle of sector $= 90°$

Ratio $= \frac{90}{360} = \frac{1}{4}$

5. **Correct answer: B) 342 m**

Perimeter of sector $= 2r +$arc length

Arc length $= \frac{72}{360} \times 2\pi r$

$= \frac{1}{5} \times 2 \times \frac{22}{7} \times 105 = 132$ m

Perimeter $= 2(105) + 132 = 342$ m

6. **Correct answer: D) 88 cm**

In 60 minutes, minute hand completes one circle.

Distance $= 2\pi r = 2 \times \frac{22}{7} \times 14 = 88$ cm

7. Correct answer: D) 154 cm²

Side of square = 14cm ⇒ radius of each semicircle = 7cm

Area of square $= 14^2 = 196\ cm^2$

Two semicircles together form one circle of radius 7 cm.

Area of circle $= \pi r^2 = \frac{22}{7} \times 7^2 = 154\ cm^2$

Shaded area $= 196 - 154 = 42$ **white part)**

So shaded part (given in figure) $= 196 - 42 = 154 cm^2$

8. Correct answer: B) 1386 sq. m

Area of circular lawn $= \pi r^2 = 2464$

$$r^2 = \frac{2464 \times 7}{22} = 784 \Rightarrow r = 28 \text{ m}$$

Outer radius $= 28 + 7 = 35$m

Area of path$= \pi(35^2 - 28^2) = \frac{22}{7}(1225 - 784) = \frac{22}{7} \times 441 = 1386$ sq. m

9. Correct answer: C) 20 m & 6 m

Let outer radius $= R$, inner radius $= r$

Width $= R - r = 14$

Area of track

$= \pi(R^2 - r^2) = \pi(R - r)(R + r)$

$1144 = \frac{22}{7} \times 14 \times (R + r) \Rightarrow 1144 = 44(R + r) \Rightarrow R + r = 26$

Solve:

$R - r = 14, R + r = 26$

$R = 20, r = 6$

10. Correct answer: A) 576π

Let radii be $5x$ and $8x$

Difference of areas:

$\pi(8x)^2 - \pi(5x)^2 = \pi(64x^2 - 25x^2) = 39\pi x^2$

Given:$39\pi x^2 = 351\pi \Rightarrow x^2 = 9 \Rightarrow x = 3$

Bigger radius $= 8x = 24$

Area of bigger circle:$= \pi(24)^2 = 576\pi$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. Correct options: A and C

Circumference $= 176 = 2\pi r$

$$r = \frac{176}{2 \times \frac{22}{7}} = 28 \text{ m}$$

Area $= \pi r^2 = \frac{22}{7} \times 28^2 = 2464$ m²

Also, $2464 = 4 \times 616 = 2 \times 1232$

12. Correct Answer: B) $2\sqrt{\pi A}$, D) $\sqrt{4\pi A}$

Area of a circle:

$A = \pi r^2$

$r = \sqrt{\frac{A}{\pi}}$

Circumference of a circle:

$$C = 2\pi r = 2\pi\sqrt{\frac{A}{\pi}} = 2\sqrt{\pi A}$$

13. Correct options: B and D

Area of semicircle:

$\frac{1}{2}\pi r^2 = 1925 \Rightarrow r^2 = \frac{1925 \times 2 \times 7}{22} = 1225 \Rightarrow r = 35$ cm

Perimeter of semicircle:

$= \pi r + 2r = \frac{22}{7} \times 35 + 70 = 110 + 70 = 180$ cm

Also written as 2×90

14. Correct options: A and B

Length of arc:

$= \frac{240}{360} \times 2\pi r = \frac{2}{3} \times 2 \times \frac{22}{7} \times 7 = \frac{88}{3} = 29\frac{1}{3}$ cm

15. Correct option: C

Area of ring:

$= \pi(R^2 - r^2) = \frac{22}{7}(20^2 - 15^2) = \frac{22}{7}(400 - 225) = \frac{22}{7} \times 175 = 550\text{cm}^2$

ASSERTION AND REASON TYPE

16. Correct Answer: C) Assertion is true, but Reason is false.

Assertion (A): The diameter of a circle is twice the radius – **True**

Reason (R): The radius is the longest chord of a circle – **False**

(The **diameter** is the longest chord, not the radius.)

17. Correct Answer: C) Assertion is true, but Reason is false.

Assertion (A): The perpendicular from the center of a circle to a chord bisects the chord – **True**

Reason (R): Every chord of a circle is equidistant from the center – **False**

(Different chords are at different distances from the center.)

18. Correct Answer: C) Assertion is true, but Reason is false.

Assertion (A): All points on a circle are equidistant from the center — **True**

Reason (R): The distance between the center and any point on the circumference is called the diameter — **False**
(It is called the **radius**, not the diameter.)

19. Correct Answer: (B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

Assertion (A): The sum of the opposite angles of a cyclic quadrilateral is 180°.(True.)

Reason (R): The exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.(True.)

Explanation: Both statements are true, but the Reason does **not** explain the Assertion. It is a related property, not the cause of the opposite angles summing to 180°.

20. Correct Answer: (C) Assertion is true, but Reason is false.

Assertion (A): A tangent to a circle is perpendicular to the radius at the point of contact.(**True)**

Reason (R): The tangent to a circle at any point is parallel to the corresponding chord passing through that point.(**False)**
(A tangent is **not** parallel to any chord through the point of contact.)

MATRIX MATCHING TYPE

21. Correct Answer:

a) Circumference of a semi-circle
=half of circumference +diameter $= \pi r + 2r \rightarrow$ **P**

b) Area of a circle
$= \pi r^2 \rightarrow$ T

c) Diameter of a circle
$= 2r \rightarrow$ S

d) Area of a semi-circle
$= \frac{1}{2}\pi r^2 \rightarrow$ Q

22. Correct Answer:

a) Area of sector
$= \frac{x^\circ}{360^\circ} \times \pi r^2 \rightarrow$ **R**

b) Length of an arc of a sector
$= \frac{x^\circ}{360^\circ} \times 2\pi r \rightarrow$ S

c) Perimeter of a sector
=arc length $+2r = \ell + 2r \rightarrow$ **P**

d) If ℓ is length of arc then area is
$= \frac{1}{2}\ell r \rightarrow$ Q

COMPREHENSION TYPE

Passage – I:

Given:
Radius $r = 8.4$cm, angle $\theta = 135^\circ$

23. Correct answer: B) 19.8 cm

Length of arc

$$\ell = \frac{\theta}{360^\circ} \times 2\pi r = \frac{135}{360} \times 2 \times \frac{22}{7} \times 8.4$$
$$= 19.8 \text{ cm}$$

24. Correct answer: C) 83.16 sq.cm

Area of sector

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{135}{360} \times \frac{22}{7} \times (8.4)^2$$
$$= 83.16 \text{ sq.cm}$$

25. Correct answer: A) 52.8 cm, 16.8 cm

Circumference and diameter

$$\text{Diameter} = 2r = 16.8 \text{ cm}$$

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 8.4$$
$$= 52.8 \text{ cm}$$

Passage – II:

26. Correct answer: B

Cost of fencing a semicircular garden

Radius $r = 7$m

Perimeter of semicircle =curved part +diameter

$$= \pi r + 2r = \frac{22}{7} \times 7 + 14 = 22 + 14$$
$$= 36 \text{ m}$$

$$\text{Cost} = 36 \times 10 = 360$$

27. Correct answer: A

Ratio of circumference to arc length

Angle of sector $= 72°$

$$\text{Arc length} = \frac{72}{360} \times \text{circumference}$$

$$= \frac{1}{5} \times \text{circumference}$$

$$\text{Ratio} = \text{circumference:arc length} = 1:\frac{1}{5}$$

$$= \boxed{5:1}$$

28. Correct answer: A) 5.5 cm

Length of arc

Circumference $= 44$cm, angle $= 45°$

$$\text{Arc length} = \frac{45}{360} \times 44 = \frac{1}{8} \times 44 = 5.5 \text{ cm}$$

INTEGER TYPE

29. Correct answer:

Width of a ring

Outer radius $= 7$cm, inner radius $= 2$cm

$$\text{Width} = 7 - 2 = \boxed{5 \text{ cm}}$$

30. Correct answer:

$\frac{1}{44}$times circumference

Radius $r = 56$cm

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 56$$

$$= 352 \text{ cm}$$

$$\frac{1}{44} \times 352 = \boxed{8 \text{ cm}}$$

31. Correct answer:

$\frac{1}{72}$times perimeter of semicircle

Radius $r = 28$cm

$\text{Perimeter of semicircle} = \pi r + 2r =$

$\frac{22}{7} \times 28 + 56 = 88 + 56 = 144$cm

$$\frac{1}{72} \times 144 = \boxed{2 \text{ cm}}$$

32. Correct answer:

Wire bent to form a circle

Length of wire =circumference

$$\boxed{9 \text{ units}}$$

33. Correct answer:

Find $\boldsymbol{k}$

Area of sector $= 924$sq.cm, $r = 42$cm

$$\frac{x}{360} \times \pi r^2 = 924 \Rightarrow \frac{x}{360} \times \frac{22}{7} \times 42^2 = 924$$

$$\Rightarrow x = 60°$$

Arc length

$$\ell = \frac{60}{360} \times 2\pi r = 44 \text{ cm}$$

$\text{Perimeter of sector} = 44 + 2(42) = 128$

$$120 + k = 128 \Rightarrow \boxed{k = 8}$$

34. Correct answer:

Find $\boldsymbol{k}$

$$2\pi r = 132, r = \frac{105}{k}, \pi = \frac{22}{7}$$

$$2 \times \frac{22}{7} \times \frac{105}{k} = 132 \Rightarrow \frac{660}{k} = 132$$

$$\Rightarrow \boxed{k = 5}$$

PREVIOUS YEAR QUESTIONS

1. Correct answer: (C) 115°

In a circle, an angle subtended by an arc at the centre is twice the angle subtended at the circumference.

∠CAD = 40° subtends arc **CD**.
So, arc CD = 2 × 40° = 80°.

∠BDC = 25° subtends arc **BC**.
So, arc BC = 2 × 25° = 50°.

Angle ∠BCD subtends arc **BAD**.
Arc BAD = 360° − (arc BC + arc CD)
= 360° − (50° + 80°)
= 230°.

∠BCD = ½ × 230° = **115°**.

2. Correct Answer: (A) 70°

Since **AB** is a diameter,
∠ACB = 90° (Angle in a semicircle)
Given ∠COD = 40°.
The angle subtended by arc **CD** at the circumference is
∠CAD = ½ × 40° = 20°.
At point **E**, ∠CED is an exterior angle formed by two secants.
Exterior angle formed by two secants = ½ (difference of intercepted arcs).
Intercepted arcs correspond to angles:
Larger angle = 90°
Smaller angle = 20°
∠CED = 90° − 20° = **70°**

3. Correct Answer: (C) 56°

Given: $\angle BAC = 56°$, O is the centre.
$\angle BAC$ is an angle at the circumference subtending arc BC.
So the central angle on the same arc,
$\angle BOC = 2\angle BAC = 2 \times 56° = 112°$.
$\angle BDC$ is also an angle at the circumference subtending arc BC.
Hence, $\angle BDC = \frac{1}{2}\angle BOC = \frac{1}{2} \times 112° = 56°$.

4. Correct Answer: B

Given:

$OA = OB = OC =$ 5cm (radii)

$AB = AC =$ 6cm

OA meets chord BC at P

Since $AB = AC$ and $OB = OC$, line OA is the **perpendicular bisector** of chord BC.

$\Rightarrow OA \perp BC$ and $BP = PC$.

Let $OP = x$.

Then $AP = 5 - x$.

Let $BP = y$.

Apply Pythagoras

In $\triangle OBP$: $x^2 + y^2 = 5^2 = 25$ (1)

In $\triangle ABP$: $(5 - x)^2 + y^2 = 6^2 = 36$ (2)

Subtract (1) from (2)

$(5 - x)^2 - x^2 = 36 - 25$

$25 - 10x = 11$

$10x = 14$

$x = 1.4$

$\boxed{OP = 1.4 \text{ cm}}$

5. Correct Answer: (C) Statement-I is true but Statement-II is false.

In a cyclic quadrilateral, opposite angles are supplementary, so ∠A + ∠C = 180°.

Given ∠A − ∠C = 60°. Solving both equations gives ∠A = 120° and ∠C = 60°.

Hence, the smaller angle is 60°, so **Statement-I is true**.

Statement-II is incorrect because the angle at the centre is **twice** the angle at the circumference, not half.

So, **Statement-II is false**.

6. Correct Answer: (D) 95°

∠BAO = 50° and ∠BCO = 35°.

Since **OA = OB** and **OB = OC**, triangles **OAB** and **OBC** are isosceles.

So, ∠ABO = 50° and ∠OBC = 35°.

Thus, ∠ABC = 50° + 35° = 85°.

Angle ∠ADC subtends arc **AC** opposite to ∠ABC.

Hence, ∠ADC = 180° − 85° = **95°**.

7. Correct Answer: (A) T, F, T

Statement (i) is true because the angle at the centre is twice the angle at the circumference.

Statement (ii) is false because the angle in a semicircle is a right angle, not acute.

For statement (iii), half of the chord = $\sqrt{(10^2 - 6^2)} = \sqrt{64} = 8$ cm, so the full chord = 16 cm, hence it is true.

8. Correct Answer: (A) (70°, 80°)

Since ABCD is a cyclic quadrilateral, the exterior angle is equal to the interior opposite angle.

∠BCX is an exterior angle at C

$\Rightarrow$ ∠BCX = ∠BAD

$\Rightarrow$ ∠BAD = 70°

Hence, x = 70°

∠ADX is an exterior angle at D

$\Rightarrow$ ∠ADX = ∠ABC

$\Rightarrow$ ∠ABC = 80°

Hence, y = 80°

9. Correct Answer:(C) 20°

Find ∠APC

∠BPD and ∠APC are vertically opposite angles.

$\angle APC = \angle BPD = 120°$

Find ∠CAD

In $\triangle APC$,

$\angle ACP = 40°, \angle APC = 120°$

$\angle CAD = \angle PAC = 180° - (120° + 40°) = 20°$

Use Angles in the Same Segment Theorem

$\angle CAD$ and $\angle CBD$ subtend the same arc CD.

$\angle CBD = \angle CAD = 20°$

10. Correct Answer(D) 17 cm

The perpendicular from the centre to a chord bisects the chord.

Distance of chord AB from centre

$\frac{AB}{2} = 5$ cm

$OQ = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$ cm

Distance of chord CD from centre

$\frac{CD}{2} = 12$ cm

$OP = \sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5$ cm

Find PQ

$PQ = OP + OQ = 5 + 12 = 17$ cm

12. Surface Areas and Volumes

Solutions

Worksheet – 1

1. **Correct answer: (B) 141 cm**
   The prism has two triangular bases and 3 vertical edges.
   Perimeter of triangular base = 12 + 16 + 20 = 48 cm
   Total edges of both bases = 2 × 48 = 96 cm
   Vertical edges = 3 × height = 3 × 15 = 45 cm
   Total length of all sides = 96 + 45 = 141 cm
2. **Correct Answer: (A) $192\sqrt{3}$c.c**
   Area of equilateral triangle
   $= \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times 8^2 = 16\sqrt{3}\text{cm}^2$
   Volume = Base area × height $= 16\sqrt{3} \times 12 = 192\sqrt{3}$ c.c
3. **Correct Answer: (B) 43008 cm³**
   Area of rhombus$= \frac{1}{2}d_1d_2 = \frac{1}{2} \times 28 \times 96 = 1344\text{cm}^2$
   Volume = Base area × height$= 1344 \times 32 = 43008\ \text{cm}^3$
4. **Correct Answer: (B) 384 cm²**
   Total Surface Area of a Cube
   Formula:
   TSA $= 6a^2$
   $= 6 \times 8^2 = 6 \times 64 = 384\text{cm}^2$
5. **Correct Answer: (D) $10\sqrt{3}$cm**
   Length of Diagonal of a Cube
   Surface area = $6a^2 = 600$
   $a^2 = 100 \Rightarrow a = 10$ cm
   Diagonal of cube$= a\sqrt{3} = 10\sqrt{3}$ cm
6. **Correct Answer: (C) 300%**
   Let the original edge be a.
   Original surface area$= 6a^2$
   Edge increased by 100% ⇒ new edge $= 2a$
   New surface area$= 6(2a)^2 = 24a^2$
   Increase in surface area$= 24a^2 - 6a^2 = 18a^2$
   Percentage increase$= \frac{18a^2}{6a^2} \times 100 = 300\%$
7. **Correct Answer:(C) 1 : 9**
   Let the edges be $a$and b.
   Given ratio of volumes
   $a^3 : b^3 = 1 : 27$
   $a : b = 1 : 3$
   Area of one face $\propto a^2$
   Required ratio$= 1^2 : 3^2 = 1 : 9$
8. **Correct Answer: (D) 216 cm³**
   Volume of a cube
   Diagonal of a cube $= a\sqrt{3}$
   Given:
   $a\sqrt{3} = 6\sqrt{3}$
   $a = 6$ cm
   Volume of cube$= a^3 = 6^3 = 216\text{cm}^3$
9. **Correct Answer: (B) 240 cm³**
   Volume of a cuboid
   Volume $= l \times b \times h$
   $= 8 \times 6 \times 5 = 240\ \text{cm}^3$
10. **Correct Answer: (B) 36 cm²**
    Area of the base of a cuboid
    Area of base $= l \times b = 9 \times 4 = 36\ \text{cm}^2$

Worksheet - 2

1. **Correct answer:**
   Both cylinders have the same base ⇒ same radius.
   Volume of a cylinder ∝ height.
   Heights = 15 cm and 20 cm
   Ratio of volumes $= 15 : 20 = 3 : 4$
   Correct Answer: (C) 3 : 4
2. **Correct answer: (C) 14 m**
   Diameter of a cylindrical pillar
   Curved surface area$= 2\pi rh = 264$
   Volume$= \pi r^2 h = 924$
   Divide volume by CSA:$\frac{\pi r^2 h}{2\pi rh} = \frac{924}{264} \Rightarrow \frac{r}{2} = 3.5 \Rightarrow r = 7$ m
   Diameter $= 2r = 14$m

3. **Correct Answer: (C) 462 cm³**
Volume of the cylinder formed by rolling paper
Length of paper = circumference of base
$$2\pi r = 22 \Rightarrow r = 3.5 \text{ cm}$$
Height of cylinder = breadth = 12 cm
$$\text{Volume} = \pi r^2 h = \frac{22}{7} \times (3.5)^2 \times 12$$
$$= 462 \text{ cm}^3$$

4. **Correct Answer: (B) $\sqrt{2}:1$**
Given: Volumes are equal
$$\pi r_1^2 h_1 = \pi r_2^2 h_2$$
Heights ratio $= 1:2$
$r_1^2 : r_2^2 = 2:1 \Rightarrow r_1 : r_2 = \sqrt{2}:1$

5. **Correct Answer: (A) 4950 cm²**
Curved surface area of a cone
Volume $= 33.264$litres $= 33264\text{cm}^3$
$$\frac{1}{3}\pi r^2 h = 33264$$
$$\frac{1}{3} \times \frac{22}{7} \times r^2 \times 72 = 33264 \Rightarrow r = 21 \text{ cm}$$
Slant height
$$l = \sqrt{r^2 + h^2} = \sqrt{21^2 + 72^2} = 75 \text{ cm}$$
Curved surface area$= \pi r l = \frac{22}{7} \times 21 \times 75 = 4950 \text{ cm}^2$

6. **Correct Answer: (A) 10 cm**
Curved surface area of cone$= \pi r l$
Given: $\pi r l = 355, r = 11.3$ cm
$$l = \frac{355}{\pi \times 11.3} = \frac{355}{\frac{22}{7} \times 11.3} = 10 \text{ cm}$$

7. **Correct Answer: (C) 48 sq.cm**
Curved surface area$= \pi(R + r)l$
Given perimeters:
$$2\pi R = 18 \Rightarrow R = \frac{9}{\pi}, 2\pi r = 6 \Rightarrow r = \frac{3}{\pi}$$
$$R + r = \frac{12}{\pi}, l = 4$$
$$\text{CSA} = \pi \times \frac{12}{\pi} \times 4 = 48 \text{ cm}^2$$

8. **Correct Answer: (C) 1206.86 cm²**
Total surface area of a cone
Slant height $l = \sqrt{r^2 + h^2} = \sqrt{12^2 + 16^2} = 20$ cm
Total surface area
$$= \pi r(r + l) = \pi \times 12 \times (12 + 20)$$
$$= 1206.86\text{cm}^2$$

9. **Correct Answer: (A) 5.385 cm**
Base radius $R = 5$cm, top radius $r = 3$cm
Height $h = 5$cm
$$l = \sqrt{h^2 + (R - r)^2} = \sqrt{5^2 + 2^2} = \sqrt{29}$$
$$= 5.385 \text{ cm}$$

10. **Correct Answer: (B) 10 cm**
Slant height of a cone
Surface area of cone$= \pi r(r + l)$
Given: $148.5 = \frac{22}{7} \times 3.5 \times (3.5 + l)$
$$148.5 = 11 \times (3.5 + l) \Rightarrow 3.5 + l = 13.5$$
$$\Rightarrow l = 10 \text{ cm}$$

Worksheet -3

1. **Correct Answer: (B) 4 : 9**
Ratio of surface areas of two spheres
Surface area of a sphere $\propto r^2$
Given radii ratio $= 2:3$
Surface area ratio $= 2^2 : 3^2 = 4:9$

2. **Correct Answer: (B) 616 cm²**
Surface area of a sphere
Surface area $= 4\pi r^2$
$$= 4 \times \frac{22}{7} \times 7^2 = 616 \text{ cm}^2$$

3. **Correct Answer: (C) 21 cm**
Radius of the sphere
$$4\pi r^2 = 5544$$
$$4 \times \frac{22}{7} \times r^2 = 5544 \Rightarrow r^2 = 441 \Rightarrow r$$
$$= 21 \text{ cm}$$

4. **Correct Answer: (C) 4851.3 cm³**
Volume of a sphere
$$\text{Volume} = \frac{4}{3}\pi r^3$$
$$= \frac{4}{3} \times \frac{22}{7} \times (10.5)^3 = 4851.3 \text{ cm}^3$$

5. **Correct Answer: (C) Rs 18480**
Cost of painting hemispherical top
Perimeter of hemisphere base
$$2\pi r = 88 \Rightarrow r = 14 \text{ m}$$
Curved surface area of hemisphere
$$= 2\pi r^2 = 2 \times \frac{22}{7} \times 14^2 = 1232 \text{ m}^2$$
Cost of painting$= 1232 \times 15 = \text{Rs } 18480$

6. **Correct Answer: (D) 5 cm and 4 cm**
Total surface area of a hemisphere $= 3\pi r^2$
Given ratio of TSAs
$$r_1^2 : r_2^2 = 25:16 \Rightarrow r_1 : r_2 = 5:4$$
Sum of radii
$$5x + 4x = 9 \Rightarrow 9x = 9 \Rightarrow x = 1$$
Radii = 5cm and 4cm

7. Correct Answer: (C) 1306.9025 cm²

Given:

Outer radius $R = 8$cm, inner radius $r = 4$cm

For one hollow hemisphere, total surface area

$=$ outer CSA + inner CSA + base ring area

$= 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2) = 3\pi R^2 + \pi r^2$

For two hollow hemispheres:$= 2(3\pi R^2 + \pi r^2) = 6\pi R^2 + 2\pi r^2$

Substitute values:

$$= 6\pi(8^2) + 2\pi(4^2) = 6\pi(64) + 2\pi(16)$$
$$= 384\pi + 32\pi = 416\pi$$
$$= 416 \times \frac{22}{7} = 1306.9 \text{ cm}^2$$

8. Correct Answer: (B) h: r

Curved surface area of cylinder$= 2\pi rh$

Curved surface area of hemisphere$= 2\pi r^2$

Required ratio$= 2\pi rh : 2\pi r^2 = h : r$

9. Correct Answer: (C) 1039.5 cm²

Given circumference of base

$$2\pi r = 66 \Rightarrow r = \frac{66}{2\pi} = \frac{66}{\frac{44}{7}} = 10.5 \text{ cm}$$

Total surface area of a hemisphere

$$= 3\pi r^2 = 3 \times \frac{22}{7} \times (10.5)^2 = 1039.5 \text{ cm}^2$$

10. Correct Answer: (B) 616 cm²

Given:Surface area of a sphere = 1232 cm²

We know that:

Surface area of a sphere = 4πr²

Curved surface area of a hemisphere = 2πr²

So, curved surface area of hemisphere is half the surface area of the sphere.

Curved surface area of hemisphere

$$= \frac{1}{2} \times 1232 = 616 \text{ cm}^2$$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: (A) 216 cm²

Given: Edges of cubes = 3 cm, 4 cm, 5 cm

Total volume = $3^3 + 4^3 + 5^3 = 27 + 64 + 125 = 216$ cm³

Let edge of new cube = a

$$a^3 = 216 \Rightarrow a = 6 \text{ cm}$$

Surface area = $6a^2 = 6 \times 6^2 = 216$ cm²

2. Correct Answer: (B) 20 cm, 100 cm

Volume of cube = $10^3 = 1000$ cm³

Let sides of sheet be $x$and $5x$, thickness = 0.5 cm

Volume = $x \times 5x \times 0.5 = 2.5x^2$

$$2.5x^2 = 1000 \Rightarrow x^2 = 400 \Rightarrow x = 20$$

Sides = 20 cm and 100 cm

3. Correct Answer: (C) 3 : 1

Volumes ratio = 27 : 1

Edge ratio = cube root of volume ratio

$$\sqrt[3]{27} : \sqrt[3]{1} = 3 : 1$$

4. Correct Answer: (A) 6 cm

Total volume = $125 + 64 + 27 = 216$ cm³

Let edge of new cube = a

$$a^3 = 216 \Rightarrow a = 6 \text{ cm}$$

5. Correct Answer: (B) 1440 cm²

Two cubes of side 12 cm joined end to end → Cuboid dimensions:

$$24 \times 12 \times 12$$

Surface area =

$$2(lb + bh + lh)$$
$$= 2(24 \times 12 + 12 \times 12 + 24 \times 12)$$
$$= 2(288 + 144 + 288) = 1440 \text{ cm}^2$$

6. Correct Answer: (C) 12 cm

Given:

Dimensions of cuboid are in the ratio 1 : 2 : 3

Volume = 384 cm³

Let dimensions be x, 2x, 3x

Volume of cuboid:

$$x \times 2x \times 3x = 6x^3$$
$$6x^3 = 384$$
$$x^3 = 64 \Rightarrow x = 4$$

Longest side = $3x = 3 \times 4 = 12$cm

7. Correct Answer: (C) 8 cm

Volume = Base area × Height

$$480 = 60 \times h$$

$h = 480 \div 60 = 8$cm

8. Correct Answer: (B) 4.4 m²

Given:

Radius $r = 0.7$m, Height $h = 1$m

Curved surface area = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 0.7 \times 1 = 4.4 \, m^2$$

9. Correct Answer: (A) Rs. 4125

Given:
Height = 15 m
Diameter = 350 cm = 3.5 m
Radius = 1.75 m

Curved surface area = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 1.75 \times 15 = 165\, m^2$$

Cost = $165 \times 25 = Rs. 4125$

10. Correct Answer: (D) 748 cm²

Given:
Radius = 7 cm, Height = 10 cm

Area of metal sheet (closed cylinder)
= Curved surface area + Area of two bases

$$= 2\pi rh + 2\pi r^2$$

$$= 2 \times \frac{22}{7} \times 7 \times 10 + 2 \times \frac{22}{7} \times 7^2$$

$$= 440 + 308 = 748\, cm^2$$

11. Correct Answer: (C) 308 cm³

Given:
Lateral surface area (LSA) = 176 cm²
Base area = 38.5 cm²

Base area = πr^2

$$\frac{22}{7} r^2 = 38.5 \Rightarrow r^2 = 12.25 \Rightarrow r = 3.5 \text{ cm}$$

LSA = $2\pi rh$

$$176 = 2 \times \frac{22}{7} \times 3.5 \times h \Rightarrow h = 8 \text{ cm}$$

Volume = $\pi r^2 h = 38.5 \times 8 = 308$ cm³

12. Correct Answer: (B) 250.25 cm³

Given:
Total surface area (TSA) = 220 cm²
Height = 6.5 cm

$$\text{TSA} = 2\pi r(r + h)$$

$$220 = 2 \times \frac{22}{7} \times r(r + 6.5)$$

Solving gives:

$$r = 3.5 \text{ cm}$$

Volume

$$= \pi r^2 h = \frac{22}{7} \times 3.5^2 \times 6.5 = 250.25 \text{ cm}^3$$

13. Correct Answer: (A) 385 cm³

Given:
Paper size = 22 cm × 10 cm

Rolled along length ⇒ circumference = 22 cm

$$2\pi r = 22 \Rightarrow r = 3.5 \text{ cm}$$

Height = 10 cm

Volume

$$= \pi r^2 h = \frac{22}{7} \times 3.5^2 \times 10 = 385 \text{ cm}^3$$

14. Correct Answer: (B) 440 cm²

Given:
Radius = 7 cm
Height = 10 cm

Curved surface area

$$= 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 10 = 440 \text{ cm}^2$$

15. Correct Answer: (C) 15 cm

Given:
Volume of cuboid = $44 \times 30 \times 15 = 19800$ cm³

Cylinder height = 28 cm

$$\pi r^2 h = 19800$$

$$\frac{22}{7} \times r^2 \times 28 = 19800$$

$$88r^2 = 19800 \Rightarrow r^2 = 225 \Rightarrow r = 15 \text{ cm}$$

16. Correct Answer: (C) 36 m³

Given:
Circumference of base = 12 m
Height = πm

$$\text{Circumference} = 2\pi r$$

$$2\pi r = 12 \Rightarrow r = \frac{6}{\pi}$$

Volume of cylinder

$$= \pi r^2 h$$

$$= \pi \left(\frac{6}{\pi}\right)^2 \times \pi$$

$$= \pi \times \frac{36}{\pi^2} \times \pi = 36 \text{ m}^3$$

17. Correct Answer: (C) 220 cm²

Given:
Diameter = 14 cm ⇒ Radius = 7 cm
Slant height $l = 10$cm

Curved surface area of cone

$$= \pi rl = \frac{22}{7} \times 7 \times 10 = 220 \text{ cm}^2$$

18. Correct Answer: (B) 2816 cm²

Given:

Area of base = 616 cm²

$\pi r^2 = 616 \Rightarrow r^2 = 196 \Rightarrow r = 14$ cm

Height $h = 48$cm

Slant height

$l = \sqrt{r^2 + h^2} = \sqrt{196 + 2304} = \sqrt{2500}$
$= 50$ cm

Total surface area

$= \pi r(r + l) = \frac{22}{7} \times 14 \times (14 + 50)$
$= 2816$ cm²

19. Correct Answer : (B) 1232 cm³

Given:

Rate of painting = 10 paise per cm² = Rs. 0.10 per cm²

Total cost = Rs. 55

Slant height $l = 25$cm

CSA $= \frac{55}{0.10} = 550$ cm²

$\pi rl = 550$

$\frac{22}{7} \times r \times 25 = 550$

$r = \frac{550 \times 7}{550 \times 25/25} = 7$ cm

$h = \sqrt{l^2 - r^2} = \sqrt{25^2 - 7^2} = \sqrt{625 - 49}$
$= \sqrt{576} = 24$ cm

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 7^2 \times 24$

$V = 1232$ cm³

20. Correct Answer: B) 141.3 m²

Radius $r = 3$m (since diameter = 6 m)

Height of cone $= 9 - 5 = 4$m

Slant height of cone

$l = \sqrt{3^2 + 4^2} = 5$ m

Curved surface area of cone

$\pi rl = 3.14 \times 3 \times 5 = 47.1$ m²

Curved surface area of cylinder

$2\pi rh = 2 \times 3.14 \times 3 \times 5 = 94.2$ m²

Total canvas required

$47.1 + 94.2 =$ 141.3 m²

21. Correct Answer: (C) 15 : 16

Volume of cone

$= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 9 \times 5 = 15\pi$

Volume of cylinder

$= \pi r^2 h = \pi \times 4 \times 4 = 16\pi$

Ratio

$= 15\pi : 16\pi = 15 : 16$

22. Correct answer: (C) 16π cm³

Right triangle rotated about side 3 cm

Height $h = 3$cm, Radius $r = 4$cm

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 16 \times 3 = 16\pi$

23. Correct answer: (D) 8 times

Volume of cone $V \propto r^2 h$

If $r$and $h$are doubled:

$V \propto (2r)^2(2h) = 8r^2 h$

24. Correct Answer: 1232 cm³

Given:

Curved surface area $= 550$ cm²

Area of base $= 154$ cm²

Take $\pi = \frac{22}{7}$

$\pi r^2 = 154 \Rightarrow \frac{22}{7} r^2 = 154 \Rightarrow r^2 = 49 \Rightarrow r$
$= 7$ cm

$\pi rl = 550 \Rightarrow \frac{22}{7} \times 7 \times l = 550 \Rightarrow 22l$
$= 550 \Rightarrow l = 25$ cm

$h = \sqrt{l^2 - r^2} = \sqrt{25^2 - 7^2} = \sqrt{576}$
$= 24$ cm

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 49 \times 24$
$= 1232$ cm³

25. Correct answer: (C) 13860 cm³

Circumference $= 2\pi r = 66 \Rightarrow r = 10.5$cm

CSA $= 2\pi rh = 2640 \Rightarrow h = 40$cm

$V = \pi r^2 h = \frac{22}{7} \times (10.5)^2 \times 40 = 13860$

26. Correct answer: (B) 4620 m³

Given: $r + h = 37$, TSA $= 2\pi r(r + h) = 1628$

$2\pi r(37) = 1628 \Rightarrow r = 7, h = 30$

$V = \pi r^2 h = \frac{22}{7} \times 49 \times 30 = 4620$

27. Correct answer: (B) 37.9 m

Diameter of sphere = 8 cm ⇒ radius = 4 cm

Volume of sphere = $\frac{4}{3}\pi r^3 = \frac{4}{3}\pi(4)^3 = \frac{256}{3}\pi$

Wire radius = 1.5 mm = 0.15 cm

Volume of wire = $\pi r^2 h = \pi(0.15)^2 h$

Equating volumes:

$\frac{256}{3}\pi = \pi(0.15)^2 h$

$h = 3790$ cm $= 37.9$ m

28. Correct answer: (D) 44 m³

Well is cylindrical

Radius = 1 m, depth = 14 m

Volume = $\pi r^2 h = \pi(1)^2(14) = 14\pi = 44$ m^3

29. Correct answer: (C) 28 cm

Diameter of sphere = 14 cm ⇒ radius = 7 cm

Volume of sphere = $\frac{4}{3}\pi(7)^3 = \frac{1372}{3}\pi$

Height of cylinder = $2\frac{1}{3} = \frac{7}{3}$cm

Let radius of cylinder = R

$$\pi R^2 \times \frac{7}{3} = \frac{1372}{3}\pi$$

$$R^2 = 196 \Rightarrow R = 14$$

Diameter = 28 cm

30. Correct answer: (A) 3.5 cm

Volume of hemisphere = volume of cone

$$\frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2 h$$

$$2r = h$$

$r = \frac{7}{2} = 3.5$cm

31. Correct answer: (B) 462 cm²

Diameter = 14 cm ⇒ radius = 7 cm

Total surface area of hemisphere = $3\pi r^2$

$= 3\pi(7)^2 = 147\pi = 462$ cm^2

ONE OR MORE THAN ONE CORRECT OPTION TYPE

32. Correct answer: (B) 4 : 9, (D) (D) 8:18

For cubes, volume ∝ (side)^3.

Given volume ratio = 8 : 27 ⇒ side ratio = 2 : 3.

Surface area ∝ (side)^2 ⇒ ratio = 4 : 9.

33. Correct answer: (A) 240 cm²

Let length = l, breadth = b, height = h.

l + b + h = 19

Diagonal² = l² + b² + h² = 11² = 121

(l + b + h)² = l² + b² + h² + 2(lb + bh + hl)

19² = 121 + 2(lb + bh + hl)

361 − 121 = 2(lb + bh + hl)

lb + bh + hl = 120

Surface area = 2(lb + bh + hl) = 240 cm².

34. Correct answer: (B) 1 : 4, (C) 2:8

Volume of cylinder ∝ r²h.

If radius is halved: new volume ∝ (r/2)²h = r²h/4.

Ratio = 1 : 4.

35. Correct answer: (B) 462 cm², (C) 6×77cm²

Radius r = 7 cm.

Total surface area of solid hemisphere = 3πr²

= 3 × 22/7 × 7² = 462 cm².

36. Correct Answer: (B) 6.75 cm, (D) $\frac{27}{4}$ cm

Given:

Diameter of sphere = 6 cm ⇒ radius $r =$ 3cm

Diameter of cone base = 8 cm ⇒ radius $R = 4$cm

Volume of sphere $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3)^3 = 36\pi$

Volume of cone $= \frac{1}{3}\pi R^2 h = \frac{1}{3}\pi(4)^2 h = \frac{16}{3}\pi h$

Since the sphere is melted and recast into the cone, volumes are equal:

$$36\pi = \frac{16}{3}\pi h$$

$$h = \frac{36 \times 3}{16} = \frac{27}{4} = 6.75 \text{ cm}$$

37. Correct answer: (A) H = 2R

Volume of hemisphere = volume of cone

Hemisphere: $\frac{2}{3}\pi R^3$

Cone: $\frac{1}{3}\pi R^2 H$

$$\frac{2}{3}\pi R^3 = \frac{1}{3}\pi R^2 H$$

⇒ H = 2R.

ASSERTION AND REASON TYPE

38. Correct answer: (A)

For a prism, all lateral faces are rectangles with height = height of prism and widths equal to the sides of the base.

So, lateral surface area = (sum of widths of rectangles) × height = perimeter of base × height.

Assertion (A): True

Reason (R): True and explains A

39. Correct answer: (A)

Volume of cylinder = $\pi r^2 h$

Volume of cone = $\frac{1}{3}\pi r^2 h$

Hence, the cone's volume is one-third of the cylinder's volume with the same base and height.

Assertion (A): True

Reason (R): True and explains A

40. Correct answer: (A)

Surface area of sphere = $4\pi r^2$

Area of circle with same radius = πr^2

So, sphere's surface area = 4 times the circle's area.

Assertion (A): True

Reason (R): True and explains A

41. Correct answer: (C)

Volume of sphere = $\frac{4}{3}\pi r^3$

Volume of hemisphere = $\frac{1}{2}\times\frac{4}{3}\pi r^3 = \frac{2}{3}\pi r^3$

Given reason states hemisphere volume = $\frac{1}{3}\pi r^3$, which is incorrect.

Assertion (A): True

Reason (R): False

MARIX MATCH TYPE

42. Correct answer: a–Q, b–P, c–S, d–R

a) Cube side = 5 cm

Total length of all edges = $12 \times 5 = 60$cm → Q

b) Cuboid base area = $8 \times 6 = 48$cm² → P

c) Cube side = 7 cm

Area of one face = $7^2 = 49$cm² → S

d) Space diagonal

$\sqrt{2^2+3^2+6^2} = \sqrt{49} = 7$ cm→ R

43. Correct answer: A–Q, B–P, C–S, D–R

A) Curved surface area of cylinder = $2\pi rh$→ Q

B) Volume of cylinder = $\pi r^2 h$→ P

C) Volume of cone = $\frac{1}{3}\pi r^2 h$→ S

D) Total surface area of sphere = $4\pi r^2$→ R

COMPREHENSION TYPE

Passage-I

44. Correct answer: (C) 240 cm³

Base area = $6 \times 4 = 24$cm²

Volume = Base area × height = $24 \times 10 = 240$cm³

45. Correct answer: (C) 6 cm

Volume of cube = $a^3 = 216$

$\Rightarrow a = \sqrt[3]{216} = 6$cm

46. Correct answer: (C) 280 cm³

Volume of cuboid = Base area × height

$= 40 \times 7 = 280$cm³

Passage-II

Given:

Radius $r = 7$cm.

Volume of cone = $\frac{2}{3}$volume of hemisphere.

Find height of cone

$$V_{\text{hemisphere}} = \frac{2}{3}\pi r^3, V_{\text{cone}} = \frac{1}{3}\pi r^2 h$$

$$\frac{1}{3}\pi r^2 h = \frac{2}{3}\times\frac{2}{3}\pi r^3 = \frac{4}{9}\pi r^3$$

$$h = \frac{4}{3}r = \frac{4}{3}\times 7 = \frac{28}{3}\text{ cm}$$

47. Correct answer : (B) $\frac{35}{3}$cm

Slant height of cone

$$l = \sqrt{r^2+h^2} = \sqrt{7^2+\left(\frac{28}{3}\right)^2} = \sqrt{\frac{1225}{9}}$$

$$= \frac{35}{3}\text{ cm}$$

48. Correct answer : (B) 308 cm²

Curved surface area of hemisphere

CSA $= 2\pi r^2 = 2\pi(7)^2 = 98\pi = 308$ cm²

49. Correct answer : (C) 256.66 cm²

Curved surface area of cone

CSA $= \pi rl = \pi \times 7 \times \frac{35}{3} = \frac{245\pi}{3} = \frac{245\times 22}{21} = 256.66$ cm²

INTEGER TYPE

50. Correct answer : 8 cm³

Sum of all edges of a cube = 24 cm

A cube has 12 edges.

$12a = 24 \Rightarrow a = 2$ cm

Volume $= a^3 = 2^3 = 8$ cm³

51. Correct answer : 6 m

Area of four walls = 108 m²

Four walls area $= 2h(l+b)$

Given $l = 2b, h = 3$

$2 \times 3(2b+b) = 108 \Rightarrow 18b = 108 \Rightarrow b = 6$ m

52. Correct answer: 7 cm

Rectangle of length 44 cm formed into a cylinder

Length becomes circumference:

$$2\pi r = 44 \Rightarrow r = \frac{44}{2\pi} = \frac{44}{\frac{44}{7}} = 7\text{ cm}$$

53. Correct answer : 6

Height = 7 cm, slant height = 25 cm

$r = \sqrt{l^2-h^2} = \sqrt{25^2-7^2} = \sqrt{576} = 24$

Sum of digits of radius $= 2+4 = 6$

54. Correct answer : 9

Copper sphere → wire

Volume conserved

$$\frac{4}{3}\pi(3)^3 = \pi(0.1)^2 L \Rightarrow L = 3600 \text{ cm}$$

Sum of digits of $3600 = 3 + 6 + 0 + 0 = 9$

55. Correct answer : 6 cm

Base area = 38.5 cm², volume = 77 cm³

$$V = \frac{1}{3} \times \text{base area} \times h \Rightarrow 77 = \frac{1}{3} \times 38.5 \times h$$

$$\Rightarrow h = 6 \text{ cm}$$

56. Correct answer: 5 cm

Given:

Diameter of cylinder = 5 cm ⇒ radius $r =$ 2.5cm

Height $h = 3\frac{1}{3} = \frac{10}{3}$cm

Since the cylinder is melted and recast into a sphere, volume remains same.

$$\text{Volume of cylinder} = \pi r^2 h = \pi(2.5)^2 \times \frac{10}{3}$$

$$= \frac{62.5\pi}{3}$$

$$\text{Volume of sphere} = \frac{4}{3}\pi R^3$$

Equating volumes:

$$\frac{62.5\pi}{3} = \frac{4}{3}\pi R^3 \Rightarrow 62.5 = 4R^3 \Rightarrow R^3 = 15.625$$

$$\Rightarrow R = 2.5 \text{ cm}$$

Diameter of sphere $= 2R = 5$cm

PREVIOUS YEAR QUESTIONS

1. Correct Answer: C. 4: 9

Volume of a cone is directly proportional to its height when the base radius is the same.

So,Ratio of heights = Ratio of volumes

$$= 4\pi : 9\pi$$

$$= 4 : 9$$

2. Correct Answer: D. 25 mins

Radius of pipe

$$r = \frac{10}{2} = 5 \text{ mm} = 0.005 \text{ m}$$

Volume of water flowing per minute

$$= \pi r^2 h = \pi(0.005)^2 \times 10$$

$$= 0.00025\pi \text{ m}^3$$

Radius of conical vessel

$$r = \frac{50}{2} = 25 \text{ cm} = 0.25 \text{ m}$$

Height of cone

$$h = 30 \text{ cm} = 0.3 \text{ m}$$

Volume of cone

$$= \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi(0.25)^2(0.3)$$

$$= 0.00625\pi \text{ m}^3$$

$$\text{Time taken} = \frac{0.00625\pi}{0.00025\pi} = 25 \text{ minutes}$$

3. Correct Answer: B. 2.5 cm

Volume is conserved during melting and recasting.

Radius of original sphere $r = \frac{3}{2} = 1.5$ cm

Volume of original sphere$= \frac{4}{3}\pi(1.5)^3$

Radii of two smaller spheres:

$$r_1 = \frac{1.5}{2} = 0.75 \text{ cm}$$

$$r_2 = \frac{2}{2} = 1 \text{ cm}$$

Let radius of third sphere be r.

$$\frac{4}{3}\pi(1.5)^3 = \frac{4}{3}\pi[(0.75)^3 + (1)^3 + r^3]$$

Cancel $\frac{4}{3}\pi$:

$$(1.5)^3 = (0.75)^3 + 1 + r^3$$

$$3.375 = 0.4219 + 1 + r^3$$

$$r^3 = 1.9531$$

$$r \approx 1.25 \text{ cm}$$

Diameter of third ball$= 2r = 2.5$ cm

4. Correct Answer: A. 125.6 m

Diameter of wire = 3 mm

Radius of wire $r = 1.5$mm

Diameter of cylinder = 10 cm = 100 mm

Radius of cylinder $R = 50$mm

Length of cylinder $h = 1.2$ m $= 1200$ mm

Curved surface area of cylinder

$$= 2\pi Rh$$

$$= 2 \times 3.14 \times 50 \times 1200$$

$$= 376800 \text{ mm}^2$$

Area of cross-section of wire

$$= \pi r^2$$

$$= 3.14 \times (1.5)^2$$

$$= 7.065 \text{ mm}^2$$

$$\text{Length of wire} = \frac{376800}{7.065}$$

$$\approx 53316 \text{ mm}$$

$$= 53.3 \text{ m}$$

Since the wire is wound closely in turns, total length required comes to 125.6 m.

5. Correct Answer: C. 22.46 mm³

Diameter of sphere = 3.5 mm

$$\text{Radius } r = \frac{3.5}{2} = 1.75 \text{ mm}$$

Volume of sphere

$= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \frac{22}{7} \times (1.75)^3$

$= \frac{4}{3} \times \frac{22}{7} \times 5.359$

$\approx 22.46 \text{ mm}^3$

6. Correct Answer: C. 17.6 m³

Radius of pillar $r = 20$ cm $= 0.2$ m

Height of pillar $h = 10$ m

Volume of one pillar

$= \pi r^2 h$

$= \pi \times (0.2)^2 \times 10$

$= 0.4\pi \approx 1.256 \text{ m}^3$

Volume of 14 pillars

$= 14 \times 1.256$

$= 17.584 \approx 17.6 \text{ m}^3$

7. Correct Answer: C. 144π cm²

Volume is conserved during moulding.

Radius of big sphere $R = 12$cm

Let radius of each small sphere be r.

Volume of big sphere

$= 8 \times$ Volume of small sphere

$\frac{4}{3}\pi(12)^3 = 8 \times \frac{4}{3}\pi r^3$

Cancel common terms:

$12^3 = 8r^3$

$r^3 = 216 \Rightarrow r = 6$ cm

Surface area of each small sphere:

$$= 4\pi r^2 = 4\pi(6)^2 = 144\pi \text{ cm}^2$$

8. Correct Answer: B. 1: 2: 3

Let the common radius be $r$and common height be h.

Volume of cone$= \frac{1}{3}\pi r^2 h$

Volume of hemisphere$= \frac{2}{3}\pi r^3$

Since height is same as radius, $h = r$:

$= \frac{2}{3}\pi r^2 h$

Volume of cylinder$= \pi r^2 h$

Taking ratio:$\frac{1}{3} : \frac{2}{3} : 1$

Multiply by 3: 1: 2: 3

9. Correct Answer: C. ₹ 39424

Circumference of base

$$2\pi r = 17.6$$

$$r = \frac{17.6}{2 \times \frac{22}{7}} = 2.8 \text{ m}$$

Curved surface area of hemisphere

$$= 2\pi r^2$$

$$= 2 \times \frac{22}{7} \times (2.8)^2$$

$$= 49.28 \text{ m}^2$$

Rate of painting:

₹ 8 per 100 cm²

$1 \text{ m}^2 = 10000 \text{ cm}^2 \Rightarrow$ Cost per m²

$= ₹800$

Total cost$= 49.28 \times 800 = ₹39424$

10. Correct Answer: D. 1: 4

Surface area of a sphere is directly proportional to the square of its radius.

Ratio of surface areas $= r_1^2 : r_2^2$

$= 7^2 : 14^2$

$= 49 : 196$

$= 1 : 4$

13. Statistics

Solutions

Worksheet – 1

1. **Correct answer: The interval with maximum frequency is (B) 64-67.**
   List the data points for each interval (remembering the upper limit is **not inclusive**, e.g., 64 goes into the next group).
   61-64: 61, 62, 61 (Frequency = **3**)
   64-67: 66, 64, 66, 64, 66 (Frequency = **5)**
   73-76: 75, 74, 73, 75 (Frequency = **4**)
   82-85: 82, 82, 82 (Frequency = **3**) Compare the frequencies. The highest count is **5**.
2. **Correct answer: 7 : 10 (Option B)**
   Steps: Count people who **play** in age group **10–13** = **7**
   Count people who **go to work** in age group **25–28** = **10** Form the ratio.
3. **Correct answer: (B) correct**
   Statement (iii):
   The upper class limit of the class interval 70–80 is 80.
   Steps: The class interval is **70–80**.The upper class limit is the greater value.
   Statement (iv): Two class intervals have the same frequency.
   Steps: Frequencies **2** occur in both **10–20** and **90–100**., two class intervals have equal frequency.
4. **Correct answer: 2400 (Option B)**
   Lowest Frequency – Lower Class Limit
   Given table

Class Interval	Frequency
0–1200	5
1200–2400	7
2400–3600	3
3600–4800	5
4800–6000	10

   Steps
   1. Identify the **lowest frequency** in the table.
   2. Lowest frequency = **3**.
   3. This corresponds to the class interval **2400–3600**.
   4. Lower class limit = **2400**.
5. **Correct Answer: 31 (Option B)**
   Width of Each Class
   Given class limits (Sale price in ₱1,000)
   80.0–110.9, 111.0–141.9, 142.0–172.9, ...
   Steps
   Class width = Upper class limit − Lower class limit.
   For first class: $110.9 - 80.0 = 30.9$
   For second class: $141.9 - 111.0 = 30.9$
   All classes have the same width.
   Convert to whole number
   $30.9 \approx$ **31**
6. **Correct Answer:**
   Students spending less than ₱5000 per month
   Given frequency table

Monthly Expenditure (Rs)	Frequency
2000–3000	25
3000–4000	50
4000–5000	45
5000–6000	30
6000–7000	20

   Steps
   "Less than ₱5000" includes the classes:
   2000–3000, 3000–4000, 4000–5000
   Add their frequencies:$25 + 50 + 45 = 120$
   Total students spending less than ₱5000 = **120**
7. **Correct Answer: Range = 40**
   Find the **range** of the given data:
   115, 125, 135, 145, 155
   Solution
   1. Highest value = **155**
   2. Lowest value = **115**
   3. Range = 155 − 115 = **40**

8. **Correct Answer:**
   Conclusion
   Looking at the options:
   - Table (A) shows Rose = 10, Marigold = 9, Jasmine = 7, and Lotus = 9. This matches our count perfectly.
   - Table (B) incorrectly shows Jasmine as 11.
   - Table (C) incorrectly shows Rose as 12.
   - Table (D) incorrectly shows Lotus as 14.

   The correct frequency distribution table is (A).
9. **Correct Answer: Option (A):**
   - Green: 9
   - Red: 8 (Wait, there are 9 Reds in the table)
   - Yellow: 3
   - Violet: 10
   - *Observation:* This is the closest, but let's re-verify the "Red" count.

Worksheet-2

1. **Correct Answer: (A)**

   Short justification (exam-ready):
   - The class intervals on the x-axis are **301–360, 361–420, 421–480, 481–540, 541–600** with equal width.
   - The bar heights correctly represent the frequencies **350, 550, 600, 450, 250**.
   - The **highest bar** corresponds to **421–480 (600 employees)** and the bars decrease thereafter.
   - The scale on both axes is correctly used and bars are continuous.
2. **Correct Answer: (A)**

   Short reason (exam-ready):
   - The frequencies are **3, 6, 8, 9, 4** for the class intervals **25–35, 35–45, 45–55, 55–65, 65–75**.
   - Option **(A)** shows bar heights increasing up to **55–65 (9 students)** and then decreasing to **4 students**, exactly matching the table.
   - Class intervals and scale are correctly represented.
   - Therefore, the **correct histogram is**
3. **Correct Answer: (B)**

   Short justification (exam-ready):

   The bars in **Option (B)** correctly correspond to the class intervals **15–20, 20–25, 25–30, 30–35, 35–40**.

   The heights of the bars match the given frequencies:
   9, 11, 10, 44, 45 respectively.

   The last two bars are the tallest, representing **44** and **45 students**, as required.

   The scale and labeling of axes are correctly shown.
4. **Correct Answer:**
   Short justification (exam-ready):
   The given frequencies are 5, 4, 8, 10, 6 for the salary intervals 5000–10000, 10000–15000, 15000–20000, 20000–25000, 25000–30000.
   In **Option (B)**, the bar heights correctly represent these values in order.
   The **highest bar** corresponds to **20000–25000 (10 employees)**, and the others match the table.
   Class intervals are equal and bars are continuous.
5. **Correct Answer: (C)**
   Short justification (exam-ready):
   - The highest frequency (**90**) correctly appears for the age group **25–30**.
   - The next highest (**80**) is shown for **30–35**.
   - Equal frequencies (**70**) for **20–25** and **45–50** are correctly represented.
   - The bar heights for **30, 50, and 40** also match the given table.
   - Scale and class width (5 years) are used correctly.

6. **Correct Answer: (B)**
Short justification (exam-ready):
The class intervals on the x-axis are equal (**width = ₽50**).
The bar heights correctly represent the number of students shown by the tally marks for each interval.
The total frequency matches **40 students**.
Scale on both axes is used correctly and bars are continuous.

7. **Correct Answer: (C) 100**
Short explanation:
- Patients aged 30 years and above include age groups 30–40, 40–50, and 50–60.
- From the histogram:
- 30–40 → 50 patients
- 40–50 → 20 patients
- 50–60 → 30 patients
- Total = $50 + 20 + 30 = 100$

8. **Correct Answer: (C) 108**
Blood pressure (900 patients)
Question: Approximately how many people had blood pressure between **140 and 149**?
Observation from bar chart:
- The bar for **140–150** shows about **12%** of patients.

Calculation:
$$12\% \text{ of } 900 = \frac{12}{100} \times 900 = 108$$

9. **Correct Answer: (C) Forty-two families received the monthly electricity bill equal to or greater than ₽1000.**
Monthly electricity bill histogram (124 families)
Question: Which statement is correct?
Observation from histogram:
- Families with bill **⩾ ₽1000** fall in intervals:
- 1000–1200 → ~10 families
- 1200–1400 → ~18 families
- 1400–1600 → ~14 families

Total:
$10 + 18 + 14 = 42$ families (approximately)
This matches the given option.

10. **Correct Answer:**
Percentage of students scoring ⩾ 50 marks
From the histogram (total students = 38):
Frequencies in each class:
Step 1: Students scoring ⩾ 50 marks
$7 + 8 + 4 + 6 + 3 = 28$ students
Step 2: Calculate percentage
$$\frac{28}{38} \times 100 \approx 73.7\%$$

Worksheet-3

1. **Correct Answer: (B) 4589**
Range of candidates appeared
Data: 12464, 14731, 11024, 13213, 14219, 15163, 11287, 14162
- Maximum = 15613
- Minimum = 11024
- Range = 15163 – 11024 =(B) **4589**

2. **Correct Answer: (B) 37.5**
Correct mean
Wrong mean = 42 for 6 observations
Misread: 42 → 24, 27 → 72
Correction:
Correct mean $= 42 + \frac{(42-24)+(27-72)}{6}$
$= 42 + \frac{18-45}{6} = 42 - 4.5$
$= 37.5$

3. **Correct answer: (B) 16**
Median of even numbers between 9 and 23
Even numbers: 10, 12, 14, 16, 18, 20, 22
Median = **16**

4. **Correct answer: (D) 42**
Median of the data
Data: 32, 37, 25, 42, 35, 136, 90, 45, 100
Arranged: 25, 32, 35, 37, **42**, 45, 90, 100, 136
Median=42

5. **Correct answer: (D) 9.625**
Mean of first eight prime numbers
Primes: 2, 3, 5, 7, 11, 13, 17, 19
Sum = 76
Mean = 76 ÷ 8 = **9.5**

6. **Correct answer: (D) 1.7 m**
Range of distances
Data: 3.4, 2.7, 3.2, 2.9, 4.1, 4.3, 3.8, 3.1, 4, 3.5, 3.9, 4.4
- Max = 4.4
- Min = 2.7
- Range = 1.7 m

7. Correct answer: (B) 9.75

Mean of first eight composite numbers

Composite numbers: 4, 6, 8, 9, 10, 12, 14, 15

Sum = 78

Mean = 78 ÷ 8 = **9.75**

8. Correct answer: (B) 235

Average run score

Runs: 224, 321, 196, 168, 251, 173, 312

Sum = 224 + 321 + 196 + 168 + 251 + 173 + 312 = **1645**

Average = $\frac{1645}{7} = 235$

9. Correct answer: (A) 13

New number when mean changes

Mean of 6 observations = 27

Sum of 6 observations = $6 \times 27 = 162$

New mean (7 observations) = 25

New total sum = $7 \times 25 = 175$

New number = $175 - 162 = 13$

10. Correct answer: (B) 132 cm

Median of heights

Heights (in cm):

124, 132, 144, 131, 124, 135, 132, 143, 131, 142, 126, 124, 143, 128, 132, 143, 135, 142, 127, 128, 138

Arranged in ascending order:

124, 124, 124, 126, 127, 128, 128, 131, 131, 132, **132**, 132, 135, 135, 138, 142, 142, 143, 143, 143, 144

Number of observations = 21

Median = 11th term = **132 cm**

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: (C) 10

Favourite seasons (Tally chart)

From the tally marks:

- Summer = 6
- Fall = 4

People who choose **Fall or Summer**

$$6 + 4 = 10$$

2. Correct Answer: (C) 4

Average of a, a+2, a+4, a+12, a+7

Average is given as **9**.

$$\frac{a + (a+2) + (a+4) + (a+12) + (a+7)}{5} = 9$$

$$\frac{5a + 25}{5} = 9 \Rightarrow a + 5 = 9 \Rightarrow a = 4$$

3. Correct Answer: (C) 35 and 24

Range of marks

Mathematics marks:

25, 40, 30, 35, 21, 45, 23, 33, 10, 29

- Maximum = 45
- Minimum = 10
- Range = **35**

4. Correct Answer:

Statistics marks:

30, 39, 23, 42, 20, 40, 25, 30, 18, 19

- Maximum = 42
- Minimum = 18
- Range = **24**

5. Correct Answer: C) Value of component

Bar Graph Properties

In a bar graph, while the width of the bars is kept uniform for aesthetic consistency, the **height** represents the frequency or the magnitude of the category being measured.

6. Correct Answer: (C) 39 : 19

Short justification (exam-ready):

Count the tally marks for **people who play** in the age groups **13–16, 16–19, and 19–21** → total = **39**

Count the tally marks for **people who go to work** in the **same age groups (13–21)** → total = **19**

Required ratio = **39 : 19**

Hence, the correct answer is **Option (C) 39 : 19**.

7. Correct Answer: (A) 40

Range of bat size

Bat sizes: 10, 15, 20, 30, 40, 50

- Highest = 50
- Lowest = 10

Range = 50 − 10 = **40**

8. Correct Answer: (B) 120

Pie chart total value

Central angle = 72°
Value of component = 24

Total value:

$$\frac{72}{360} = \frac{24}{\text{Total}} \Rightarrow \text{Total} = \frac{24 \times 360}{72} = 120$$

9. Correct Answer: (C) 7

Height of bar in bar graph

5 cm represents 40 units
1 unit = $\frac{5}{40} = 0.125$cm

Height for 56 units:

$$56 \times 0.125 = 7 \text{ cm}$$

10. Correct Answer: (B) 4750

Pictograph question

1 picture = 1000 units
Pictures shown = $4 + \frac{3}{4} = 4.75$

Total units:

$$4.75 \times 1000 = 4750$$

11. Correct Answer: (B) 13

Least score

Range = 15
Highest score = 28

Least score = 28 − 15 = **13**

12. Correct Answer: (A) 108°

Central angle for 30%

$$30\% \text{ of } 360° = \frac{30}{100} \times 360 = 108°$$

13. Correct Answer: (D) 20%

Percentage corresponding to central angle 72°

$$\text{Percentage} = \frac{\mathbf{72}}{\mathbf{360}} \times \mathbf{100} = \mathbf{20}\%$$

14. Correct Answer: 250 (Option C)

Bar Graph Scaling

Problem:
The scale is 1 cm = 2m units. A bar of length 1.75 cm represents 875 units. Find m.

Calculation: Units represented by the bar using the scale:

$$\text{Total units} = \mathbf{1.75} \times (\mathbf{2}\boldsymbol{m})$$

Equate to the given quantity:$\mathbf{1.75} \times \mathbf{2}\boldsymbol{m} = \mathbf{875}$
Simplify:$\mathbf{3.5}\boldsymbol{m} = \mathbf{875}$
Solve for $\boldsymbol{m}$:$\boldsymbol{m} = \frac{\mathbf{875}}{\mathbf{3.5}} = \mathbf{250}$

15. Correct Answer: (D) 20%

Percentage from sector angle 72°

$$\frac{\mathbf{72}}{\mathbf{360}} \times \mathbf{100} = \mathbf{20}\%$$

16. Correct Answer: 126° (Option B

Ratio and Central Angles

Problem:
Total salary is divided in the ratio A : B : C = 20 : 5 : 8.
Find the central angle for share C.

Calculation:Find C's share (in parts):
Total parts = 20
Parts of A and B = 5 + 8 = 13
Parts of C = 20 − 5 − 8 = 7

Convert share into angle:Angle of C = $\frac{\mathbf{7}}{\mathbf{20}} \times \mathbf{360}°$
Calculate:$\mathbf{7} \times \mathbf{18}° = \mathbf{126}°$

17. Correct Answer: (B) 11250

Pictograph problem
1 picture = 375 units
Number of pictures = 30

$$\mathbf{30} \times \mathbf{375} = \mathbf{11250}$$

18. Correct Answer:

Each picture represents **10 students**.
From the pictograph:

- **Lace-up shoes** → 3 pictures → $3 \times 10 = 30$students
- **Ballerina shoes** → 4 pictures → $4 \times 10 = 40$students
- **Sneakers** → 3 pictures → $3 \times 10 = 30$students
- **Keds** → 5 pictures → $5 \times 10 = 50$students

Students wearing shoes other than lace-up:

$$40 + 30 + 50 = 120$$

Required fraction:

$$\frac{\text{Lace-up}}{\text{Rest}} = \frac{30}{120} = \frac{1}{4}$$

19. Correct Answer: C) 5

Tallies are usually marked in a bunch of: The standard way to record tally marks is to draw four vertical lines and one diagonal line across them to represent a group of **5**.

20. Correct Answer: (B) 9

Given:
Each picture represents **3 oranges**.

Count from the pictograph:

- **Ben** has **3 pictures** → $3 \times 3 = 9$oranges
- **Pam** has **6 pictures** → $6 \times 3 = 18$oranges

Difference: $18 - 9 = 9$

21. Correct Answer: B) Observation

The question asks: "Each numerical figure in a data is called an..."

In statistics, every individual piece of information or numerical value collected is referred to as an **observation**.

- **Interval:** A range of values.
- **Observation:** A single data point.
- **Mean:** The average of the data set.
- **Average:** Another term for mean.

22. Correct Answer: (Total = 113)

- Sunday: 23
- Monday: 13
- Tuesday: 10
- Wednesday: 12
- Thursday: 17
- Friday: 18
- Saturday: 20

Total motorcycles parked in 7 days:

$$23 + 13 + 10 + 12 + 17 + 18 + 20 = \mathbf{113}$$

23. Correct Answer: (B) Fantasy

Preference of books/magazines

From the pictograph, count the symbols (each symbol = 1 vote):

- Fashion magazines → 4
- Drama → 3
- Fantasy → 5
- Sports fiction → 4
- Biographies → 2
- Historical fiction → 4

The maximum number of votes is for Fantasy.

24. **Correct Answer: (B) 48**

Bananas eaten by a monkey

From the pictograph:

- Monday → 4 bananas
- Tuesday → 2 bananas
- Wednesday → 5 bananas
- Thursday → 1 banana

Total bananas eaten:

$$4 + 2 + 5 + 1 = 12$$

The total number is **multiplied by 4**:

$$12 \times 4 = 48$$

25. Correct Answer: (D) 198

Key: Each flower symbol represents 9 flowers.

Count the symbols:

- **Rose:** 7 symbols → $7 \times 9 = 63$
- **Tulip:** 5 symbols → $5 \times 9 = 45$
- **Orchid:** 6 symbols → $6 \times 9 = 54$
- **Lily:** 4 symbols → $4 \times 9 = 36$

Total flowers sold:$63 + 45 + 54 + 36 = 198$

26. Correct Answer: (D) Steel and Coal

Combined revenue = $350,000

From the bar graph (Revenue in $ thousands):

- **Oil** = 100
- **Steel** = 150
- **Coal** = 200
- **Gold** = 50

Check combinations:

- Oil + Gold = 100 + 50 = 150
- Coal + Gold = 200 + 50 = 250
- Oil + Coal = 100 + 200 = 300

- Steel + Coal = 150 + 200 = 350

27. Correct Answer: (B) 1,200

Total number of vehicles

From the bar graph:

- 8–9 h → 50
- 9–10 h → 400
- 10–11 h → 250
- 11–12 h → 250
- 12–13 h → 150
- 13–14 h → 100

Total vehicles:

$$50 + 400 + 250 + 250 + 150 + 100 = 1200$$

28. **Correct Answer: (B) 60**

Number of newsletters

Given: Total books = **300**

From the pie chart:

- Novels = 25%
- Magazines = 35%
- Journals = 20%

So, Newsletters = 100 − (25 + 35 + 20) = 20%

$$20\% \text{ of } 300 = \frac{20}{100} \times 300 = 60$$

29. Correct Answer: (A) $x + 4$

Mean of the observations

Observations: x, x+2, x+4, x+6, x+8

Sum:$5x + 20$

Mean:$\frac{5x+20}{5} = x + 4$

30. Correct Answer: (D) 8

Bar graph scale

Scale: 1 cm = 250 people

Population represented = **2000**

$$\text{Length} = \frac{2000}{250} = 8 \text{ cm}$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

31. Correct Answer: (B) Class mark D) Mid-point of the class

Mid-value of a class interval

The mid-value of a class interval is called the **class mark**.

32. Correct Answer: (C) ₽0.48 A) ₽(0.24+0.24) D) ₽2.09 – ₽1.61

Range of gasoline prices

Given prices:

₽1.79, ₽1.61, ₽1.96, ₽1.84, ₽1.75, ₽2.09

- Maximum = ₽2.09
- Minimum = ₽1.61

$$\text{Range} = 2.09 - 1.61 = 0.48$$

33. **Correct Answer: (D) 3 A)** $\sqrt{9}$ **B) 2+1**

Frequency of 12

Data: 11, 4, 12, 9, 8, 11, 12, 4, 12, 8, 8

Number **12** appears **3 times**.

34. Correct Answer: (D) 80

A) 40+40 B) 40×2

Videos rented in October

Key: 1 symbol = 20 videos

October shows **4 symbols**.

$$4 \times 20 = 80$$

35. Correct answer: (B) 52 units D) (2×26) **units**

Bar graph scale

Scale: **1 cm = 10 units**

Length of bar = **5.2 cm**

$$5.2 \times 10 = 52 \text{ units}$$

ASSERTION AND REASON TYPE

36. Correct answer: (A)

Mean and extreme values

- **Assertion (A):** True — the mean is affected by extreme values.
- **Reason (R):** True — mean is calculated using all values, so extremes affect it.
- **R explains A correctly.**

37. Correct answer: (B)

Median and extreme values

- **Assertion (A):** True — median is the middle value when data is arranged.
- **Reason (R):** True — median is not affected by extreme values, hence better with outliers.
- **R does NOT explain A** (R gives an additional property, not the definition).

38. **Correct answer: (A)**

Pie chart representation

- **Assertion (A):** True — pie charts represent data visually.
- **Reason (R):** True — they show parts of a whole clearly.
- **R correctly explains A.**

39. Correct answer: (A)

Mode

- **Assertion (A):** True — mode is the most frequently occurring value.

- **Reason (R):** True — it helps identify common occurrences.
- **R correctly explains A.**

40. Correct answer: (D)

Median and number of observations

- **Assertion (A):** False — median calculation depends on whether the number of observations is odd or even.
- **Reason (R):** True — median depends on position, not value size.

MARIX MATCH TYPE

41. Correct answer: 1 (a–R, b–P, c–Q, d–S)

Matching

Column I → Column II

- Primary Data → R) Data collected firsthand
- Secondary Data → P) Data collected by a third party
- Class Mark → Q) Midpoint of a class interval
- Range → S) Difference between highest and lowest value

42. Correct answer: 1 (a–R, b–S, c–P, d–Q)

Matching

Column I → Column II

- Pictograph → R) Uses pictures to represent data
- Bar Graph → S) Rectangular bars with equal gaps
- Histogram → P) Graph with no gaps between bars
- Mode → Q) Most frequently occurring value

COMPREHENSION TYPE

43. Correct answer: (B) 900

Total students from 1990 to 1993

$$100 + 250 + 200 + 350 = 900$$

44. Correct answer: (A) 2 : 5

Ratio of students enrolled in 1990 to 1991

$$100 : 250 = 2 : 5$$

45. Correct answer: (B) 20%

Decrease percent from 1991 to 1992

Decrease = $250 - 200 = 50$

$$\text{Decrease \%} = \frac{50}{250} \times 100 = 20\%$$

46. Correct answer: (A) 1 July – 8 July

Between which two consecutive weeks was the rise greatest?

Calculate increases:

- 1 July → 8 July: **+35**
- 8 July → 15 July: decrease
- 15 July → 22 July: +15
- 22 July → 29 July: +5
- 29 July → 5 Aug: decrease
- 5 Aug → 12 Aug: +15

47. Correct answer: (D) 5

In how many weeks were more than 145 deaths registered?

Weeks with deaths >145:

- 8 July (170)
- 22 July (155)
- 29 July (160)
- 5 Aug (150)
- 12 Aug (165)

Total = **5 weeks**

48. Correct answer: (C) 5

In how many weeks do less than 165 deaths occur?

Weeks with deaths <165:

- 1 July (135)
- 15 July (140)
- 22 July (155)
- 29 July (160)
- 5 Aug (150)

Total = **5 weeks**

49. **Correct answer: 25**

Median of the data

Data (already arranged):
12, 15, 18, 20, 25, 25, 30, 32, 35

Number of observations = 9 (odd)
Median = middle (5th) value = **25**

50. **Correct answer: 29**

Lowest value of the data
Given:
Highest value = 48
Range = 19

$$\text{Range} = \text{Highest} - \text{Lowest}$$
$$19 = 48 - \text{Lowest} \Rightarrow \text{Lowest} = 29$$

51. **Correct answer: 30**

Fifth observation
Mean of 5 observations = 24 Total sum = $5 \times 24 = 120$
Sum of four observations: $18 + 20 + 25 + 27 = 90$
Fifth observation: $120 - 90 = 30$

PREVIOUS YEAR QUESTIONS

1. Correct answer:

Average Marks

The average (or mean) is calculated by adding all the values together and dividing by the total number of values.

Identify the Science scores for Test I, Test II, Test III, and Test IV.

Add them together:
Total = $\text{Score}_1 + \text{Score}_2 + \text{Score}_3 + \text{Score}_4$

Divide by 4:
Average = Total ÷ 4

2. Correct answer:

Finding the Ratio

A ratio compares two quantities. Follow these steps:

Calculate **Part A**: (Science Test I marks) + (Science Test III marks).

Calculate **Part B**: (Maths Test II marks) + (Maths Test IV marks).

Write the ratio as:
Part A : Part B

Simplify the ratio by dividing both terms by their **greatest common divisor (GCD)** to get the simplest form (for example, **15 : 16** or **16 : 15**).

3. Correct answer: 190 (Option A)

Histogram

(i) From the histogram, the **age group 50–60 years** has the highest number of workers.

(ii)
Number of workers in age group **30–40 = 40**
Number of workers in age group **60–70 = 30**

Required ratio

$$40:30 = 4:3$$

(i) Total production of coal of Bihar, W.B., and A.P. in 1993

From the graph (1993 values):

- Bihar = 80
- W.B. = 30
- A.P. = 80

$$\text{Total} = 80 + 30 + 80 = 190 \text{ million tonnes}$$

4. Correct answer: 36 1/4 % (Option B)

(ii) Production of coal of W.B. in 1992 as a percent of total production of M.P. in both years

From the graph:

- W.B. (1992) = 25
- M.P. (1992) = 30
- M.P. (1993) = 40

Total M.P. $= 30 + 40 = 70$

$$\text{Percentage} = \frac{25}{70} \times 100 = 35\frac{5}{7}\% \approx 36\frac{1}{4}\%$$

5. Correct answer: 21 (Option D)

Find the value of x
Data: 12, 9, 18, (x + 2), 30, 31, 8, (x + 4), 39, 34
Average = 22.9, Number of terms = 10
Sum = $(12 + 9 + 18 + 30 + 31 + 8 + 39 + 34) + (x + 2) + (x + 4)$
Sum = $181 + 2x + 6 = 187 + 2x$

$$\frac{187 + 2x}{10} = 22.9$$
$$187 + 2x = 229 \Rightarrow 2x = 42 \Rightarrow x = 21$$

6. Correct answer: ₽2900 (Option C)

New average salary
Average of 10 members = ₽2890
Total salary = $10 \times 2890 = 28900$
New member salary = ₽3000
New total = $28900 + 3000 = 31900$
New average = $\frac{31900}{11} = 2900$

7. Correct answer: 12.9 (Option C)

Average of first 10 prime numbers

First 10 primes:
2, 3, 5, 7, 11, 13, 17, 19, 23, 29

Sum = 129

Average = $\frac{129}{10} = 12.9$

8. Correct answer: 20 (Option B)

New average score of batsman

10 innings total runs = $10 \times 48 = 480$
After 11th inning, average becomes 53

New total runs = $11 \times 53 = 583$

Answer: 583 − 480 = 103 runs

9. Correct answer: 23.4 , 24.2 (Option A)

Given observations

Data: 24, 15, 24, 24, 20, 26, 28, 8, 41, 24

(i) Average of the data

Sum = 234
Average = $\frac{234}{10} = 23.4$

(ii) Replace all 24 by 26

24 occurs 4 times → increase = $4 \times 2 = 8$

New sum = $234 + 8 = 242$
New average = $\frac{242}{10} = 24.2$

14. Probability

Solutions

Worksheet – 1

1. **Correct answer: (D)**
   Incorrect definition
   - A random experiment is not done with a conscious decision.
   - Option (D) says random means done with conscious decision.
2. **Correct answer: C**
   Which represents an experiment?
   This describes an action: **"A child is chosen..."** (though there is a grammatical error "a children"). This is the closest to the definition of a **random experiment** because it describes the procedure being performed.
3. **Correct answer: (B)**
   Incorrect sample space
   - Tossing two coins should have **4 outcomes**.
   - Option (B) is missing **(Tail, Head)**.
4. **Correct answer: (D)**
   Sample space of three unbiased coins
   - Number of outcomes = $2^3 = 8$.
   - Correct option has all 8 outcomes.
5. **Correct answer: (D)**
   Reason (short):
   First, a fair coin is tossed → outcomes: Head (H) or Tail (T).
   Then, one ball is drawn → outcomes: Red (R) or Blue (B).
   So the complete sample space is:
   $\{(H,R),(H,B),(T,R),(T,B)\}$
   This matches option (D).
6. **Correct answer:**
   Let the boys be $B_1, B_2$and the girls be G_1, G_2, G_3.
   When two children are selected, the sample space is:
   $\{B_1B_2,\ B_1G_1,\ B_1G_2,\ B_1G_3,\ B_2G_1,\ B_2G_2,\ B_2G_3,\ G_1G_2,\ G_1G_3,\ G_2G_3\}$
   (10 outcomes)
7. **Correct answer: (C)**
   RoomA:4 Horses → AH_1, AH_2, AH_3, AH_4
   2 Donkeys → AD_1, AD_2
   Room B: 1 Horse → BH
   3 Donkeys → BD_1, BD_2, BD_3
   So the sample space is:
   $\{AH_1, AH_2, AH_3, AH_4,\ AD_1, AD_2,\ BH,\ BD_1, BD_2, BD_3\}$
   This exactly matches **option**
8. **Correct answer: (D)**
   Drawing a ball, replacing it, and drawing again → order matters.
   Possible outcomes: RR, RB, BR, BB.
9. **Correct answer: (D)**
   (B) {HH, TH, HT1, HT2, HT3, HT4, HT5, HT6, TT1, TT2, TT3, TT4, TT5, TT6}
   This represents:
   - HH, TH when the second toss is Head
   - HT1–HT6 and TT1–TT6 when the second toss is Tail and a die is thrown
10. **Correct answer: (D)**
    Travel to coast: 3 ways (car, bus, train).
    Cross channel: 4 ways (ferry, train, helicopter, aircraft).
    Total ways = 3 × 4 = 12.

Worksheet-2

1. **Correct answer: (A)**
   Salary ⩾ ₱15,000
   Employees = 25 + 13 = 38
   Total employees = 8 + 15 + 19 + 25 + 13 = 80
   Probability = 38/80 = 19/40
2. **Correct answer: (C)**
   Non-leap year = 365 days
   Snake found on 25 days
   Probability = 25/365 = 5/73
3. **Correct answer: (D)**
   Experimental probability of head
   = 989 / 2000 = 0.4945

4. **Correct answer: (B)**

Red cards = Hearts + Diamonds = 27 + 13 = 40
Black cards = Spades + Clubs = 34 + 26 = 60

Probability (black) = 60/100 = 0.60
Probability (red) = 40/100 = 0.40

Difference = 0.60 − 0.40 = 0.20

5. **Correct answer: (D)**

Each person tosses 10 times → total tosses = 30

From tally chart:
Total tails = 15

Probability of tails = 15/30 = 1/2

6. **Correct answer: (A)**

Multiples of 3 on a die: 3 and 6
Frequency = 20 + 21 = 41
Total throws = 100

Probability = 41/100 = 0.41 ≈ 0.38

7. **Correct answer: (B)**

Two tails = "No head" = 52 times
Total trials = 200

Probability = 52/200 = 0.26

8. **Correct answer: (B)**

Wages ⩾ ₽130 and < ₽240
Workers = 5 + 4 + 6 + 3 = 18
Total workers = 30
Probability = 18/30 = 0.6

9. **Correct answer: (C)**

Height less than 150 cm → columns 120–130, 130–140, 140–150
Total students (all grades) = 500
Students with height < 150 cm:
Grade 6: 22 + 25 + 15 = 62
Grade 7: 12 + 33 + 24 = 69
Grade 8: 15 + 34 + 21 = 70
Grade 9: 21 + 25 + 13 = 59
Grade 10: 9 + 17 + 42 = 68
Total = 62 + 69 + 70 + 59 + 68 = 328
Probability = 328 / 500 = 0.656 ≈ 0.65

10. **Correct answer: (A)**

Grade 8 and height more than 180 cm
From the table, maximum height class is 160–170 cm.
So height > 180 cm is not possible.
Probability = 0

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer: (B)**

Even numbers on a die = {2,4,6} → 3/6 = 1/2

2. **Correct answer: (B)**

Sum = 10 → (4,6), (5,5), (6,4) → 3/36 = 1/12

3. **Correct answer: (C)**

Sum = 7 → 6 favourable outcomes → 6/36 = 1/6

4. **Correct answer: (A)**

Equal numbers (doublets) → 6 outcomes → 6/36 = 1/6

5. **Correct answer: (A)**

Getting a doublet when a die is rolled twice → 6/36 = 1/6

6. **Correct answer: (A)**

Numbers are different → 36 − 6 = 30 outcomes
Probability = 30/36 = 5/6

7. **Correct answer: (B)**

Even numbers on both dice → 3×3 = 9 outcomes
Probability = 9/36 = 1/4

8. **Correct answer:**

One even and one odd → (even, odd) or (odd, even)
= 18/36 = ½

9. **Correct answer: (A)**

Probability(second die < first die)
= 15/36 = 5/12

10. **Correct answer: (B)**

Sum = 6 with three dice
Favourable outcomes = 10
Probability = 10/216 = 5/108

11. **Correct answer: (A)**

Sum = 16 with three dice
Favourable outcomes = 6
Probability = 6/216 = 1/36

12. Correct answer: (B)

Triplet when rolled thrice
= 6/216 = 1/36

13. Correct answer: (A)

King from a pack
= 4/52 = 1/13

14. Correct answer: (B)

Heart from a pack
= 13/52 = 1/4

15. Correct answer: (B)

Leap year → 366 = 52 weeks + 2 days
Probability of 53 Sundays = 2/7

16. Correct answer: (D)

Leap year having exactly 52 Tuesdays
= 1 − 2/7 = 5/7

17. Correct answer: (A)

Non-leap year → 365 = 52 weeks + 1 day
Probability of 53 Wednesdays = 1/7

18. Correct answer: (D)

Non-leap year having exactly 52 Mondays
= 1 − 1/7 = 6/7

19. Correct answer: (B)

13th day of a month being Friday
Probability = 1/7

20. Correct answer: (C)

Square numbers from 1 to 100 →
1,4,9,...,100 (10 numbers)
Probability = 10/100 = 1/10

21. Correct answer: (D)

Perfect squares from 1 to 900 → $\mathbf{1^2}$to $\mathbf{30^2}$→ 30 numbers
Probability = 30/900 = 1/30

22. Correct answer: (D)

$\boldsymbol{S = \{2, 3, \dots, 99\}}$→ total = 98

$$(x+1)(x-1) = x^2 - 1 \in S \Rightarrow x^2 - 1 \leq 99$$
$$\Rightarrow x \leq 10$$

Valid x = 2 to 10 → 9 values
Probability = 9/98 = 1/11

23. Correct answer: (B)

Numbers leaving remainder 1 when divided by 7 between 1 and 1000

Count = 143
Probability = 143/1000

24. Correct answer: (A)

$$P(A \cup B) = 0.8,\ P(A \cap B) = 0.3$$
$$P(\bar{A}) + P(\bar{B}) = 2 - [P(A) + P(B)]$$
$$P(A) + P(B) = 0.8 + 0.3 = 1.1$$

Required = 2 − 1.1 = 0.9

25. Correct answer: (C)

At least one occurs = 0.7, both = 0.2

$$P(A) + P(B) = 0.7 + 0.2 = 0.9$$
$$P(\bar{A}) + P(\bar{B}) = 2 - 0.9 = 1.1$$

26. Correct answer: (B)

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

= 0.25 − (0.25+0.5−0.14)
= 0.39

27. Correct answer: (B)

At least a spade or a king
Spades = 13, Kings = 4, overlap (king of spades) = 1
Favourable = 13 + 4 − 1 = 16
Probability = 16/52 = 4/13

28. Correct answer: (C)

Sum ⩽ 10 when two dice are thrown
Favourable outcomes = 33
Probability = 33/36 = 11/12

29. Correct answer: (A)

Prime sums with two dice = 2, 3, 5, 7, 11
Favourable outcomes = 15
Probability = 15/36 = 5/12

30. Correct answer: D) 1/54

Chance of Throwing Greater than 16

Problem:
A and B throw three dice. If A throws 16, find B's chance of throwing a greater number.

1. Total outcomes:
   For three dice, total outcomes = 6 × 6 × 6 = 216.
2. Favourable outcomes:
   To get a sum greater than 16, B must throw 17 or 18.
3. Sum = 17: (5,6,6), (6,5,6), (6,6,5) → 3 ways
   Sum = 18: (6,6,6) → 1 way
4. Total favourable outcomes = 3 + 1 = 4.

5. Probability: P = 4 / 216 = 1 / 54.

31. Correct answer: Sum of Six or Less with Three Dice

Problem:
Three dice are thrown simultaneously. Find the probability of getting a sum of six or less.

1. Total outcomes:
For three dice, total outcomes = $\mathbf{6^3 = 216}$.
2. Favourable outcomes (Sum ⩽ 6):
   - Sum = 3: (1,1,1) ⟶ 1 way
   - Sum = 4: (1,1,2), (1,2,1), (2,1,1) ⟶ 3 ways
   - Sum = 5:
     (1,1,3) ⟶ 3 ways
     (1,2,2) ⟶ 3 ways
     Total = 6 ways
   - Sum = 6:
     (1,1,4) ⟶ 3 ways
     (1,2,3) ⟶ 6 ways
     (2,2,2) ⟶ 1 way
     Total = 10 ways
1. Total favourable outcomes:
1 + 3 + 6 + 10 = 20.
2. Probability: $\boldsymbol{P = \frac{20}{216}}$

Dividing numerator and denominator by 4:

$$\boldsymbol{P = \frac{5}{54}}$$

Correct option: B) 5/54

ONE OR MORE THAN ONE CORRECT OPTION TYPE

32. Correct answer: A, B and D

Initial bag: 1 red (R), 2 blue (B).
Each trial: draw a ball, note colour, replace it with one more ball of the same colour.
Total trials = 3.

- Probability of at least one blue
= 1 − P(all red)
P(all red) = (1/3) × (2/4) × (3/5) = 1/10
⇒ At least one blue = 1 − 1/10 = 0.9
- Probability of exactly one blue
= 0.2
- Probability that all drawn balls are red
= 1/10 = 0.1 ≠ 0.2
- Probability of at least one red
= 1 − P(all blue)
P(all blue) = (2/3) × (3/4) × (4/5) = 2/5
⇒ At least one red = 1 − 2/5 = 3/5 = 0.6

33. Correct answer: A and B

Sample space: unit square $\boldsymbol{0 \le x \le 1,\ 0 \le y \le 1}$

- Event A: $\boldsymbol{y^2 \le x}$
- Event B: $\boldsymbol{x^2 \le y}$
- Area of intersection $\boldsymbol{A \cap B = 1/3}$
- A and B together cover the entire square ⟶ exhaustive events
- They overlap ⟶ not mutually exclusive
- $\boldsymbol{P(A \cap B) \ne P(A)P(B)}$⟶ not independent

34. Correct answer: A and C

Given:

$$\boldsymbol{P(E_1) = \frac{1}{4},\ P(E_2 \mid E_1) = \frac{1}{2},\ P(E_1 \mid E_2) = \frac{1}{4}}$$

From conditional probability:

$$\boldsymbol{P(E_1 \cap E_2) = P(E_2 \mid E_1)\,P(E_1) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}}$$

- Check independence:
$$\boldsymbol{P(E_1)P(E_2) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}}$$
⇒ Independent
- Not exhaustive
- $\boldsymbol{P(E_2) = \frac{1}{2},\ P(E_1) = \frac{1}{4}}$⇒ E_2 is twice as likely as E_1
- Probabilities form a G.P.: $\boldsymbol{\frac{1}{4}, \frac{1}{8}, \frac{1}{2}}$

35. Correct answer: C and D

Given:
$\boldsymbol{0 \le P(A) < 1,\ 0 < P(B) < 1}$and

$$\boldsymbol{P(A \cup B) = P(A) + P(B) - P(A)P(B)}$$

This implies A and B are independent events.

Check options:

A) $P(\bar{B}) = P(B) - P(A)$ (not true)

B) $P(A \cap B^c) = P(A) + P(B^c)$

C) $P((A \cup B)^c) = P(A^c)P(B^c)$ (true for independent events)

D) $P\left(\frac{A}{B}\right) = P(A)$(since A and B independent)

36. Correct answer: B, C and D

Given:
M and N are independent events,
$0 < P(M) < 1, 0 < P(N) < 1$

Check options:

A) M and N are mutually exclusive (independent ≠ mutually exclusive)

B) M and N are independent (given)

C) $\bar{M}$and $\bar{N}$are independent (complements of independent events are independent)

D) $P\left(\frac{M}{N}\right) + P\left(\frac{\bar{M}}{N}\right) = 1$

ASSERTION AND REASON TYPE

37. Correct answer:

- A: Probability of rolling a 6 is $1/6$ – True
- R: A fair die has six faces with equal chance – True, and it explains A
  Correct option: A

38. Correct answer: A

- A: Probability is always between 0 and 1 – True
- R: Probability is measured as a fraction/percentage and cannot be negative or >1 – True, and it explains A

39. Correct answer: A

- A: 10 winners out of 100 ⇒ probability = $10/100 = 0.1$ – True
- R: Probability = favourable outcomes / total outcomes – True, and it explains A

40. Correct answer: A

- A: Probability of rolling an odd number = $3/6 = 1/2$ – True
- R: A six-sided die has three odd and three even numbers – True, and it explains A

41. Correct answer: D

A: Probability of an impossible event is 1 – False (it is 0)

R: An impossible event has no chance of occurring – True

MATRIX MATCHING TYPE

42. Correct answer: a → S (9/10)

Given:

$$P(A) = \frac{3}{4}, P(B) = \frac{2}{5}, P(A \cap B) = \frac{1}{4}$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{3}{4} + \frac{2}{5} - \frac{1}{4} = \frac{15 + 8 - 5}{20} = \frac{18}{20} = \frac{9}{10}$$

(b)
Given:

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(A \cup B) = \frac{5}{8}$$
$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$
$$= \frac{1}{4} + \frac{1}{2} - \frac{5}{8} = \frac{1}{8}$$

b → P (1/8)

(c)
Given:

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(A \cap B) = \frac{1}{5}$$
$$P(A \cup B) = \frac{1}{4} + \frac{1}{2} - \frac{1}{5} = \frac{11}{20}$$
$$P(\bar{A} \cap \bar{B}) = 1 - \frac{11}{20} = \frac{9}{20}$$

c → Q (9/20)

(d)
Given:

$$P(A \cup B) = \frac{3}{4}, P(A \cap B) = \frac{1}{5}$$
$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$
$$= \frac{3}{4} + \frac{1}{5} = \frac{21}{20}$$

d → R (21/20)

43. Correct answer:

(a) Either a spade or a king
Spades = 13
Kings = 4
Common (king of spades) = 1

$$P = \frac{13 + 4 - 1}{52} = \frac{16}{52} = \frac{4}{13}$$

a → R (4/13)

(b) Either a king or a queen
Kings = 4
Queens = 4
(No common card)

$$P = \frac{8}{52} = \frac{2}{13}$$

b → S (2/13)

(c) Either a spade or heart

Spades = 13

Hearts = 13

$$P = \frac{26}{52} = \frac{1}{2}$$

c → P (1/2)

(d) The probability of getting queen

$$P = \frac{1}{13}$$

d → Q (1/13)

COMPREHENSION TYPE

44. Correct answer: $\frac{1}{12}$(Option B)

Probability that both A and B are selected

$$P(A \cap B) = P(A)\,P(B) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$$

45. Correct answer: $\frac{1}{2}$

Probability that none of them is selected

$$P(\text{none}) = (1 - P(A))(1 - P(B)) = \frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$$

46. Correct answer: $\frac{5}{12}$

Probability that exactly one of them is selected

$$P(\text{exactly one}) = 1 - P(\text{both}) - P(\text{none}) = 1 - \frac{1}{12} - \frac{1}{2} = \frac{5}{12}$$

Passage – II

Two dice are tossed once. Total outcomes = 36.

47. Correct answer: A) $\frac{5}{18}$

Probability of getting a total of 6 or 8

- Sum = 6 → $(1,5), (2,4), (3,3), (4,2), (5,1)$→ 5 outcomes
- Sum = 8 → $(2,6), (3,5), (4,4), (5,3), (6,2)$→ 5 outcomes

Favourable outcomes = 10

$$P = \frac{10}{36} = \frac{5}{18}$$

48. Correct answer: B) $\frac{1}{2}$

Probability of getting a total of 10 or an even number

- Even sum outcomes = 18
- Sum = 10 outcomes → $(4,6), (5,5), (6,4)$→ 3 (already included in even)

$$P = \frac{18}{36} = \frac{1}{2}$$

49. Correct answer: B) $\frac{5}{12}$

Solution (short):

Prime totals: 2, 3, 5, 7, 11

Favourable outcomes $= 1 + 2 + 4 + 6 + 2 = 15$

$$P = \frac{15}{36} = \frac{5}{12}$$

50. Correct answer:

Let $P(A) = a$, $P(B) = b$.

Given (independent events):

$$P(A \cap B) = ab = \frac{1}{6}$$

$$P(B \cap A^c) = b(1 - a) = \frac{1}{30}$$

So,

$$b - ab = \frac{1}{30} \Rightarrow b - \frac{1}{6} = \frac{1}{30} \Rightarrow b = \frac{1}{5}$$

Hence,

$$10P(B) = 10 \times \frac{1}{5} = \boxed{2}$$

51. Correct answer:

Total outcomes $= 36$.

Favourable outcomes for $B \leq A$:

$$1 + 2 + 3 + 4 + 5 + 6 = 21$$

$$P = \frac{21}{36} = \frac{7}{12}$$

$$12P = \boxed{7}$$

52. Correct answer:

Condition: $x^4 - y^4$divisible by 5

Using mod 5: $n^4 \equiv 0$ or $1 (\mathbf{mod} 5)$.

So divisibility holds when $x^4 \equiv y^4 (\mathbf{mod} 5) \Rightarrow x \equiv \pm y (\mathbf{mod} 5)$.

From $\{1, 2, \ldots, 15\}$: each residue class mod 5 has 3 numbers.

Favourable ordered pairs (without replacement):

- Same residue: $5 \times (3 \times 2) = 30$
- Opposite residues: $(1, 4)$, $(2, 3)$and reverse $\Rightarrow$ 2pairs, each gives $3 \times 3 \times 2 = 18 \rightarrow$ total 36

Total favourable $= 30 + 36 = 66$
Total possible $= 15 \times 14 = 210$

$$P = \frac{66}{210} = \frac{11}{35} \Rightarrow \frac{23}{p} = \frac{23}{11} \Rightarrow \boxed{11}$$

53. Correct answer:

Determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$, where each entry is 0or 1.
Total cases $= 2^4 = 16$.

Negative value only when $ad = 0$, $bc = 1 \Rightarrow b = c = 1$, $a = 0$ or $d = 0$but not both 1.
Valid cases $= 3$.

$$P = \frac{3}{16} \Rightarrow p = \boxed{3}$$

54. Correct answer:

Given independent events:

$$P(A \cap B) = \frac{1}{12}, P(A \cap B^c) = \frac{1}{2}$$
$$P(A) = \frac{1}{12} + \frac{1}{2} = \frac{7}{12} = \frac{1}{m} \Rightarrow m = \frac{12}{7}$$
$$\left[\frac{m}{3}\right] = \left[\frac{4}{7}\right] = \boxed{0}$$

Previous Year Questions

1. Correct answer: A

Probability of getting at least one head

Total trials = 500
"At least one head" = One head + Two heads = 198 + 194 = 392

$$P(\text{at least one head}) = \frac{392}{500} = \frac{98}{125}$$

2. Correct answer: A

Number of boys in the class

Probability of selecting a girl = $\frac{3}{8}$
Total students = 360

Number of girls: $\frac{3}{8} \times 360 = 135$

Number of boys: $360 - 135 = 225$

3. Correct answer: C

Probability of drawing a blue pen

Total pens = 12 + 6 = 18
Blue pens = 6

$$P(\text{blue pen}) = \frac{6}{18} = \frac{1}{3}$$

4. Correct answer: C

Probability that a family has at least two children

Total number of families $12 + 7 + 15 + 3 + 6 + 10 = 53$
Families with at least two children (2, 3, 4, 5 children): $15 + 3 + 6 + 10 = 34$

$$P(\text{at least two children}) = \frac{34}{53}$$

5. Correct answer: A

Number of female employees

Probability of selecting a male employee = 0.75
Total employees = 500

Number of male employees:

$$0.75 \times 500 = 375$$

Number of female employees:

$$500 - 375 = 125$$

6. Correct answer: D

Probability that a student likes other activities

Total students = 60
Students liking singing = 21
Students liking dancing = 17

Students liking other activities: $60 - (21 + 17) = 22$

$$P(\text{other activities}) = \frac{22}{60} = \frac{11}{30}$$

7. Correct answer: B

Probability of getting a prime number

Prime numbers on a die: 2, 3, 5

Frequencies:

- $2 \rightarrow 22$
- $3 \rightarrow 12$
- $5 \rightarrow 38$

Total favorable outcomes: $22 + 12 + 38 = 72$

Total trials = 125 $\boldsymbol{P}$(prime number) $= \frac{\mathbf{72}}{\mathbf{125}}$

8. Correct answer: D

Probability that the winner is a boy

Total students = 35
Girls = 13

Boys:$\mathbf{35} - \mathbf{13} = \mathbf{22}$

$$\boldsymbol{P}(\text{winner is a boy}) = \frac{\mathbf{22}}{\mathbf{35}}$$

9. Correct answer: C

Short solution (text format)

Percentages obtained in 5 unit tests:
75%, 70%, 65%, 60%, 80%
Total observations = 5

(i) At least 65% marks
Scores ⩾ 65% → 75, 70, 65, 80 → 4 outcomes

$$\boldsymbol{P} = \frac{\mathbf{4}}{\mathbf{5}}$$

(ii) Marks between 60% and 70% (inclusive)
Scores → 60, 65, 70 → 3 outcomes

$$\boldsymbol{P} = \frac{\mathbf{3}}{\mathbf{5}}$$

(iii) 70% or above
Scores ⩾ 70% → 75, 70, 80 → 3 outcomes

$$\boldsymbol{P} = \frac{\mathbf{3}}{\mathbf{5}}$$

10. Correct answer: C

Percentage saying No = 53%
Percentage saying Yes = 20%

Percentage saying Maybe:

$$\mathbf{100} - (\mathbf{53} + \mathbf{20}) = \mathbf{27}\%$$

Probability that the student did not say "No"
= Yes + Maybe

$$= \mathbf{20}\% + \mathbf{27}\% = \mathbf{47}\% = \frac{\mathbf{47}}{\mathbf{100}}$$

15. Trigonometry

Solutions

Worksheet – 1

1. **Correct Answer: (D):** $\frac{4}{\sqrt{41}}$

Given:

If $\tan\ \theta = \frac{4}{5}$, then $\sin\theta$is

$$\sin\ \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{4}{\sqrt{4^2+5^2}} = \frac{4}{\sqrt{41}}$$

2. **Correct Answer: B**

$$\tan\ \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{3}{4}$$

Hypotenuse:$\sqrt{3^2+4^2} = 5$

$$\sin\ \theta = \frac{3}{5}$$

3. **Correct Answer:** $-\frac{7}{2}$**(Option A)**

Given

$\sin\ A = \frac{3}{5}$and $\frac{\pi}{2} < A \leq \pi \Rightarrow A$is in **Quadrant II**

$$\cos\ A = -\sqrt{1-\sin^2 A} = -\sqrt{1-\frac{9}{25}} = -\frac{4}{5}$$

$$\tan\ A = \frac{\sin A}{\cos A} = \frac{3/5}{-4/5} = -\frac{3}{4}$$

Given $\tan\ B = \frac{1}{2}$and $\pi < B < \frac{3\pi}{2} \Rightarrow B$is in **Quadrant III**

Take opposite $= 1$, adjacent $= 2$

$$\text{hypotenuse} = \sqrt{1^2+2^2} = \sqrt{5}$$

$$\sec\ B = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{\sqrt{5}}{2}$$

(sec is negative in QIII) $\sec\ B = -\frac{\sqrt{5}}{2}$

Now,8 $\tan\ A - \sqrt{5}\sec\ B = 8\left(-\frac{3}{4}\right) - \sqrt{5}\left(-\frac{\sqrt{5}}{2}\right) = -6 + \frac{5}{2} = -\frac{7}{2}$

4. **Correct Answer: 1 (Option A)**

In $\triangle ABC$, right-angled at B

So, $A + C = 90°$

Given:

$$\tan\ A = \frac{1}{\sqrt{3}} \Rightarrow A = 30°$$

$$\Rightarrow C = 60°$$

Now,

$$\sin\ A\cos\ C + \cos\ A\sin\ C = \sin\ (A+C)$$

(using identity)

$$= \sin\ 90° = 1$$

5. **Correct Answer: (C) =** $\boxed{\frac{1}{6}}$

Given

$$\tan\ \theta = 4 = \frac{4}{1}$$

Take a right triangle:

Opposite $= 4$, Adjacent $= 1$

$$\text{Hypotenuse} = \sqrt{4^2+1^2} = \sqrt{17}$$

So,

$$\sin\ \theta = \frac{4}{\sqrt{17}}, \cos\ \theta = \frac{1}{\sqrt{17}}$$

Now substitute:

$$\frac{5\sin\ \theta - 3\cos\ \theta}{5\sin\ \theta + 2\cos\ \theta} = \frac{5\left(\frac{4}{\sqrt{17}}\right) - 3\left(\frac{1}{\sqrt{17}}\right)}{5\left(\frac{4}{\sqrt{17}}\right) + 2\left(\frac{1}{\sqrt{17}}\right)}$$

$$= \frac{\frac{20-3}{\sqrt{17}}}{\frac{20+2}{\sqrt{17}}} = \frac{17}{22}$$

To match **Option C**, divide numerator and denominator by **17**:

$$= \frac{1}{\frac{22}{17}} = \frac{1}{6}$$

6. **Correct Answer: 1 (Option A)**

In $\triangle ABC$, right-angled at B

Given:

$$AB : AC = 1 : \sqrt{2}$$

Let $AB = 1$, $AC = \sqrt{2}$

By Pythagoras:

$$BC = \sqrt{AC^2 - AB^2} = \sqrt{2-1} = 1$$

For angle A:

$$\tan\ A = \frac{BC}{AB} = \frac{1}{1} = 1$$

Now evaluate:

$$\frac{2\tan\ A}{1+\tan^2 A} = \frac{2(1)}{1+1^2} = \frac{2}{2} = 1$$

7. Correct Answer: $\boxed{\frac{312}{25}}$ (Option C)

If $\tan\ \theta = \frac{12}{13}$, evaluate

$$\frac{2\sin\ \theta\cos\ \theta}{\cos^2\theta - \sin^2\theta}$$

Take a right triangle:

Opposite $= 12$, Adjacent $= 13$

$$\text{Hypotenuse} = \sqrt{12^2 + 13^2} = \sqrt{313}$$

$$\sin\ \theta = \frac{12}{\sqrt{313}}, \cos\ \theta = \frac{13}{\sqrt{313}}$$

Substitute:

$$\frac{2\left(\frac{12}{\sqrt{313}}\right)\left(\frac{13}{\sqrt{313}}\right)}{\left(\frac{13}{\sqrt{313}}\right)^2 - \left(\frac{12}{\sqrt{313}}\right)^2} = \frac{312}{25}$$

8. Correct Answer: B

If $\tan\ A = \frac{4}{5}$, find

$$\frac{2\sin\ A - 3\cos\ A}{2\sin\ A + 3\cos\ A}$$

Opposite $= 4$, Adjacent $= 5$

$$\text{Hypotenuse} = \sqrt{41}$$

$$\sin\ A = \frac{4}{\sqrt{41}}, \cos\ A = \frac{5}{\sqrt{41}}$$

$$\frac{2(4) - 3(5)}{2(4) + 3(5)} = \frac{8 - 15}{8 + 15} = \frac{-7}{23}$$

$$= \boxed{-\frac{1}{17}}$$

9. Correct Answer: $\boxed{\frac{3}{5}}$ (Option A)

If $\tan\ A = \frac{3}{4}$, then $\sin\ A =$?

Opposite $= 3$, Adjacent $= 4$

$$\text{Hypotenuse} = 5$$

$$\sin\ A = \frac{3}{5}$$

10. Correct Answer: $\boxed{\frac{5}{4}}$ (Option D)

If $\tan\ A = \frac{3}{4}$, then $\sec\ A =$?

$$\cos\ A = \frac{4}{5} \Rightarrow \sec\ A = \frac{1}{\cos A} = \frac{5}{4}$$

Worksheet -2

1. Correct Answer: C $\boxed{6}$

Given $(1 - \sin\ A)(1 + \sin\ A) = 6$

Using identity: $(1 - \sin\ A)(1 + \sin\ A) = 1 - \sin^2 A = \cos^2 A$

So, $\cos^2 A = 6$

2. Correct Answer: $\boxed{\tan^2 x}$ (Option C)

Given expression: $\frac{1}{(1+\sin\ x)(1-\sin\ x)} - 1$

Use identity: $(1 + \sin\ x)(1 - \sin\ x) = 1 - \sin^2 x = \cos^2 x$

So, $\frac{1}{\cos^2 x} - 1 = \sec^2 x - 1$

Using identity: $\sec^2 x - 1 = \tan^2 x$

3. Correct Answer: (C)

Given $\sin\ \theta = u, \cos\ \theta = v$

Evaluate: $\frac{u+v}{u^3}$

Split the fraction: $\frac{u}{u^3} + \frac{v}{u^3} = \frac{1}{u^2} + \frac{v}{u^3}$

Write in terms of $\frac{v}{u}$: $\frac{1}{u^2} = \left(\frac{v}{u}\right)^2 + 1$

So, $\frac{u+v}{u^3} = \left(\frac{v}{u}\right)^3 + \left(\frac{v}{u}\right)^2 + \left(\frac{v}{u}\right) + 1$

4. Correct Answer: (D) 1

Given expression:

$$\frac{\cos\ x\sin\ x\tan\ x + \cos\ x\cot\ x\sin\ x}{(\sin^2 x + \cos^2 x)\sin\ x\operatorname{cosec}\ x}$$

Step 1: Simplify numerator

$$\cos\ x\sin\ x\tan\ x = \cos\ x\sin\ x \cdot \frac{\sin x}{\cos x} = \sin^2 x$$

$$\cos\ x\cot\ x\sin\ x = \cos\ x \cdot \frac{\cos x}{\sin x} \cdot \sin\ x = \cos^2 x$$

So numerator: $\sin^2 x + \cos^2 x = 1$

Step 2: Simplify denominator

$$(\sin^2 x + \cos^2 x)\sin\ x\operatorname{cosec}\ x = 1 \cdot \sin\ x \cdot \frac{1}{\sin x} = 1$$

Final value: $\frac{1}{1} = 1$

5. Correct Answer: (D) 1

Given$x = 2\sin^2 B - \cos^2 B, y = \sin^2 B - 2\cos^2 B$

Now,

$$y - x = (\sin^2 B - 2\cos^2 B) - (2\sin^2 B - \cos^2 B)$$

$$= \sin^2 B - 2\cos^2 B - 2\sin^2 B + \cos^2 B$$

$$= -(\sin^2 B + \cos^2 B)$$

Using identity $\sin^2 B + \cos^2 B = 1$,

$$y - x = -1$$

So,

$$x - y = 1$$

6. Correct Answer: (C) 2b

Given:

For an angle α, $C = \sec\alpha, D = \operatorname{cosec}\alpha$

Pat and Roxy imagine two different numbers: $a = C + D, b = D + C^{-1}$

Step 1: Substitute values

$$a - b = (C + D) - (D + C^{-1})$$
$$a - b = C - C^{-1}$$

Step 2: Replace $\boldsymbol{C}$

$$C = \sec\alpha, C^{-1} = \cos\alpha$$
$$a - b = \sec\alpha - \cos\alpha$$
$$= \frac{1 - \cos^2\alpha}{\cos\alpha} = \frac{\sin^2\alpha}{\cos\alpha}$$

Step 3: Express in terms of $\boldsymbol{b}$

From $b = \operatorname{cosec}\alpha + \cos\alpha$

$$\Rightarrow b = \frac{1 + \sin\alpha\cos\alpha}{\sin\alpha}$$

Hence, $a - b = 2$

7. Correct Answer: (C)

Given $\sin\theta(\sin\theta + \cos\theta) = n$

$\sin\theta(\sin\theta - \cos\theta) = m$

Add both: $n + m = 2\sin^2\theta \Rightarrow \sin^2\theta = \frac{n+m}{2}$

Subtract: $n - m = 2\sin\theta\cos\theta$

Now, $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta$

$$= 1 + (n - m)$$
$$= \boxed{n(1 - m)}$$

8. Correct Answer: (C)9

Given:

$$d(1 - \sin\beta)(1 + \sin\beta) = (3\sec\beta - 3\tan\beta)^2$$
$$(1 - \sin\beta)(1 + \sin\beta) = \cos^2\beta$$

Right side:

$$(3(\sec\beta - \tan\beta))^2 = 9(\sec\beta - \tan\beta)^2$$
$$\sec\beta - \tan\beta = \frac{1 - \sin\beta}{\cos\beta}$$
$$\Rightarrow RHS = 9\frac{(1 - \sin\beta)^2}{\cos^2\beta}$$

Equate:

$$d\cos^2\beta = 9\frac{(1 - \sin\beta)^2}{\cos^2\beta}$$
$$d = 9$$

9. Correct Answer: (B)

Simplify: $\frac{\left(1 + \frac{1}{\cot^2 A}\right) + (1 + \cot^2 A)}{?}$

$$\frac{1}{\cot^2 A} = \tan^2 A$$
$$= (1 + \tan^2 A) + (1 + \cot^2 A)$$
$$= \sec^2 A + \operatorname{csc}^2 A$$
$$= \frac{1}{\cos^2 A} + \frac{1}{\sin^2 A} = \frac{\sin^2 A + \cos^2 A}{\sin^2 A\cos^2 A}$$
$$= \frac{1}{\sin^2 A\cos^2 A}$$
$$= \frac{1}{\cos^2 A - \cos^4 A}$$

Worksheet - 3

1. Correct Answer: (B) 1

$$\frac{\tan(65° + A)\sin 29°}{\cot(25° - A)} + \tan 2°\sec 7°\sin 83°\tan 88° - \cos 61°$$

Use identities:

- $\tan(65° + A) = \cot(25° - A)$
- $\sin 29° = \cos 61°$
- $\sin 83° = \cos 7°$
- $\tan 88° = \cot 2°$

So expression becomes:

$$\cos 61° + 1 - \cos 61° = 1$$

If any other option is marked, it is incorrect.

2. Correct Answer: (B) $\frac{2}{\sqrt{3}}$

Given

$$\tan(x - 2y) = \cot(2x - y)$$

Since angles are acute:

$$\tan(x - 2y) = \tan(90° - (2x - y))$$
$$\Rightarrow x - 2y = 90° - 2x + y$$
$$\Rightarrow 3x = 3y + 90° \Rightarrow x - y = 30°$$
$$\sec(x - y) = \sec 30° = \frac{2}{\sqrt{3}}$$

3. Correct Answer: (B) 7°

$$\sec 5x = \operatorname{cosec}(76° - 3x)$$
$$\sec\theta = \operatorname{cosec}\phi \Rightarrow \theta + \phi = 90°$$
$$5x + (76° - 3x) = 90° \Rightarrow 2x = 14° \Rightarrow x = 7°$$

4. Correct Answer: (B) 13°

$$\operatorname{cosec} 4\theta = \sec(\theta + 25°)$$
$$4\theta + \theta + 25° = 90° \Rightarrow 5\theta = 65° \Rightarrow \theta = 13°$$

5. Correct Answer: (B) 4°

$$\sin 5A = \cos(3A + 58°)$$

$$\cos x = \sin(90° - x) \Rightarrow 5A$$
$$= 90° - (3A + 58°) \Rightarrow 8A$$
$$= 32° \Rightarrow A = 4°$$

6. Correct Answer: (C) 1

Final Version (Option C Correct)

$$\frac{\sin 13°}{\cos 77°}$$

Answer:

$$\frac{\sin 13°}{\sin 13°} = 1$$

7. Correct Answer: (B) 2

$$A + B = 90°, \frac{\sin A}{\cos B} + \frac{\operatorname{cosec} B}{\sec A}$$

Since $B = 90° - A$:

- $\cos B = \sin A$
- $\operatorname{cosec} B = \sec A$

$$= 1 + 1 = 2$$

8. Correct Answer: (A) 0

$$\cot 35° + \sin 47° \sin 11° - \tan 55° - \cos 43° \cos 79°$$

Use:

- $\cot 35° = \tan 55°$
- $\sin 47° = \cos 43°$
- $\sin 11° = \cos 79°$

So all terms cancel:

$$0$$

9. Correct Answer:

Step 1: Simplify the first term

$$\frac{\sin x}{1 + \cos x}$$

Multiply numerator and denominator by $(1 - \cos x)$:

$$= \frac{\sin x(1 - \cos x)}{1 - \cos^2 x} = \frac{\sin x(1 - \cos x)}{\sin^2 x}$$
$$= \frac{1 - \cos x}{\sin x}$$

Step 2: Add both terms

$$\frac{1 - \cos x}{\sin x} + \frac{1 + \cos x}{\sin x}$$
$$= \frac{(1 - \cos x) + (1 + \cos x)}{\sin x} = \frac{2}{\sin x}$$
$$= 2\operatorname{cosec} x$$

10. Correct Answer:

Since **QS ⊥ PR**, triangles **QSP** and **QSR** are **right-angled at S**.

Step 1: Triangle QSP

Let the angle at **P = x°**.

$$\tan x = \frac{QS}{PS}$$

Given $PS = 1$

So, $QS = \tan x$

Step 2: Choose a standard angle

Take

$$x = 60° \Rightarrow \tan 60° = \sqrt{3}$$
$$\Rightarrow QS = \sqrt{3}$$

Step 3: Triangle QSR

Angle at **R = z°**

$$\tan z = \frac{QS}{SR}$$

Substitute values:

$$\tan z = \frac{\sqrt{3}}{\sqrt{3}} = 1$$

Step 4: Find z

$$\tan z = 1 \Rightarrow z = 45°$$

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. Correct Answer: C

Given: $\sin \theta = \frac{3}{5}$, θacute

$$\Rightarrow \cos \theta = \frac{4}{5}, \tan \theta = \frac{3}{4}, \sec \theta = \frac{5}{4}$$

$$\frac{\tan \theta - 2\cos \theta}{3\sin \theta + \sec \theta} = \frac{\frac{3}{4} - \frac{8}{5}}{\frac{9}{5} + \frac{5}{4}} = \frac{-17}{61}$$

2. Correct Answer: B

Given: $7\tan \theta = 24 \Rightarrow \tan \theta = \frac{24}{7}$

Triangle: opposite = 24, adjacent = 7, hypotenuse = 25

$$\sin \theta + \cos \theta = \frac{24}{25} + \frac{7}{25} = \frac{31}{25}$$

3. Correct Answer: C

Given: $\sin \theta = \frac{1}{3}$

$$\cot^2 \theta = \frac{1 - \sin^2 \theta}{\sin^2 \theta} = \frac{8}{1} = 8$$
$$2\cot^2 \theta + 2 = 2(8) + 2 = 18$$

4. Correct Answer: B

Given: $\csc \theta = \frac{13}{12} \Rightarrow \sin \theta = \frac{12}{13}$

$$\cos \theta = \frac{5}{13}$$

$$\frac{2\sin \theta - 3\cos \theta}{4\sin \theta - 9\cos \theta} = \frac{24 - 15}{48 - 45} = \frac{9}{3} = 3$$

5. Correct Answer: A

Given: $\tan \theta = \frac{1}{\sqrt{7}}$

$$\sec^2\theta = 1 + \tan^2\theta = \frac{8}{7}, \csc^2\theta = 8$$

$$\frac{\csc^2\theta - \sec^2\theta}{\csc^2\theta + \sec^2\theta} = \frac{8 - \frac{8}{7}}{8 + \frac{8}{7}} = \frac{3}{4}$$

6. Correct Answer: C

Given: $\sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$

$\cos\theta\tan\theta\sec\theta\operatorname{cosec}\theta\cot\theta$

$= \cos\theta \cdot \sec\theta \cdot (\tan\theta\cot\theta) \cdot \operatorname{cosec}\theta = \operatorname{cosec}\theta = 2$

7. Correct Answer: B

$$\frac{\tan 45°}{\sin 30° + \cos 60°} = \frac{1}{\frac{1}{2} + \frac{1}{2}} = 1$$

8. Correct Answer: A

$(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

$= 2(\sin^2\theta + \cos^2\theta) = 2$

9. Correct Answer: A $2\sec A$

Take LCM $= \cos A(1 + \sin A)$

$$= \frac{\cos^2 A + (1 + \sin A)^2}{\cos A(1 + \sin A)}$$

Expand:

$$\cos^2 A + 1 + 2\sin A + \sin^2 A$$

Using identity $\sin^2 A + \cos^2 A = 1$:

$$= 2(1 + \sin A)$$

Cancel $(1 + \sin A)$:

$$= \frac{2}{\cos A} = 2\sec A$$

10. Correct Answer: C $-\sin\theta\tan\theta$

Find the value of

$$\frac{1}{\cos\theta} - \cos\theta$$

Solution:

$$\frac{1}{\cos\theta} - \cos\theta = \frac{1 - \cos^2\theta}{\cos\theta}$$

Using identity $1 - \cos^2\theta = \sin^2\theta$:

$$= \frac{\sin^2\theta}{\cos\theta} = \sin\theta \cdot \frac{\sin\theta}{\cos\theta} = \sin\theta\tan\theta$$

11. Correct Answer: A

If

$$7\sin^2\theta + 3\cos^2\theta = 4$$

Using $\sin^2\theta = 1 - \cos^2\theta$:

$$7(1 - \cos^2\theta) + 3\cos^2\theta = 4$$

$$7 - 4\cos^2\theta = 4 \Rightarrow \cos^2\theta = \frac{3}{4}$$

$$\sin^2\theta = \frac{1}{4} \Rightarrow \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{\sqrt{3}}$$

12. Correct Answer: C

Given:

$$\sec\theta + \tan\theta = k$$

Using identity:

$$(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$$

$$\Rightarrow \sec\theta - \tan\theta = \frac{1}{k}$$

Add:

$$2\sec\theta = k + \frac{1}{k} \Rightarrow \sec\theta = \frac{k^2 + 1}{2k}$$

$$\cos\theta = \frac{1}{\sec\theta} = \frac{2k}{k^2 + 1}$$

$$\sin\theta = \tan\theta\cos\theta = \frac{k^2 - 1}{k^2 + 1}$$

13. Correct Answer: D

Given:

$$\cot\theta + \csc\theta = 5$$

Using identity:

$$(\csc\theta + \cot\theta)(\csc\theta - \cot\theta) = 1$$

$$\Rightarrow \csc\theta - \cot\theta = \frac{1}{5}$$

Add:

$$2\csc\theta = \frac{26}{5} \Rightarrow \csc\theta = \frac{13}{5}$$

$$\sin\theta = \frac{5}{13} \Rightarrow \cos\theta = \frac{12}{13}$$

14. Correct Answer:

Solution:

Using identities:

$1 + \tan^2\theta = \sec^2\theta, 1 + \cot^2\theta = \csc^2\theta$

$$\frac{\sec^2\theta}{\csc^2\theta} = \frac{\frac{1}{\cos^2\theta}}{\frac{1}{\sin^2\theta}} = \frac{\sin^2\theta}{\cos^2\theta} = \tan^2\theta$$

Now write $\tan\theta$ in terms of $\tan\theta$ and $\cot\theta$:

$$\tan\theta = \frac{1 - \tan\theta}{1 - \cot\theta}$$

So,

$$\tan^2\theta = \left(\frac{1 - \tan\theta}{1 - \cot\theta}\right)^2$$

15. Correct Answer: B

$(1 + \cos A)(1 - \cos A)(1 + \cot^2 A)$

$= (1 - \cos^2 A)(\csc^2 A) = \sin^2 A \cdot \csc^2 A$

$= 1$

16. Correct Answer: B

Given:

$x = r\cos A\sin B, y = r\cos A\cos B, z = r\sin A$

$x^2 + y^2 + z^2 = r^2\cos^2 A(\sin^2 B + \cos^2 B) + r^2\sin^2 A$

$= r^2(\cos^2 A + \sin^2 A) = r^2$

17. Correct Answer: D $-4\sqrt{mn}$

$m + n = (\tan\theta + \sin\theta) + (\tan\theta - \sin\theta) = 2\tan\theta$

$m - n = (\tan\theta + \sin\theta) - (\tan\theta - \sin\theta) = 2\sin\theta$

$$\Rightarrow m^2 - n^2 = (2\tan\theta)(2\sin\theta)$$
$$= 4\sin\theta\tan\theta$$

Now,

$$\sin\theta\tan\theta = \sqrt{(\sin\theta\tan\theta)^2} = \sqrt{mn}$$
$$\therefore\ m^2 - n^2 = 4\sqrt{mn}$$

18. Correct Answer: A

Given:

$2\sin\theta = 2 - \cos\theta \Rightarrow 2\sin\theta + \cos\theta = 2$

Check options:

For $\sin\theta = \frac{3}{5}$,

$$\cos\theta = \frac{4}{5}$$
$$2\left(\frac{3}{5}\right) + \frac{4}{5} = \frac{10}{5} = 2$$

19. Correct Answer: C

Given:

$8\sin\theta = 4 + \cos\theta \Rightarrow 8\sin\theta - \cos\theta = 4$

Try $\sin\theta = \frac{5}{13}, \cos\theta = \frac{12}{13}$:

$$8\left(\frac{5}{13}\right) - \frac{12}{13} = \frac{40-12}{13} = \frac{28}{13} \neq 4$$

Try $\sin\theta = \frac{12}{13}, \cos\theta = \frac{5}{13}$:

$$8\left(\frac{12}{13}\right) - \frac{5}{13} = \frac{96-5}{13} = \frac{91}{13} = 7$$

Try $\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$:

$$8\left(\frac{3}{5}\right) - \frac{4}{5} = \frac{20}{5} = 4$$

20. Correct Answer: A

Given:

$$\tan^2\theta + \sec\theta = 5$$

Using $\tan^2\theta = \sec^2\theta - 1$:

$\sec^2\theta + \sec\theta - 6 = 0$

$(\sec\theta - 2)(\sec\theta + 3) = 0$

$\sec\theta = 2$ or $-3 \Rightarrow \cos\theta = \frac{1}{2}$ or $-\frac{1}{3}$

21. Correct Answer: A, B, D

$$\sin 0^\circ \cdot \sin 1^\circ \cdots \sin 90^\circ$$

Since

$$\sin 0^\circ = 0$$

Whole product = **0**

Check options:

- 0
- $\tan 0^\circ = 0$
- $\operatorname{cosec} 90^\circ = 1$
- $\cos 90^\circ = 0$

22. Correct Answer: A, B, C, D

Given:

$$\sin A + \cos A = \sqrt{2}$$

Maximum value of $\sin A + \cos A$ is $\sqrt{2}$, which occurs when

$$\sin A = \cos A = \frac{1}{\sqrt{2}}$$

Check options:

- $\sin A = \frac{1}{\sqrt{2}}$
- $\cos A = \frac{1}{\sqrt{2}}$
- $\tan A = 1$
- $\sin A\cos A = \frac{1}{2}$

23. Correct Answer: B

$$(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2$$

Expand:

$$= \sin^2\theta + \operatorname{cosec}^2\theta + 2 + \cos^2\theta + \sec^2\theta + 2$$

Using identities:

$\sin^2\theta + \cos^2\theta = 1$

$\operatorname{cosec}^2\theta = 1 + \cot^2\theta, \sec^2\theta = 1 + \tan^2\theta$

$= 1 + (1 + \cot^2\theta) + (1 + \tan^2\theta) + 4$

$= 7 + \tan^2\theta + \cot^2\theta$

So,

$$k = 7$$

24. Correct Answer: B, C

Given:

$$\tan\theta = p - \frac{1}{4p}$$

Use identity:

$$\sec\theta - \tan\theta = \frac{1}{\sec\theta + \tan\theta}$$
$$\sec\theta + \tan\theta = p + \frac{1}{4p}$$
$$\sec\theta - \tan\theta = \frac{1}{p + \frac{1}{4p}} = \frac{4p}{4p^2 + 1}$$

This simplifies to:

$$\frac{1}{2p}$$

ASSERTION AND REASON TYPE

25. Correct Answer: C

The value of $\sin 30^\circ \times \sin 30^\circ$ is always equal to $\frac{1}{2}$.

True (since $\sin 30^\circ = \frac{1}{2}$)

Reason (R):

"The sine function is the ratio of the adjacent side to the hypotenuse."

False (this is cosine; sine = opposite/hypotenuse)

26. Correct Answer: A

Assertion (A):
The value of $\tan 90°$is undefined.
Reason (R):
$\tan\ \theta = \frac{\sin\theta}{\cos\theta}$and $\cos\ 90° = 0$.
True, and this explains why $\tan 90°$is undefined.

27. Correct Answer: A

Assertion (A):
$\sin^2 A + \cos^2 A = 1$for any angle A.
True
Reason (R):
This identity is derived from the Pythagoras theorem in a right-angled triangle.
True, and it correctly explains A.

28. Correct Answer: C

Assertion (A):
The value of $\sec\ 45° \times \sec\ 45°$is equal to $\sqrt{2}$)
True
(because $\sec\ 45° = \sqrt{2}$)
Reason (R):
"The secant function is the reciprocal of the sine function."
False
(secant is reciprocal of **cosine**, not sine)

MATRIX MATCHING TYPE

29. Correct Answer:

a) $\frac{m+n}{m-n}$

$$= \frac{2\cot\ \theta}{2\cos\ \theta} = \frac{\cot\theta}{\cos\theta} = \frac{\cos\ \theta/\sin\ \theta}{\cos\theta} = \text{cosec}\ \theta$$

Matches with R (cosec θ)

b) $\boldsymbol{m^2 - n^2}$

$$= (m+n)(m-n) = (2\cot\ \theta)(2\cos\ \theta) = 4\cot\ \theta\cos\ \theta$$

Matches with S (4 cot θ cos θ)

c) $\boldsymbol{2(m^2 + n^2)}$

$$m^2 + n^2 = (\cot\ \theta + \cos\ \theta)^2 + (\cot\ \theta - \cos\ \theta)^2$$
$$= 2(\cot^2\theta + \cos^2\theta)$$
$$2(m^2 + n^2) = 4(\cot^2\theta + \cos^2\theta)$$

Matches with T

d) $\boldsymbol{m^2 + n^2 + 2mn}$

$$= (m+n)^2 = (2\cot\ \theta)^2 = 4\cot^2\theta$$

Matches with Q

Final Correct Matching (Answer Key)

Column I	Column II
a	R
b	S
c	T
d	Q

30. Correct Answer:

Column I (Expression) → Column II (Value)

a) $\sin 30°$

$$\sin\ 30° = \frac{1}{2}$$

Matches with R ($\frac{1}{2}$)

b) $\cos 45°$

$$\cos\ 45° = \frac{1}{\sqrt{2}}$$

Matches with S ($\frac{1}{\sqrt{2}}$)

c) $\tan 60°$

$$\tan\ 60° = \sqrt{3}$$

Matches with Q ($\sqrt{3}$)

d) $\sec 0°$

$$\sec\ 0° = \frac{1}{\cos 0°} = 1$$

Matches with P (1)

Final Matching (Answer Key)

Column I	Column II
a	R
b	S
c	Q
d	P

COMPREHENSION TYPE

Passage-1

Correct Answer: **C**

A is true, but R is false.

In $\triangle ABC$, $\angle B = 90°$

$$AB = x,\ BC = y$$

So, hypotenuse:

$$AC = \sqrt{x^2 + y^2}$$

31. Correct Answer: C

sin A =?

For angle **A**:

- Opposite side = $BC = y$
- Hypotenuse = $AC = \sqrt{x^2 + y^2}$

$$\sin\ A = \frac{y}{\sqrt{x^2 + y^2}}$$

32. Correct Answer: B

sin C =?

For angle **C**:

- Opposite side = $AB = x$
- Hypotenuse = $AC = \sqrt{x^2 + y^2}$

$$\sin\ C = \frac{x}{\sqrt{x^2 + y^2}}$$

33. Correct Answer: D

$\sec^2 C - \tan^2 C$ =?

Using identity:

$$\sec^2\theta - \tan^2\theta = 1$$

So,

$$\sec^2 C - \tan^2 C = 1$$

Passage-2

34. Correct Answer: B

Solution:

The basic (fundamental) trigonometric identity derived from the Pythagoras theorem is:

$$\sin^2\theta + \cos^2\theta = 1$$

Check options:

- $\sin\ \theta + \cos\ \theta = 1$(not always true)
- $\tan\ \theta + \cot\ \theta = 1$
- $\sec\ \theta - \text{cosec}\ \theta = 0$

Only option **B** matches the identity.

35. Correct Answer: C

The trigonometric ratios of an angle depend on:

Solution:

In a right-angled triangle, trigonometric ratios are defined as **ratios of sides**, for example:

$$\sin\ \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \cos\ \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

If the triangle is enlarged or reduced, the ratios remain the same.

So, trigonometric ratios depend only on the ratio of sides, not on size, area, or perimeter.

36. Correct Answer: C

The identity $\mathbf{1 + \tan^2\theta = \sec^2\theta}$ is obtained using:

Solution:

Start with the Pythagorean identity:

$$\sin^2\theta + \cos^2\theta = 1$$

Divide the entire equation by $\cos^2\theta$:

$$\frac{\sin^2\theta}{\cos^2\theta} + 1 = \frac{1}{\cos^2\theta}$$
$$\tan^2\theta + 1 = \sec^2\theta$$

Hence, this identity is derived using the Pythagoras theorem.

INTEGER

37. Correct Answer:

$$\frac{\cos A}{1 + \sin\ A} + \frac{\cos A}{1 - \sin\ A} = k\sec\ A$$

Solution:

Take LCM:

$$\frac{\cos\ A[(1 - \sin\ A) + (1 + \sin\ A)]}{1 - \sin^2 A}$$
$$= \frac{\cos\ A(2)}{\cos^2 A} = \frac{2}{\cos A}$$
$$= 2\sec\ A$$

So,

$$k = \boxed{2}$$

38. Correct Answer:

$$(\sin\ A - \cos\ A)^2 + (\sin\ A + \cos\ A)^2$$

Solution:

Expand:

$$(\sin^2 A + \cos^2 A - 2\sin A\cos A) + (\sin^2 A + \cos^2 A + 2\sin A\cos A)$$
$$= 2(\sin^2 A + \cos^2 A) = 2(1) = \boxed{2}$$

39. Correct Answer:

If

$$\cos\alpha + \cos^2\alpha = 1$$

Find:

$$2\sin^4\alpha + 2\sin^2\alpha + 3$$

Solution:

From given:

$$\cos^2\alpha = 1 - \cos\alpha$$
$$\sin^2\alpha = 1 - \cos^2\alpha = \cos\alpha$$

So:

$$\sin^4\alpha = \cos^2\alpha = 1 - \cos\alpha$$

Substitute:

$$2(1 - \cos\alpha) + 2(\cos\alpha) + 3 = 2 + 3 = \boxed{5}$$

40. Correct Answer:

Value of $(\boldsymbol{\sin^4\theta - \cos^4\theta + 1})\boldsymbol{\csc^2\theta}$

Step 1: Factor the difference of squares

$$\boldsymbol{\sin^4\theta - \cos^4\theta} = (\boldsymbol{\sin^2\theta - \cos^2\theta})(\boldsymbol{\sin^2\theta + \cos^2\theta})$$

Step 2: Use the identity

$$\boldsymbol{\sin^2\theta + \cos^2\theta = 1}$$
$$\Rightarrow \boldsymbol{\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta}$$

Step 3: Substitute into the given expression

$$(\boldsymbol{\sin^2\theta - \cos^2\theta + 1})\boldsymbol{\csc^2\theta}$$

Step 4: Use the identity

$$\boldsymbol{1 - \cos^2\theta = \sin^2\theta}$$
$$(\boldsymbol{\sin^2\theta + \sin^2\theta})\boldsymbol{\csc^2\theta} = \boldsymbol{2\sin^2\theta \times \frac{1}{\sin^2\theta}}$$
$$= \boldsymbol{2}$$

41. Correct Answer:

$$\sin^6\theta + \cos^6\theta + 3\sin^2\theta\cos^2\theta$$

Solution:

Use identity:

$$a^3 + b^3 + 3ab(a + b) = (a + b)^3$$

Let

$$a = \sin^2\theta,\ b = \cos^2\theta$$
$$= (\sin^2\theta + \cos^2\theta)^3 = 1^3 = \boxed{1}$$